SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)
(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi)
Accredited by NAAC with 'A' Grade
Recognised as Scientific and Industrial Research Organisation
CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

ESTD: 1980

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R17)

I/IV B.TECH

(With effect from **2017-2018** Admitted Batch onwards**)**

Under Choice Based Credit System

GROUP-B (CSE, ECE & IT)

I-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 BS 1101	English – I*	3	3	1	--	4	30	70	100
B17 BS 1102	Mathematics – I*	3	3	1	--	4	30	70	100
B17 BS 1103	Mathematics-II	3	3	1	--	4	30	70	100
B17 BS 1104	Engineering Physics	3	3	1	--	4	30	70	100
B17 CS 1101	Computer Programming Using C	3	3	1	--	4	30	70	100
B17 CE 1101	Environmental Studies *	2	2	1	--	3	30	70	100
B17 BS 1106	Engineering Physics Lab	2	--	--	3	3	50	50	100
B17 BS 1108	English Communication Skills Lab – I *	2	--	--	3	3	50	50	100
# DL1	Department Lab	2	--	--	3	3	50	50	100
B17 BS 1110	Engineering Physics Virtual Labs-Assignments	--	--	--	2	2	--	--	--
B17 BS 1112	NCC	--	--	--	2	2	--	--	--
	Total	**23**	**17**	**6**	**13**	**36**	**330**	**570**	**900**

*** Common to both Group - A and Group - B**

#DL 1	CSE & IT	B17 CS 1102	C Programming Lab & Hardware Fundamentals
	ECE	B17 CS 1103	C Programming Lab

Code: B17 BS 1101

ENGLISH - I
(Common to all Branches)

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. To improve the language proficiency of the students in English with emphasis on LSRW skills.
2. To enable the students to study and comprehend the prescribed lessons and subjects more effectively relating to their theoretical and practical components.
3. To develop the communication skills of the students in both the formal and informal situations.
4. To expose the students to various forms of writing in formal settings in order to build confidence in standard grammar
5. To introduce the students various types of reading comprehension practices.

Course Outcomes: At the end of the Course, Student will be able to:

1. Understand the rudiments of LSRW Skills, comprehension and fluency of speech.
2. Gain confidence and competency in vocabulary and grammar.
3. Listen, speak, read and write effectively in both the academic and non- academic environment.
4. Extend his/her reading skills towards literature.
5. Strengthen his/her analytical and compositional skills.

SYLLABUS

Life through Language: An Effective Learning Experience
Life through Language has a systematic structure that builds up communicative ability progressively through the chapters. It will enable the learner to manage confusion; frame question for themselves and others; develop new ideas; support ideas with evidence; express themselves with poise and clarity; and think critically. Acquisition of skill leads to confidence.

UNIT-I
People and Places:-Word search - Ask yourself-Self-assessment-I -Self-assessment-II - Sentence and its types- Describing people, places and events-Writing sentences-Self-awareness-Self-motivation, Dialogue writing.

UNIT-II
Personality and Lifestyle:- Word quiz – Verbs-Adverbs-Negotiations-Proving yourself-Meeting Carl Jung- Describing yourself- Living in the 21st century- Using your dictionary-Communication-Adaptability.

UNIT-III

Media and Environment: - A list of 100 basic words – Nouns- Pronouns- Adjectives-News report- Magazine article- User's Manual for new iPod- A documentary on the big cat- Why we need to save our tigers: A dialogue- Global warming- Paragraph Writing-Arguing a case-Motivation- Problem solving.

UNIT-IV

Entertainment and Employment:- One word substitutes- Parts of speech- Gerunds and infinitives- An excerpt from a short story an excerpt from a biography- A consultant interviewing employees- Your first interview- Reality TV- Writing an essay-Correcting sentences- Integrity Sense of humor.

UNIT-V

Work and Business:- A list of 100 difficult words- Articles, Quantifiers- Punctuation - Open letter to the Prime Minister Business dilemmas: An email exchange- A review of *IPL: The Inside Story,* Mark Zuckerberg: World's Youngest Billionaire- A conversation about a business idea-Pair work: Setting up a new business- Recession- Formal letters-Emails- Reports-Professionalism-Ethics, Fill in the blanks.

Text Book:

1. *Life through Language: A Holistic Approach to Language Learning.*Board of Editors, Pearson Publishers, India. 2013.

Reference Books:

1. Basic Vocabulary. Edgar Thorpe, Showick Thorpe. Pearson P. 2008.
2. Quick Solutions to Common Errors in English, Angela Bunt. MacMillan P. 2008.
3. Know Your English (Volume 1&2), by Dr. S. Upendra, Universities Press, India 2012
4. Business Communication Strategies. Mathukutty Monippally. Tata Mc Grahill P. 2009.

Code: B17 BS 1102

MATHEMATICS - I
(Common to all Branches)

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. The course is designed to equip the students with the necessary mathematical skills and techniques that is essential for an engineering course.
2. The skills derived from the course will help the student form a necessary base to develop analytic and design concepts.
3. Learn about linear first order and higher order ordinary differential equations and their applications.
4. Acquire knowledge of Laplace transform, partial differentiation and their applications.
5. Learn certain first order and higher order partial differential equations.

Course Outcomes: At the end of the Course, Student will be able to:

1. Solve linear ordinary differential equations of first order and first degree. Also will be able to apply the knowledge in simple applications such as Newton's law of cooling, orthogonal trajectories and simple electrical circuits.
2. Solve linear ordinary differential equations of second order and higher order. Also will be able to apply the knowledge in simple applications such as LCR circuits and Simple harmonic motion.
3. Determine Laplace transform and inverse Laplace transform of various functions.
4. Use Laplace transforms to solve a linear ODE.
5. Calculate total derivative, Jocobian and maxima/minima of functions of two variables.
6. Form partial differential equations and solve some standard types of first order PDEs. Find complimentary function and particular integral of linear higher order homogeneous and non-homogeneous PDEs.

SYLLABUS

UNIT I: Differential equations of first order and first degree:

Linear, Bernoulli, Exact, Reducible to exact types.

Applications: Newton's Law of cooling, Law of natural growth and decay, Orthogonal trajectories, Simple electrical circuits, Chemical reactions.

UNIT II: Linear differential equations of higher order:

Non-homogeneous equations of higher order with constant coefficients with RHS term of the type e^{ax}, sin ax, cos ax, polynomials in x, $e^{ax}V(x)$, $xV(x)$, Method of Variation of parameters.

Applications: LCR circuit, Simple Harmonic motion.

UNIT III: Laplace transforms:

Laplace transforms of standard functions, transforms of tf(t), f(t)/t, properties, transforms of derivatives and integrals, transforms of unit step function, Dirac delta function, Inverse Laplace transforms, convolution theorem (without proof).

Applications: Solving ordinary differential equations (initial value problems) using Laplace transforms.

UNIT IV: Partial differentiation:

Introduction, Homogeneous functions, Euler's theorem, Total derivative, Chain rule, which variable is to be treated as constant, Functional dependence, Jacobians, Taylor series for a function of two variables, Leibnitz rules for differentiation under the integral sign.

Applications: Errors and Approximations, Maxima and Minima of functions of two variables without constraints, Lagrange's method (with constraints)

UNIT V: First order and higher order partial differential equations:

Formation of partial differential equations by elimination of arbitrary constants and arbitrary functions, solutions of first order Lagrange linear equation and nonlinear equations of standard types (excluding Charpit's method).

Solutions of Linear homogeneous and non-homogeneous Partial differential equations with constant coefficients - RHS terms of the type e^{ax+by}, sin (ax+by), cos (ax+by), $x^m y^n$.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali & Manish Goyal, A Text book of Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
2. Michael Greenberg, Advanced Engineering Mathematics, 9th edition, Pearson.
3. Dean G. Duffy, Advanced engineering mathematics with MATLAB, CRC Press.
4. Peter O'Neil, Advanced Engineering Mathematics, Cengage Learning.
5. Srimanta Pal, Subodh C.Bhunia, Engineering Mathematics, Oxford University Press.
6. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

Code: B17 BS 1103

MATHEMATICS – II
(Common to CSE, ECE& IT)

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. The course is designed to equip the students with the necessary mathematical skills and techniques that are essential for an engineering course.
2. The skills derived from the course will help the student form a necessary base to develop analytic and design concepts.
3. Understand some basic numerical methods to solve algebraic and transcendental equations.
4. Learn interpolation methods for equally spaced and unequally spaced data points.Learn methods for numerical evaluation of integrals and for solving first order ODEs.
5. Learn concepts of a Fourier series and Fourier transform.

Course Outcomes: At the end of the Course, Student will be able to:

1. Find a real root of algebraic and transcendental equations using different methods.
2. Know the relation between the finite difference operators. Determine interpolation polynomial for a given data.
3. Evaluate numerically certain definite integrals applying Trapezoidal and Simpson's rules.
4. Solve a first order ordinary differential equation by Euler and RK methods.
5. Find Fourier series of a given function satisfying Dirichlet conditions. Find half range cosine and sine series for appropriate functions.
6. Find Fourier transforms, Fourier cosine and sine transforms of appropriate functions and evaluate certain integrals using inverse transforms and Fourier integral.

SYLLABUS

UNIT I: Solution of Algebraic and Transcendental Equations:

Introduction, Bisection method, Method of false position, Iteration method, Newton- Raphson method (One variable and simultaneous Equations).

UNIT II: Interpolation:

Introduction, Errors in polynomial interpolation, Finite differences, Forward differences, Backward differences, Central differences and Symbolic relations between the operators, Differences of a polynomial, Newton's formulae for interpolation, Interpolation with unequal intervals, Lagrange's interpolation formula.

UNIT III: Numerical Integration and solution of Ordinary Differential equations:

Trapezoidal rule, Simpson's $1/3^{rd}$ and $3/8^{th}$ rules, Solution of ordinary differential equations by Taylor series method, Picard's method of successive approximations, Euler's method, Runge-Kutta methods (second order and fourth order).

UNIT IV: Fourier Series:
Introduction, Periodic functions, Fourier series of a periodic function, Dirichlet's conditions, Even and odd functions, Change of interval, Half-range sine and cosine series, Parseval's formula.

UNIT V: Fourier Transforms:
Fourier integral theorem (without proof), Complex form of Fourier integral, Fourier sine and cosine integrals, Fourier transform, Fourier sine and cosine transforms, properties, inverse transforms, Parseval's identities, Finite Fourier transforms.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali& Manish Goyal, A Text book of Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Dean G. Duffy, Advanced engineering mathematics with MATLAB, CRC Press.
2. V.Ravindranath and P. Vijayalakshmi, Mathematical Methods, Himalaya Publishing House.
3. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
4. David Kincaid, Ward Cheney, Numerical Analysis-Mathematics of Scientific Computing, 3rd Edition, Universities Press.
5. Srimanta Pal, SubodhC.Bhunia, Engineering Mathematics, Oxford UniversityPress.
6. Dass H.K., RajnishVerma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

Code: B17 BS 1104

ENGINEERING PHYSICS
(Common to CSE, ECE& IT)

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

To re-orient Physics curriculum to the needs of the circuital and non circuital branches of Engineering / Technology courses offered by SRKREC(A) Bhimavaram that serves to understand the general and branch specific basic topics and introduce the related advanced technologies to the stake holders.

1. To impart the knowledge of physical optics phenomena like Interference and Diffraction required to understand the design and working of optical instruments with high resolution.
2. To understand the concept of coherence and generation of high intensity light sources and propagation of light waves in fibers for technological applications.
3. To study and analyze the behavior of electric and magnetic fields and their mutual interaction, and to understand the science of ultrasonics.
4. To understand the basic concepts of Quantum mechanics to know the behavior of fundamental particles in materials and to understand the basic nature of materials in general and classify them.
5. To study the structure property relationship of crystalline solids and understand basics of nanotechnology.

Course Outcomes: At the end of the course the students-

1. Learn the basic concepts of interference and diffraction of light and its applications.
2. Understand the science of producing high intensity light beams for technological applications and also understand the propagation of light waves in optical fibers in various applications.
3. Understand the inter relationship of electric and magnetic fields and learn ultrasonic’s as a tool for technological applications
4. Learn the behavior of particles at the very microscopic level by using wave nature of particles and understand the behavior of materials and be able to classify them using the band theory of solids.
5. Learn the basics of structures of solid materials and nano material preparation Techniques/methods.

SYLLABUS

UNIT I: Interference and Diffraction

Principle of superposition-coherence-interference in thin films (reflected system) – Wedge shaped film-Newton’s rings-Michelson’s interferometer. Fraunhoffer’s diffraction at single slit, Diffraction grating-Resolving power of a grating.

UNIT- II: Lasers and Optical Fibers

Introduction, Spontaneous emission and Stimulated emission – Einstein’s relation – Requirements of Laser device- Ruby laser- He-Ne gas laser- Characteristics of laser-Applications.

Description of optical fiber, Principle of light propagation- Optical fiber –Acceptance angle-Numerical aperture of optical fiber- Modes of propagation- Classification of fibers- Applications of fiber.

UNIT- III: Electro Magnetic Fields and Ultrasonics

Concept of Electromagnetic induction , Faraday's law, Lenz's law, Electric fields due to time varying magnetic fields, Magnetic fields due to time varying electric fields, Displacement current, Modified Ampere's law, Maxwell's equations and their significance (without derivation).

Definition of Ultrasonics-Methods of Producing Ultrasonics- Detection of Ultrasonics- Applications of Ultrsonics.

UNIT- IV: Quantum Mechanics and Band Theory of Solids

Introduction, de Broglie matter waves- properties-Experimental confirmation, wave function-significance- Schrodinger's time dependent and time independent wave equations- Eigen values and functions, Particle in a box.

Band theory of Solids- Introduction- Kronig Penney model (Qualitative)- Energy bands of crystalline solids- Distinction between Conductors, Semi conductors and insulators.

UNIT-V: Crystallo Graphy and Nano Materials

Basis and Lattice, Crystal systems, Bravais lattice, Unit cell Co ordination number – Packing fraction for SC ,FCC, and BCC lattices, Miller indices- Diffraction of X rays from crystals- Bragg's law.

Introduction to Nanomaterials – Synthesis methods : Condensation, ball milling, sol-gel, chemical vapour deposition methods, properties and applications.

Text Books:

1. Physics by Resnick & Halliday. Wiley - Eastern (India) Ltd.
2. Engineering Physics By M.N. Avadhanulu & PG Kshirasagar, S.Chand & Co.
3. Engineering Physics by V. Rajendran, Mc Graw Hill Education (India) Pvt.Ltd.

Reference Books:

1. Engineering Physics by MR Srinivasan, New International Publishers.
2. Solid State Physics – A.J. Dekkar, MacMillan (India) Ltd.
3. Engineering Physics by Gaur Gupta, Dhanpat Rai Publications, e meerut, India.
4. Engineering Physics by P k Palanisamy, Scitech Publications (India) Pvt.Ltd

(Note: Assignment Marks of **Engineering Physics** are to be considered from the Internal marks of **Engineering Physics-- Virtual Labs – Assignments B17 BS 1110**)

Code: B17 CS 1101

COMPUTER PROGRAMMING USING C
(Common to CSE, ECE & IT)

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

Formulating algorithmic solutions to problems and implementing algorithms in C

1. Notion of Operation of a CPU, Notion of an algorithm and computational procedure, editing and executing programs in Linux
2. understanding branching, iteration and data representation using arrays
3. Modular programming and recursive solution formulation
4. Understanding pointers and dynamic memory allocation
5. Understanding miscellaneous aspects of C
6. Comprehension of file operations

Course Outcomes:

1. Understand the basic terminology used in computer programming
2. Write, compile and debug programs in C language.
3. Use different data types in a computer program.
4. Design programs involving decision structures, loops and functions.
5. Explain the difference between call by value and call by reference
6. Understand the dynamics of memory by the use of pointers
7. Use different data structures and create/update basic data files.

SYLLABUS

UNIT I:

Unit objective: Notion of Operation of a CPU, Notion of an algorithm and computational procedure, editing and executing programs in Linux

Introduction: Computer systems, Hardware and Software Concepts**.**

Problem Solving: Algorithm / Pseudo code, flowchart, program development steps, computer languages: machine, symbolic and highlevel languages, Creating and Running Programs: Writing, Editing(vi/emacs editor), Compiling(gcc), Linking and Executing in under Linux.

BASICS OF C: Structure of a c program, identifiers, basic data types and sizes. Constants, Variables, Arthmetic , relational and logical operators, increment and decrement operators, conditional operator, assignment operator, expressions, type conversions, Conditional Expressions, precedence and order of evaluation, Sample Programs.

UNIT II:

Unit objective: understanding branching, iteration and data representation using arrays

SELECTION – MAKING DECISION: TWO WAY SELECTION: if-else, null else, nested if,examples, Multi-way selection: switch, else-if, examples**.**

ITERATIVE: loops- while, do-while and for statements , break, continue, initialization and updating, event and counter controlled loops, Looping applications: Summation, powers, smallest and largest.

ARRAYS: Arrays- concepts, declaration, definition, accessing elements, storing elements, Strings and String Manipulations, 1-D arrays, 2-D arrays and character arrays, string manipulations, Multidimensional arrays, array applications: Matrix operations, checking the symmetricity of a Matrix**. STRINGS: concepts, c strings.**

UNIT III:

Objective: Modular programming and recursive solution formulation

FUNCTIONS- MODULAR PROGRAMMING: functions, basics, parameter passing, storage classes extern, auto, register, static, scope rules, block structure, user defined functions, standard library functions, recursive functions, Recursive solutions for fibonacci series, towers of Hanoi, header files, C Preprocessor, example c programs, Passing 1-D arrays, 2-D arrays to functions.

UNIT IV:

Objective: Understanding pointers and dynamic memory allocation

POINTERS: pointers- concepts, initialization of pointer variables, pointers and function arguments, passing by address- dangling memory, address arithmetic, character pointers and functions, pointers to pointers, pointers and multi-dimensional arrays, dynamic memory management functions, command line arguments

UNIT V:

Objective: Understanding miscellaneous aspects of C

ENUMERATED, STRUCTURE AND UNION TYPES: Derived types- structures declaration, definition and initialization of structures, accessing structures, nested structures, arrays of structures, structures and functions, pointers to structures, self referential structures, unions, typedef, bit-fields, program applications

BIT-WISE OPERATORS: logical, shift, rotation, masks.

Objective: Comprehension of file operations

FILEHANDLING: Input and output- concept of a file, text files and binary files, Formatted I/O, File I/O operations, example programs

Text Books:

1. Problem Solving and Program Design in C, Hanly, Koffman, 7^{th}ed, PERSON
2. Programming in C, Second Edition PradipDey and Manas Ghosh, OXFORD Higher Education
3. Programming in C, A practical approach Ajay Mittal PEARSON
4. The C programming Language by Dennis Richie and Brian Kernighan
5. Programming in C, B. L. Juneja, Anith Seth, Cengage Learning.

Reference Books:

1. C Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE
2. Programming with C, Bichkar, Universities Press
3. Programming in C, ReemaThareja, OXFORD
4. C by Example, Noel Kalicharan, Cambridge

Code: B17 CE 1101

ENVIRONMENTAL STUDIES
(Common to all Branches)

Lecture	**: 2 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 2**

Course Objectives:

1. Developing an awareness and sensitivity to the total environment and its related problems.
2. Motivating students for active participation in environmental protection and improvement.
3. Developing skills for active identification and development of solutions to environmental problems
4. Evaluation of environment programmes in terms of social, economic, ecological and aesthetic factors.
5. Creating a “CONCERN AND RESPECT FOR THE ENVIRONMENT”.

Course Outcomes:

1. To bring awareness among the students about the nature and natural ecosystems
2. Sustainable utilization of natural resources like water, land, energy and air
3. Resource pollution and over exploitation of land, water, air and catastrophic (events) impacts of climate change, global warming, ozone layer depletion, marine, radioactive pollution etc to inculcate the students about environmental awareness and safe transfer of our mother earth and its natural resources to the next generation
4. Safe guard against industrial accidents particularly nuclear accidents
5. Constitutional provisions for the protection of natural resources

SYLLABUS

UNIT – I

Global Environmental Crisis:

Environmental Studies - Definition, Scope and importance, Need for public awareness. Global Environmental Crisis

Ecosystems:

Basic Concepts - Structure and Functions of Ecosystems: Producers, Consumers and Decomposers. Types of Ecosystems: Forest Ecosystems, Grassland Ecosystems Desert Ecosystems and Aquatic Ecosystems

UNIT-II

Biodiversity:

Introduction to Biodiversity**,** Values of Bio-diversity, Bio-geographical classification of India, India as a Mega-diversity habitat, Threats to biodiversity, Hotspots of Biodiversity, Conservation of Biodiversity: In-situ and Ex-situ conservation of Biodiversity.

UNIT-III

Environmental and Natural Resources Management:

Land Resources: Land degradation, soil erosion and desertification, Effects of modern agriculture. **Forest Resources:** Use and over exploitation-Mining and Dams-their effects on forest and tribal people. **Water resources**: Use and over utilization of surface and ground water, Floods, droughts, conflict over water, water logging and salinity, dams – benefits and problems. **Energy Resources:** Renewable and non-renewable energy sources, use of alternate energy sources-impact of energy use on environment.

UNIT-IV

Environmental Pollution:

Causes, Effects and Control measures of - Air pollution, Water pollution, Soil pollution, Marine Pollution, Thermal pollution, Noise pollution, Nuclear Hazards; Climate change and Global warming, Acid rain and Ozone layer depletion. Solid Waste Management: Composting, Vermiculture, Urban and Industrial Wastes, Recycling and Reuse.

Environmental Problems in India:

Drinking water, Sanitation and Public health, Population growth and Environment; Water Scarcity and Ground Water Depletion; Rain water harvesting, Cloud seeding and Watershed management.

UNIT-V

Institutions and Governance:

Regulations by Government- Environmental Protection Act, Air (Prevention & Control of Pollution) Act, Water (Prevention and control of Pollution) Act, Wildlife Protection Act, Forest Conservation Act. Environmental Impact Assessment (EIA)

Case Studies:

Chipko Movement, Narmada Bachao Andolan, Silent Valley Project, Mathura Refinery and Taj Mahal, Industrialization of Patancheru, Nuclear reactor at Nagarjuna Sager, Tehri Dam, Ralegaon Siddhi (Anna Hazare), Kolleru lake – Aquaculture, Fluorosis in Andhra Pradesh & Telangana.

Field Work:

Visit to a local area to document and mapping environmental assets. Visits to Industries, Water Treatment Plants, Affluent Treatment Plants.

Text Books:

1. Environmental Studies, K. V. S. G. Murali Krishna, VGS Publishers, Vijayawada
2. Environmental Studies, R. Rajagopalan, 2nd Edition, 2011, Oxford University Press.
3. Environmental Studies, Dr. M. Sita Rama Reddy, Dr. K. Anji Reddy, Silicon Publications, ASR Nagar, Bhimavaram
4. Environmental Studies, P. N. Palanisamy, P. Manikandan, A. Geetha, and K. Manjula Rani; Pearson Education, Chennai.

Reference Books:

1. Text Book of Environmental Studies, Deeshita Dave & P. Udaya Bhaskar, Cengage Learning.
2. A Textbook of Environmental Studies, Shaashi Chawla, TMH, New Delhi
3. Environmental Studies, Benny Joseph, Tata McGraw Hill Co, New Delhi
4. Perspectives in Environment Studies, Anubha Kaushik, C P Kaushik, New Age International Publishers, 2014

Code: B17 BS 1106

ENGINEERING PHYSICS LAB
(Common to CSE, ECE& IT)

Lab	**: 3 Periods**	**Int. Marks**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

Training the field oriented Engineering graduates to handle instruments and their design methods to improve the accuracy of measurements.

1. To impart hands-on experience to the students entering engineering/Technology education about handling sophisticated equipment/ instruments.
2. To make the students understand the theoretical aspects of various phenomena experimentally.

Course Outcomes:

Physics lab curriculum gives fundamental understanding of design of an instrument with targeted accuracy for physical measurements.

1. Students get hands on experience in setting up experiments and using the instruments/equipment individually.
2. Get introduced to using new/ advanced technologies and understand their significance.

LIST OF EXPERIMENTS
(Any 10 of the following listed experiments)

1. Determination of wavelength of a source-Diffraction Grating-Normal incidence.
2. Newton's rings – Radius of Curvature of Plano - Convex Lens.
3. Determination of thickness of a spacer using wedge film and parallel interference fringes.
4. Determination of Rigidity modulus of a material- Torsional Pendulum.
5. Determination of Acceleration due to Gravity and Radius of Gyration-Compound Pendulum.
6. Melde's experiment – Transverse and Longitudinal modes.
7. Verification of laws of vibrations in stretched strings – Sonometer.
8. Determination of velocity of sound – Volume Resonator.
9. L- C- R Series Resonance Circuit.
10. Study of I/V Characteristics of Semiconductor diode.
11. I/V characteristics of Zener diode.
12. Characteristics of Thermistor – Temperature Coefficients.
13. Magnetic field along the axis of a current carrying coil – Stewart and Gee's apparatus.
14. Energy Band gap of a Semiconductor p - n junction.
15. Hall Effect in semiconductors.
16. Time constant of CR circuit.
17. Determination of wavelength of laser source using diffraction grating.
18. Determination of Young's modulus by method of single cantilever oscillations.
19. Determination of lattice constant – lattice dimensions kit.
20. Determination of Planck's constant using photocell.
21. Determination of surface tension of liquid by capillary rise method.

Reference Book:

1. Advanced Practical Physics –Vol 1 & 2 By S.Venkata Raman, S.Chand & Co.

Code: B17 BS 1108

ENGLISH COMMUNICATIONSKILS LAB- I
(Common to All Branches)

Lab	**: 3 Periods**	**Int.Marks**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

1. To enable the students to learn through practice the communication skills of listening, speaking, reading and writing.
2. To make students recognize the sounds of English through Audio- Visual aids.
3. To familiarize the students with stress and intonation.
4. To help the students build their confidence in speaking skills.

Course Outcomes:

1. A study of the communicative items in the laboratory will help the students become successful in the competitive world.
2. Students improve their speaking skills in real contexts.
3. Students learn standard pronunciation and practice it daily discourse.
4. Students give up their communicative barriers.

SYLLABUS

- ❖ WHY study Spoken English?
- ❖ Making Inquiries on the phone, thanking and responding to Thanks - Practice work.
- ❖ Responding to Requests and asking for Directions - Practice work.
- ❖ Asking for Clarifications, Inviting, Expressing Sympathy, Congratulating
- ❖ Apologising, Advising, Suggesting, Agreeing and Disagreeing - Practice work.
- ❖ Letters and Sounds-Practice work.
- ❖ The Sounds of English-Practice Work
- ❖ Pronunciation
- ❖ Stress and Intonation-Practice work.

Lab Manual:

1. **'INTERACT:** English Lab Manual for Undergraduate Students' Published byOrient Blackswan Pvt Ltd.

Reference Books:

1. Strengthen your communication skills by Dr M Hari Prasad, Dr Salivendra Raju and Dr G Suvarna Lakshmi, Maruti Publications.
2. English for Professionals by Prof Eliah, B.S Publications, Hyderabad.
3. Unlock, Listening and speaking skills 2, Cambridge University Press
4. Spring Board to Success, Orient BlackSwan
5. A Practical Course in effective english speaking skills, PHI
6. Word power made handy, Dr shalini verma, Schand Company
7. Let us hear them speak, Jayashree Mohanraj, Sage texts
8. Professional Communication, Aruna Koneru, Mc Grawhill Education
9. Cornerstone, Developing soft skills, Pearson Education.

Code: B17 CS 1102

C PROGRAMMING LAB& HARDWARE FUNDAMENTALS
(Common to CSE & IT)

Lab	**: 3 Periods**	**Int.Marks**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

1. Understand the basic concept of C Programming, and its different modules that includes conditional and looping expressions, Arrays, Strings, Functions, Pointers, Structures and File programming.
2. Acquire knowledge about the basic concept of writing a program.
3. Role of constants, variables, identifiers, operators, type conversion and other building blocks of C Language.
4. Use of conditional expressions and looping statements to solve problems associated with conditions and repetitions.
5. Role of Functions involving the idea of modularity.

Course Outcomes:

1. Apply and practice logical ability to solve the problems.
2. Understand C programming development environment, compiling, debugging, and linking and executing a program using the development environment.
3. Analyzing the complexity of problems, Modularize the problems into small modules and then convert them into programs.
4. Understand and apply the in-built functions and customized functions for solving the problems.
5. Understand and apply the pointers, memory allocation techniques and use of files for dealing with variety of problems.
6. Document and present the algorithms, flowcharts and programs in form of user manuals.
7. Identification of various computer components, Installation of software

List of Programs

Exercise - 1 Basics

a) What is an OS Command, Familiarization of Editors - vi, Emacs
b) Using commands like mkdir, ls, cp, mv, cat, pwd, and man
c) C Program to Perform Adding, Subtraction, Multiplication and Division of two numbers From Command line

Exercise - 2 Basic Math

a) Write a C Program to Simulate 3 Laws at Motion
b) Write a C Program to convert Celsius to Fahrenheit and vice versa

Exercise - 3 Control Flow - I

a) Write a C Program to Find Whether the Given Year is a Leap Year or not.
b) Write a C Program to Add Digits & Multiplication of a number

Exercise – 4 Control Flow - II

a)Write a C Program to Find Whether the Given Number is

i) Prime Number
ii) Armstrong Number

b) Write a C program to print Floyd Triangle
c) Write a C Program to print Pascal Triangle.

Exercise – 5 Functions

a) Write a C Program demonstrating of parameter passing in Functions and returning values.
b) Write a C Program illustrating Fibonacci, Factorial with Recursion without Recursion.

Exercise – 6 Control Flow – III

a) Write a C Program to make a simple Calculator to Add, Subtract, Multiply or Divide Using switch…case
b) Write a C Program to convert decimal to binary and hex (using switch call function the function)

Exercise – 7 Functions - Continued

a) Write a C Program to compute the values ofsin x and cos x and e^x values using Series expansion. (use factorial function)

Exercise – 8 Arrays

Demonstration of arrays

a) Search-Linear.
b) Sorting-Bubble, Selection.
c) Operations on Matrix.

Exercises - 9 Structures

a)Write a C Program to Store Information of a Movie Using Structure
b)Write a C Program to Store Information Using Structures with Dynamically Memory Allocation
c) Write a C Program to Add Two Complex Numbers by Passing Structure to a Function

Exercise - 10 Arrays and Pointers

a)Write a C Program to Access Elements of an Array Using Pointer
b) Write a C Program to find the sum of numbers with arrays and pointers.

Exercise – 11 Dynamic Memory Allocations

a) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using malloc () function.
b) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using calloc () function.

Understand the difference between the above two programs

Exercise – 12 Strings

a) Implementation of string manipulation operations **with** library function.
- i) copy
- ii) concatenate
- iii) length
- iv) compare

b) Implementation of string manipulation operations **without** library function.
- i) copy
- ii) concatenate
- iii)length
- iv) compare

Exercise -13 Files

a)Write a C programming code to open a file and to print it contents onscreen.
b)Write a C program to copy files

Exercise - 14 Files Continued

a) Write a C program merges two files and stores their contents in another file.
b)Write a C program to delete a file.

Exercise - 15

a) System Assembling, Disassembling and identification of Parts / Peripherals.
b) Operating System Installation-Install Operating Systems like Windows, Linux along with necessary Device Drivers.

Exercise - 16

a) MS-Office / Open Office
   - i. Word - Formatting, Page Borders, Reviewing, Equations, symbols
   - ii. Spread Sheet-Organize data, usage of formula, graphs, charts.
   - iii. Powerpoint - features of power point, guidelines for preparing an effective presentation.

b) Network Configuration & Software Installation-Configuring TCP/IP, Proxy, and firewall settings. Installing application software, system software & tools.

Note:

a) All the Programs must be executed in the Linux Environment. (Mandatory)
b) The Lab record must be a print of the LATEX (.tex) Format.

Code: B17 CS 1103

C PROGRAMMING LAB
(For ECE)

Lab	**: 3 Periods**	**Int.Marks**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

1. Understand the basic concept of C Programming, and its different modules that includes conditional and looping expressions, Arrays, Strings, Functions, Pointers, Structures and File programming.
2. Acquire knowledge about the basic concept of writing a program.
3. Role of constants, variables, identifiers, operators, type conversion and other building blocks of C Language.
4. Use of conditional expressions and looping statements to solve problems associated with conditions and repetitions.
5. Role of Functions involving the idea of modularity.

Course Outcomes:

1. Apply and practice logical ability to solve the problems.
2. Understand C programming development environment, compiling, debugging, and linking and executing a program using the development environment.
3. Analyzing the complexity of problems, Modularize the problems into small modules and then convert them into programs.
4. Understand and apply the in-built functions and customized functions for solving the problems.
5. Understand and apply the pointers, memory allocation techniques and use of files for dealing with variety of problems.
6. Document and present the algorithms, flowcharts and programs in form of user manuals.
7. Identification of various computer components, Installation of software

Programming

Exercise - 1 Basics

a) What is an OS Command, Familiarization of Editors - vi, Emacs
b) Using commands like mkdir, ls, cp, mv, cat, pwd, and man
c) C Program to Perform Adding, Subtraction, Multiplication and Division of two numbers From Command line

Exercise - 2 Basic Math

a) Write a C Program to Simulate 3 Laws at Motion
b) Write a C Program to convert Celsius to Fahrenheit and vice versa

Exercise - 3 Control Flow - I

a) Write a C Program to Find Whether the Given Year is a Leap Year or not.
b) Write a C Program to Add Digits & Multiplication of a number

Exercise – 4 Control Flow - II

a)Write a C Program to Find Whether the Given Number is

i. Prime Number
ii. Armstrong Number

b) Write a C program to print Floyd Triangle

c) Write a C Program to print Pascal Triangle.

Exercise – 5 Functions

a) Write a C Program demonstrating of parameter passing in Functions and returning values.

b) Write a C Program illustrating Fibonacci, Factorial with Recursion without Recursion.

Exercise – 6 Control Flow – III)

a) Write a C Program to make a simple Calculator to Add, Subtract, Multiply or Divide Using switch…case

b) Write a C Program to convert decimal to binary and hex (using switch call function the function)

Exercise – 7 Functions - Continued

a) Write a C Program to compute the values ofsin x and cos x and e^x values using Series expansion. (use factorial function)

Exercise – 8 Arrays

Demonstration of arrays

a) Search-Linear.
b) Sorting-Bubble, Selection.
c) Operations on Matrix.

Exercises - 9 Structures

a)Write a C Program to Store Information of a Movie Using Structure

b)Write a C Program to Store Information Using Structures with Dynamically Memory Allocation

c) Write a C Program to Add Two Complex Numbers by Passing Structure to a Function

Exercise - 10 Arrays and Pointers

a)Write a C Program to Access Elements of an Array Using Pointer

b) Write a C Program to find the sum of numbers with arrays and pointers.

Exercise – 11 Dynamic Memory Allocations

a) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using malloc () function.

b) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using calloc () function.

Understand the difference between the above two programs.

Exercise – 12 Strings

a) Implementation of string manipulation operations **with** library function.
   i) copy
   ii) concatenate
   iii) length
   iv) compare

b) Implementation of string manipulation operations **without** library function.
   i) copy
   ii) concatenate
   iii)length
   iv) compare

Exercise -13 Files

a)Write a C programming code to open a file and to print it contents onscreen.
b)Write a C program to copy files

Exercise - 14 Files Continued

a) Write a C program merges two files and stores their contents in another file.
b) Write a C program to delete a file.

Note:

a) All the Programs must be executed in the Linux Environment. (Mandatory)
b) The Lab record must be a print of the LATEX (.tex) Format.

Code: B17 BS 1110

ENGINEERING PHYSICS - VIRTUAL LABS – ASSIGNMENTS
(Common to CSE, ECE & IT)

Lab : 2 Periods **Int.Marks: 5**

Course Objective: Training Engineering students to prepare a technical document andimproving their writing skills.

Course Outcome: Physics Virtual laboratory curriculum in the form of assignment ensuresan engineering graduate to prepare a /technical/mini-project/ experimental report with scientific temper.

LIST OF EXPERIMENTS

1. Hall Effect
2. Crystal Structure
3. Hysteresis
4. Brewster's angle
5. Magnetic Levitation / SQUID
6. Numerical Aperture of Optical fiber
7. Photoelectric Effect
8. Simple Harmonic Motion
9. Damped Harmonic Motion
10. LASER – Beam Divergence and Spot size
11. B-H curve
12. Michelson's interferometer
13. Black body radiation

URL: www.vlab.co.in

(Note: Internal Marks of **Engineering Physics - Virtual Labs – Assignments** are to be considered as Assignment marks in the Internal Marks of **Engineering Physics- B17 BS 1104**)

Code: B17 BS 1112

NCC
(Common to CSE, ECE & IT)

Practice : 2 Periods

The NCC- National Integration and Awareness- Drill- Personality Development Life Skills-Leadership- Disaster Management-Social Awareness and Community Development- Health and Hygiene- Environment Awareness and Conservation.

(**Note**: It is an uncredited course. It will not be included in the Grade Memo / Certificate. The Certificate will be issued based on the performance and attendance. This course attendance will be counted in the semester overall attendance.)

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R17)

I/IV B.TECH
(With effect from **2017-2018** Admitted Batch onwards**)**
Under Choice Based Credit System

GROUP-B (CSE, ECE & IT)
II-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 BS 1201	English – II*	3	3	1	--	4	30	70	100
B17 BS 1203	Mathematics – III*	3	3	1	--	4	30	70	100
B17 BS 1205	Engineering Chemistry	3	3	1	--	4	30	70	100
B17 ME 1201	Engineering Drawing	3	1	--	3	4	30	70	100
# DS 2	Department Subject	3	3	1	--	4	30	70	100
# DS 3	Department Subject	3	3	1	--	4	30	70	100
B17 BS 1207	Engineering Chemistry Lab	2	--	--	3	3	50	50	100
B17 BS 1208	English Communication Skills Lab – II*	2	--	--	3	3	50	50	100
# DL2	Department Lab	2	--	--	3	3	50	50	100
B17 BS 1212	Inner Engineering	--	--	--	2	2	--	--	--
	Total	**24**	**16**	**5**	**14**	**35**	**330**	**570**	**900**

*** Common to both Group - A and Group - B**

# DS 2	CSE & IT	B17 CS 1202	Object Oriented Programming Through C++
	ECE	B17 CS 1203	Data Structures
#DS 3	CSE & IT	B17 EC 1201	Elements of Electronics Engineering
	ECE	B17 EE 1203	Elements of Electrical Engineering
#DL2	CSE & IT	B17 CS 1205	Object Oriented Programming Lab
	ECE	B17 BS 1209	Engineering Workshop & IT Workshop

I/IV B.Tech II Sem
Group-B

Code: B17 BS 1201

ENGLISH - II
(Common to all Branches)

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. To improve the language proficiency of the students in English with emphasis on LSRW skills.
2. To enable the students to study and comprehend the prescribed lessons and subjects more effectively relating to their theoretical and practical components.
3. To develop spoken and written forms of the students in both formal and informal situations.
4. To enrich the vocabulary of the students through the usage of the lexis in appropriate contexts.
5. To expose them to the correct structural patterns of English.

Course Outcomes: At the end of the Course, Student will be able to:

1. To comprehend the speech of people belonging to different backgrounds and regions.
2. Understand the importance of speaking and writing for personal and professional communication and practice it in real contexts.
3. To express fluently and accurately in social discourse.
4. Participate in group activities like role-plays, discussions and debates.
5. Identify the discourse features, and improve intensive and extensive reading skills.

SYLLABUS

UNIT I:
A. Detailed-Text: Unit 1: ' The Greatest Resource- Education'
B. Non-Detailed Text: Lesson 1: ' A P J Abdul Kalam' from The Great Indian Scientists.

UNIT II:
A. Detailed-Text: Unit 2: ' A Dilemma'
B. Non-Detailed Text: Lesson 2:'C V Raman' from The Great Indian Scientists.

UNIT III:
A. Detailed-Text: Unit 3: 'Cultural Shock': Adjustments to new Cultural Environments
B. Non-Detailed Text: Lesson 3:'Homi Jehangir Bhabha' from The Great Indian Scientists.

UNIT IV:
A. Detailed-Text: Unit 4: 'The Lottery'
B. Non-Detailed Text: Lesson 4: 'Jagadish Chandra Bose' from The Great Indian Scientists.

UNIT V:
A. Detailed-Text: Unit 5: ' The Chief Software Architect'
B. Non-Detailed Text: Lesson 5: ' Prafulla Chandra Ray' from The Great Indian Scientists

Detailed Textbook:

1. English Encounters Published By Maruthi Publishers.

Non-Detailed Text Book:

1. The Great Indian Scientists Published by Cenguage learning.

Code: B17 BS 1203

MATHEMATICS - III
(Common to all Branches)

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:
1. The course is designed to equip the students with the necessary mathematical skills and techniques that are essential for an engineering course.
2. The skills derived from the course will help the student form a necessary base to develop analytic and design concepts.
3. Understand important concepts of Linear algebra including solution of linear simultaneous equations, eigen values and eigen vectors.
4. Learn Beta, Gamma functions and how to evaluate double and triple integrals.
5. Learn fundamental concepts of vector calculus that help understand other Engineering courses.

Course Outcomes: At the end of the Course, student will be able to:
1. Determine rank, and solve a system of linear simultaneous equations numerically using various matrix methods.
2. Determine Eigen values and Eigen vectors of a given matrix, Reduce a Quadratic form to its canonical form and classify.
3. Evaluate double integrals over a region and triple integral over a volume.
4. Use the knowledge of Beta and Gamma functions in evaluation of different integrals.
5. Find gradient of a scalar function, divergence and curl of a vector function. Use vector identities for solving problems.
6. Evaluate line, surface and volume integrals by the use of Green's, Stokes' and Gauss divergence theorems.

SYLLABUS

UNIT I: Linear systems of equations:
Rank, Echelon form, Normal form, Solution of linear systems, Gauss elimination, Gauss-Jordan, Jacobi and Gauss-Seidel methods.
Applications: Finding the current in electrical circuits.

UNIT II: Eigen values - Eigen vectors and Quadratic forms:
Eigen values, Eigen vectors, Properties, Cayley-Hamilton theorem, Inverse and powers of a matrix by using Cayley-Hamilton theorem, Diagonalization, Quadratic forms, Reduction of a Quadratic form to Canonical form, Rank, Positive, Negative, Semi-Definite and indefinite forms of a Quadratic form, Index and Signature of a Quadratic form.
Applications: Free vibration of a two-mass system.

UNIT III: Multiple integrals:
Double and triple integrals, Change of variables, Change of order of integration. Application to finding Areas, Moment of Inertia and Volumes.
Beta and Gamma functions, Properties, Relation between Beta and Gamma functions, Application to evaluation of improper integrals. The error function and the complimentary error function.

UNIT IV: Vector Differentiation:
Gradient, directional derivative, Divergence, Curl, Incompressible flow, solenoidal and irrotational vector fields, second order operators, vector identities.

UNIT V: Vector Integration:
Line integral, Work done, Potential function; Area, Surface and volume integrals, Flux, Vector integral theorems: Greens, Stokes and Gauss Divergence theorems (without proof) and related Problems.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali & Manish Goyal, Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Michael Greenberg, Advanced Engineering Mathematics, 9th edition, Pearson edn.
2. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
3. Peter O'Neil, Advanced Engineering Mathematics,7th edition, Cengage Learning.
4. D.W. Jordan and T. Smith, Mathematical Techniques, Oxford University Press.
5. Srimanta Pal, Subodh C.Bhunia, Engineering Mathematics, Oxford University Press.
6. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

Code: B17 BS 1205

ENGINEERING CHEMISTRY
(Common to CSE, ECE & IT)

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Knowledge of basic concepts of chemistry for engineering students will help them as professional engineers later in design and material selection as well as utilizing the available resources.

Course Objectives:

1. To understand the physical and mechanical properties of Polymers/Plastics/elastomers helps in selecting suitable materials for different purpose.
2. To create awareness on fuels as a source of energy for industries like thermal power stations, steel industry, fertilizer industry etc.
3. To understand the concepts of galvanic cells and corrosion with theories like electro chemical theory.
4. To understand the importance of water.
5. To understand about the materials which are used in major industries like steel and metallurgical manufacturing industries, construction and electrical equipment manufacturing industries.

Course Outcomes:

1. At the end of the course the students learn the advantages and limitations of plastic materials and their use in design.
2. Fuels which are used commonly and their economics, advantages and limitations are discussed.
3. Students gained Knowledge reasons for corrosion and some methods of corrosion control.
4. Students understands the impurities present in raw water, problems associated with them and how to avoid them.
5. Similarly students understand liquid crystals and semi conductors. Students can gain the building materials , solar materials, lubricants and energy storage devices.

SYLLABUS

UNIT-I: High Polymers and Plastics; Rubbers & Elastomers:

Polymerization Definition, Types of Polymerization, Mechanism of addition polymerization, Plastics as engineering materials, Thermoplastics and Thermosetting plastics, Compounding of plastics, Fabrication of plastics (4 techniques); Preparation, Properties and applications of Polyethylene, PVC, Bakelite, Nylon - 6,6, Bullet Proof plastics -polycarbonate and Kelvar; Fiber reinforced plastics, conducting polymers, Biodegradable Polymers - PHBV, Nylon 2, Nylon 6.

Natural rubber – Vulcanization – Compounding of Rubber; Preparation, properties and applications of Buna – S; Buna – N;

UNIT-II :Fuel Technology& Lubricants:

Fuels: - Introduction – Classification of fuels, Calorific value – HCV and LCV, Determination of Calorific value by bomb calorimeter; Proximate and ultimate analysis of coal, coke: manufacture of coke by Otto – Hoffmann's by-product coke oven process; Refining of Petroleum, Knocking-octane number of gasoline, cetane number of diesel oil. Synthetic Petrol; LPG , CNG.

Lubricants:- Definition, Mechanism of Lubrication, Properties of Lubricants (Definition and significance)

UNIT-III: Electrochemical cells and Corrosion:
Galvanic cell, single electrode potential, Calomel electrode; Modern batteries: - Lead – Acid battery; Fuel cells- Hydrogen – Oxygen cell, Lithium battery Theories of corrosion (i) dry Corrosion (ii) wet corrosion. Types of corrosion - differential aeration corrosion, pitting corrosion, galvanic corrosion, stress corrosion, Factors influencing corrosion, Protection from corrosion-material selection & design, cathodic protection, Protective coatings- metallic coatings – Galvanizing, Tinning, Electroplating; Electrroless plating ; Paints.

UNIT-IV: Water technology:
Sources of water – Hardness of water – Estimation of hardness of water by EDTA method; Boiler troubles – sludge and scale formation, Boiler corrosion, caustic embrittlement, Priming and foaming; Softening of water by Lime – Soda Process, Zeolite Process, Ion – Exchange Process; Municipal water treatment; Desalination of sea water by Electrodialysis and Reverse osmosis methods.

UNIT-V :Chemistry of Engineering Materials& Advanced Engineering materials
Cement:- Manufacture of Portland cement, setting and hardening of cement, Deterioration of cement concrete.
Refractories: - Definition, Characteristics, classification, Properties and failure of refactories.
Solar Energy: - Construction and working of Photovoltaic cell, applications.
Solid State Materials: Crystal imperfections, Semi Conductors, Classification and chemistry of semi conductors: Intrinsic semiconductors; Extrinsic semiconductors; Defect semiconductors, Compound Semiconductors and Organic Semiconductors.
Liquid Crystals: - Definition – Classification with examples – Applications

Text Books:

1. Engineering Chemistry by Jain and Jain, Dhanpat Rai publishing co.
2. Engineering Chemistry by Willy India Pvt Ltd.
3. Engineering Chemistry by Dr.K.Anji Reddy and Dr.M.S.R.Reddy; Silicon Publications.

Reference Books:

1. Engineering Chemistry by Shikha Agarwal; Cambridge University Press,2015 edition.
2. A text of Engineering Chemistry by S.S.Dara; S.Chand & Co Ltd.

Code: B17 ME 1201

ENGINEERING DRAWING
(Common to CSE, ECE & IT)

Lecture	**: 1 Period.**	**Int.Marks**	**: 30**
Practice	**: 3 Periods.**	**Ext. Marks**	**: 70**
Exam	**: 3Hrs.**	**Credits**	**: 3**

Course Objectives:

1. To highlight the significance of universal language of engineers.
2. To impart basic knowledge and skills required to prepare engineering drawings.
3. To impart knowledge and skills required to draw projections of solids in different contexts.
4. To visualize and represent the pictorial views with proper dimensioning and scaling.

Course Outcomes:

Students will be able to

1. Apply principles of drawing to represent dimensions of an object.
2. Construct polygons and engineering curves.
3. Draw projections of points, lines, planes and solids.
4. Represent the object in 3D view through isometric views.
5. Convert the isometric view to orthographic view and vice versa.

SYLLABUS

UNIT I

Polygons: Constructing regular polygons by general methods, inscribing and describing polygons on circles.

Curves: Parabola, Ellipse and Hyperbola by general methods, cycloids, involutes, tangents & normals for the curves.

UNIT II

Orthographic Projections: Horizontal plane, vertical plane, profile plane, importance of reference lines, projections of points in various quadrants, projections of lines, lines parallel either to one of the reference planes (HP,VP or PP)

Projections of straight lines inclined to both the planes, determination of true lengths, angle of inclination and traces- HT, VT

UNIT III

Projections of planes: regular planes perpendicular/parallel to one plane and inclined to the other reference plane; inclined to both the reference planes.

UNIT IV

Projections of Solids – Prisms, Pyramids, Cones and Cylinders with the axis inclined to one of the planes.

UNIT V

Conversion of isometric views to orthographic views; Conversion of orthographic views to isometric views.

Text Books:

1. Engineering Drawing by N.D. Butt, Chariot Publications.
2. Engineering Drawing by Agarwal & Agarwal, Tata McGraw Hill Publishers

Reference Books:

1. Engineering Drawing by K.L.Narayana & P. Kannaiah, Scitech Publishers.
2. Engineering Graphics for Degree by K.C. John, PHI Publishers.
3. Engineering Graphics by PI Varghese, McGrawHill Publishers.
4. Engineering Drawing + AutoCad – K Venugopal, V. Prabhu Raja, New Age

Code: B17 CS 1202

OBJECT-ORIENTED PROGRAMMING THROUGH C++
(Common to CSE & IT)

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. This course is designed to provide a comprehensive study of the C++ programming language. students are able to write efficient, maintainable and portable code to design algorithmic solutions to problems.
2. Expertise in object oriented principles and their implementation in C++
3. To acquire an understanding of basic object-oriented concepts and the issues involved in effective class design.
4. To Write C++ programs that use arrays, structures, pointers, object oriented concepts such as information hiding, constructors, destructors, inheritance.

Course Outcomes:

1. Write, compile and debug programs in C++ language. Use different data types in a computer program.
2. Design programs involving decision structures, loops and functions.
3. Explain classes and abstract classes and objects, abstraction and encapsulation, inheritance, polymorphism, constructors, access control and overloading.
4. Solve a given application problem by going through the basic steps of program specifications, analysis, design, implementation and testing within the context of the object-oriented paradigm.

SYLLABUS

UNIT-I: Introduction to C++, Classes and Objects.

Difference between C and C++, Disadvantage of Conventional Programming, Basic Concepts of Object Oriented Programming, Advantage of OOP, Object Oriented Languages, Functions in C++, Operators in C++. **Classes and Objects:** Declaring Objects, Access Specifiers and their Scope, Static data members, static member functions, arrays of objects, local classes, Nested classes.

UNIT-II: Constructors, Destructors and Operator Overloading.

Constructors and Destructors: Introduction- Constructors and Destructor- types of constructors, Constructors with default Arguments, Dynamic initialization of objects, Dynamic constructors. **Operator Overloading** Introduction, Overloading Unary Operators and Binary Operators, Overloading Unary Operators and Binary Operators using friend function, Overloading Assignment Operator (=), Overloading insertion(<<) and extraction(>>) operators, Manipulation of Strings using Operators, Rules for Overloading Operators, Type Conversions.

UNIT-III: Inheritance , Pointers, Virtual Functions and Polymorphism.

Inheritance: Reusability, Types of Inheritance, Virtual Base Classes, Abstract Classes, Advantages of Inheritance, Disadvantages of Inheritance, and constructors in derived classes. **Pointers Introduction:** Pointers to Objects, “**this**” Pointer, Pointers to Derived Classes, including Polymorphisms and Virtual Functions, Rules for Virtual Functions, pure virtual functions.

UNIT-IV: Manipulating Strings, Managing console I/O operations and Exception Handling.

Strings: Creating String Objects, Manipulating String Objects, Relational operations, String Characteristics, Accessing Characters in Strings. **C++ Stream** Classes, Unformatted I/O operations, Formatted I/O operations, managing output with Manipulators, **Exception Handling**: Principles of Exception Handling, Exception Handling Mechanism, throwing and catching Mechanism.

UNIT-V: Generic Programming with Templates, Standard Template Library and Files.

Generic Programming with Templates, Need for Templates, Definition of class Templates, Normal Function Templates, Over Loading of Template Function-Bubble Sort Using Function Templates, Difference between Templates and Macros, Overview of **Standard Template Library**, STL Programming Model, Containers, Algorithms, Iterators, Vectors, Lists, Maps. **FILES:** Introduction, File Stream Classes, File Operations, File Pointers and Manipulators, Sequential Access Files, Random File Access Operation, Detecting End-of File, Command-Line Arguments.

Text Books:

1. A complete Guide to programming in C++, Ulla Kirch-Prinz, Peter Prinz, Jones and Bartlett Publishers (2002).
2. Programming in C++, 2nd Edition ,Ashok N Kamthane, and Pearson.
3. Object Oriented Programming C++, 4th Edition, Joyce Farrell, Cengage Learning.

Reference Books:

1. C++ Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE
2. The Complete Reference C++, 4th Edition, Herbert Schildt, TMH.
3. Programming solving with C++,9th Edition Walter Savitch.
4. Object Oriented Programming in C++, 4th Edition, Robert Lafore

Code: B17 CS 1203

DATA STRUCTURES
(For ECE)

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. To be familiar with basic techniques handling problems with Data structures
2. Solve problems using data structures such as linear lists, stacks, queues, hash tables
3. Create and traverse different types of trees and graphs.
4. To practice different searching algorithms.

Course Outcomes:

Students will be able to

1. Apply advanced data structure strategies for exploring complex data structures.
2. Compare and contrast various data structures and design techniques in the area of Performance.
3. Implement all data structures like stacks, queues, trees, lists and graphs and compare their performance and tradeoffs.
4. Implement different operations on trees.
5. Apply graphs to real time applications.
6. Perform sorting using different algorithms.

SYLLABUS

UNIT-I

Arrays and Structures

Array as an Abstract Data Type, Polynomial Abstract Data Type, Introduction to Sparse Matrix, Sparse Matrix Abstract Data Type, Representation of Multidimensional Arrays, Structures and Unions, Internal Implementation of Structures, Self-Referential Structures.

Recursion, Simple Searching and Sorting Techniques

Recursive functions, Introduction to Searching, Sequential Search, Binary Search, Interpolation Search, Selection Sort, Bubble Sort, Insertion Sort, Quick Sort, Introduction to Merge Sort, Iterative Merge Sort, Recursive Merge Sort, Heap sort.

UNIT-II

Stacks and Queues

Stack Abstract Data Type, Queue Abstract Data Type, Stacks and Queues using arrays, , Introduction to Evaluation of Expressions, Evaluating Postfix Expressions, Infix to Postfix and Prefix conversion, Multiple Stacks and Queues, Circular Queues using arrays.

UNIT-III

Linked Lists

Pointers, Dynamically Allocated Storage using pointers, Singly Linked Lists, Dynamically Linked Stacks and Queues, Polynomials, Representing Polynomials as Singly Linked Lists, Adding Polynomials, Erasing Polynomials, Polynomials as Circularly Linked Lists, Additional List Operations, Operations for Singly Linked Lists, Operations for Doubly Linked Lists, Radix Sort.

UNIT-IV

Trees

Representation of Trees, Binary Trees Abstract Data Type,Properties of Binary Trees, Binary Tree Representations, Binary Tree Traversals, Additional Binary Tree Operations, Threaded Binary Trees, Heap Abstract Data Type, Priority Queues, Insertion into a max heap, Deletion from a max heap, Heap Sort, Introduction to Binary Search Trees, Searching a Binary Search Tree, Inserting an Element into a Binary Search Tree, Deleting an Element from a Binary Search Tree, Height of a Binary Search Tree, Counting Binary Trees.

UNIT-V

Graphs

Graph Abstract Data Type, Definitions, Graph Representations, Elementary Graph Operations, Depth First Search, Breadth First Search, Connected Components, Spanning Trees, Minimum Cost Spanning Trees, Prim's and Kruskal's Algorithms, Shortest Paths and Transitive Closure, Single Source All Destination - Dijkstra's Algorithm, All Pairs Shortest Paths - Floyd's Algorithm, Transitive Closure using Warshall's Algorithm.

Text Books:

1. Fundamentals of Data Structures in C, 2nd edition, Horowitz, Sahni and Anderson-Freed, Universities Press, 2008.

Reference Books:

1. Data Structures using C by Aaron M. Tenenbaum,Y. Langsam and M.J. Augenstein, Pearson Education, 2009.
2. Data Structures with C by Seymour lipschutz, Schaum Outline series, 2010.
3. Data Structures using C by R. KrishnaMoorthy G. Indirani Kumaravel, TMH, New Delhi,2008.

Code: B17 EC 1201

ELEMENTS OF ELECTRONICS ENGINEERING
(Common to CSE & IT)

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. To give the exposure to the students on semiconductor physics of the intrinsic and extrinsic semiconductors and basics of P-N junction diode.
2. To give the exposure to the students on the basics of special purpose diodes like Zener diode, photo diode, LED, and tunnel diode and rectifier circuits using diodes.
3. To give the exposure to the students on basics of BJT, and transistor circuit configurations.
4. To give exposure to the students on transistor biasing and thermal stabilization.
5. To give exposure tothe students on the basics of JFET, MOSFET and FET biasing.

Course Outcomes:

After completion of the course the students will be able to

1. Understand the basic concepts of transport of charge carriers in semiconductors,drift and diffusion currents, physical structure,operation, V-I characteristics of semiconductor diode.
2. Understand the basic concepts ofspecial types of diodes like Zener Diode, LED, Photo Diode and tunnel diode, rectifier circuits with and without filters.
3. Understand the physical structure, operation, input and output characteristics of BJT in CE,CB,CC circuit configurations.
4. Understand the basic concepts oftransistor biasing and thermal stabilization.
5. Understand the physical structure, operation, characteristics and circuit models of JFET's and MOSFET's.

SYLLABUS

UNIT I: Semiconductors and P-N junction diode:
Intrinsic and extrinsic semiconductors, charge densities in semiconductors, Drift and Diffusion currents, Hall Effect, Mass action law.Basic operation and V-I Characteristics of semiconductor diode, Diode current equation, Avalanche breakdown and Zener breakdown phenomenon.

UNIT II: Special Diodes and Diode Rectifiers:
Zener Diode, LED, Photo Diode and tunnel diode, Half wave and Full wave Rectifiers- with and without filters, Bridge Rectifier,Expressions - Ripple factor,Efficiency, Capacitor filters
.
UNIT III: Bipolar Junction Transistor:
Introduction, construction, basic operation of npn and pnp transistors, Transistor circuit configurations- CE,CB, CC- Input and output Characteristics in various configurations.h-parameter model for transistor amplifier.*(Introductory Treatment only).*

UNIT IV:Transistor Biasing and Thermal Stabilization:
Transistor Biasing, Thermal runaway, stabilization, Different methods of Biasing-Fixed Bias, collector feedback bias, self-bias, Bias compensation.

UNIT V: Field Effect Transistors: Junction field Effect Transistors (JFET)- JFET characteristics, JFET Parameters, Small Signal model of FET, Depletion and Enhancement type MOSFET's.

Text Books:

1. Electronic Device and Circuits by SanjeevGuptha, Dhanpatrai&Co.Pvt.Ltd.
2. Electronic Device and Circuits by K.Satya Prasad, VGS.

Reference Books:

1. Integrated Electronics- Millman&Halkias, TMH.

Code: B17 EE 1203

ELEMENTS OF ELECTRICAL ENGINEERING
(For ECE)

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. To learn the fundamentals of electrical and magnetic circuits
2. To understand the principle of operation and construction details of DC machines.
3. To understand the principle of operation of Transformers.
4. To understand the principle of operation and construction details of three phase Induction motor
5. To understand the principle of operation of synchronous generator and different measuring instruments

Course Outcomes:

1. Able to understand the basics of Magnetic Circuits and Kirchhoff's laws.
2. Able to understand the operation of DC Machines and to conduct different Tests
3. Able to analyze the Performance of Transformers.
4. Able to explain the operation of three phase induction motor.
5. Able to analyze the operation of alternator and different measuring instruments.

SYLLABUS

UNIT I: Electrical and Magnetic Circuits:
Basic definitions, Types of network elements, Ohm's Law, Kirchhoff's Laws, Series and parallel Circuits and star-delta and delta-star transformations-simple problems. Magnetic flux, MMF, Reluctance, Faraday's laws, Lenz's law, statically induced EMF, dynamically induced EMF.

UNIT – II: DC Machines:
Principle of operation of DC Generator - EMF equation – Construction-Types of DC generator-OCC of DC Generator-DC motor types - Torque equation –Losses-Efficiency-speed control methods- applications

UNIT – III: Transformers:
Principle of operation of single phase transformer - EMF equation - equivalent circuit –losses - efficiency and regulation- Open circuit and Short circuit tests.

UNIT – IV: Induction Motors:
Construction-Principle of operation of induction motor-slip- rotor frequency, slip - torque characteristics - Power flow diagram-Efficiency-Applications

UNIT – V: Synchronous Generator and Measuring Instruments:

Construction-Principle of operation of alternator-EMF equation of alternator- Regulation by Synchronous impedance method.

Classification –Deflecting, controlling, damping Torque, ammeter, voltmeter, wattmeter, MI, MC instruments-Energy meter

Text Books:

1. Electrical Machinery By Dr.P.S Bhimbra,Khanna Publications, January 2011
2. Electrical Technology by Surinder Pal Bali, Pearson Publications. January 1, 2013
3. Principles of Electrical Engineering, V.K mehta, Rohit Mehta, S. Chand Publications. Revised Edition 2012
4. A Course in Electronic Measurements and Instrumentation by A.K. Sawhney Dhanpat Rai & Co. (P) Limited; 2014 edition (2015)

Reference Books:

1. Basic Electrical Engineering, by J.B Gupta, K Kataria and Sons; Reprint 2013 edition.
2. Fundamentals of Electrical Engineering and Electronics, by Theraja B.L, S Chand Multicolor edition (1 December 2006)
3. A Course in Electronics and Electrical Measurements and Instrumentation by J.B. Gupta S K Kataria and Sons; Reprint 2013 edition (2013)

Code: B17 BS 1207

ENGINEERING CHEMISTRY LAB
(Common to CSE,ECE& IT)

Lab	**: 3 Periods**	**Int.Marks**	**: 50**
Exam	**: 3Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

1. To investigate and understand Physical behavior in the laboratory using scientific reasoning and logic and interpret the result of simple experiments and demonstration of chemical Principle and also evaluate the impact of chemical discoveries on how we view the world.
2. Effectively communicate experimental results and solutions to application problems through oral and written reports.
3. Understand the basic concepts, definitions, characteristics and phenomena.
4. Recognize the classical ideas and chemical phenomena and also define and analyse the concepts.

Course Outcomes:

1. An understanding of Professional and develop confidence on recent trends.
2. Able to gain technical knowledge of measuring, operating and testing of chemical instruments and equipments.
3. Acquire ability to apply knowledge of chemistry.
4. Exposed to the real time working environment.
5. Demonstrate the ability to learn Principles, design and conduct experiments.
6. Ability to work on laboratory and multidisciplinary tasks.

List of Experiments

Introduction to chemistry Laboratory

1. Estimation of HCl using standard Sodium Hydroxide.
2. Determination of total hardness of water by EDTA method.
3. Estimation of Ferrous Iron by $KMnO_4$.
4. Estimation of oxalic acid by $KMnO_4$
5. Estimation of Mohr's salt by $K_2Cr_2O_7$
6. Estimation of Dissolved oxygen by Winkler's method.
7. Determination of pH by pH meter and universal indicator method.
8. Conductometric titration of strong acid Vs strong base
9. Conductrometric titration of strong acid Vs weak base.
10. Potentiometric titration of strong acid Vs strong base
11. Potentiometric titration of strong acid Vs weak base
12. Preparation of Phenol formaldehyde resion.
13. Determination of saponification value of oils
14. Determination of pour and cloud points of lubricating oil.
15. Determination Acid value of oil.

Text Books:

1. Engineering Chemistry Lab Manual Prepared by Chemistry Faculty of S.R.K.R.Engineering College.

Reference Books:

1. Laboratory manual on Engineering Chemistry by Dr.Sudha Rani ;Dhanpat Rai Publishing Company.
2. Engineering Chemistry Laboratory manual – I & II by Dr. K.Anji Reddy; Tulip Publications.

Code: B17 BS 1208

ENGLISH COMMUNICATION SKILS LAB- II
(Common to All Branches)

Lab	**: 3 Periods**	**Int.Marks**	**: 50**
Exam	**: 3Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

1. To enable the students to learn demonstratively the communication skills of listening, speaking, reading and writing.
2. To enable the students participate in group interactions.
3. To improve the presentation skills of the students.
4. To help the students gain their confidence in attending the interviews.

Course Outcomes:

1. A study of the communicative items in the laboratory will help the students become successful in the competitive world.
2. Students enhance their presentation skills.
3. Students participate in group discussions and improve their team skills.
4. Students confidently face the interviews.

SYLLABUS

- ❖ Debating & Practice.
- ❖ Group Discussions & Practice.
- ❖ Presentation Skills & Practice
- ❖ Interview Skills & Practice
- ❖ Email
- ❖ Curriculum Vitae & Practice
- ❖ Idiomatic Expressions
- ❖ Common Errors in English & Practice

LAB MANUAL:

1. **‘INTERACT:** English Lab Manual for Undergraduate Students’ Published byOrient Blackswan Pvt Ltd.

Reference Books:

1. Strengthen your communication skills by Dr M Hari Prasad, Dr Salivendra Raju and Dr G Suvarna Lakshmi, Maruti Publications.
2. English for Professionals by Prof Eliah, B.S Publications, Hyderabad.
3. Unlock, Listening and speaking skills 2, Cambridge University Press
4. Spring Board to Success, Orient BlackSwan
5. A Practical Course in effective english speaking skills, PHI
6. Word power made handy, Dr shalini verma, Schand Company
7. Let us hear them speak, Jayashree Mohanraj, Sage texts
8. Professional Communication, Aruna Koneru, Mc Grawhill Education
9. Cornerstone, Developing soft skills, Pearson Education

Code: B17 CS 1205

OBJECT ORIENTED PROGRAMMING LAB
(Common to CSE & IT)

Lab	**: 3 Periods**	**Int.Marks**	**: 50**
Exam	**: 3Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

1. To strengthen their problem solving ability by applying the characteristics of an object-oriented approach.
2. To introduce object oriented concepts in C++ and Java.

Course Outcomes:

1. Explain what constitutes an object-oriented approach to programming and identify potentialbenefits of object-oriented programming over other approaches.
2. Apply an object-oriented approach to developing applications of varying complexities.

LIST OF PROGRAMS

1. Write a Programme that computes the simple interest and compound interest payable on principal amount (in Rs.) of loan borrowed by the customer from a bank for a giver period of time (in years) at specific rate of interest. Further determine whether the b bank will benefit by charging simple interest or compound interest

2. Write a Programme to calculate the fare for the passengers traveling in a bus. When a Passenger enters the bus, the conductor asks "What distance will you travel?" On knowing distance from passenger (as an approximate integer), the conductor mentions the fare to the passenger according to following criteria.

3. Write a C++ Program to illustrate Enumeration and Function Overloading
4. Write a C++ Program to illustrate Scope and Storage class
5. Implementation of ADT such as Stack and Queues
6. Write a C++ Program to illustrate the use of Constructors and Destructors and Constructor Overloading
7. Write a Program to illustrate Static member and methods
8. Write a Program to illustrate Bit fields
9. Write a Program to overload as binary operator, friend and member function
10. Write a Program to overload unary operator in Postfix and Prefix form as member and friend function
11. Write a C++ Program to illustrate Iterators and Containers
12. Write a C++ Program to illustrate function templates
13. Write a C++ Program to illustrate template class
14. Write C++ Programs and incorporating various forms of Inheritance
15. Write a C++ Program to illustrate Virtual functions
16. To write a C++ program to find the sum for the given variables using function with default arguments.
17. To write a C++ program to find the value of a number raised to its power that demonstrates a function using call by value.
18. To write a C++ program and to implement the concept of Call by Address
19. To write a program in C++ to prepare a student Record using class and object

20. To implement the concept of unary operator overloading by creating a C++ program.
21. Write a C++ program for swapping two values using function templates
22. Write a C++ program to implement a file handling concept using sequential access.

Reference Books:

1. C++ Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE
2. Programming in C++, 2nd Edition, Ashok N Kamthane, and Pearson.
3. Programming solving with C++, 9th Edition Walter Savitch.

Code: B17 BS 1209

ENGINEERING WORKSHOP & IT WORKSHOP
(For ECE)

Lab	**: 3 Periods**	**Int.Marks**	**: 50**
Exam	**: 3Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

PART-A ENGINEERING WORKSHOP

Course Objective:

To impart hands-on practice on basic engineering trades and skills.

Course Outcomes:

Student will be able to
1. Use various tools to prepare basic carpentry and fitting joints.
2. Prepare jobs of various shapes using black smithy.
3. Make basic house wire connections.
4. Fabricate simple components using tin smithy.

SYLLABUS

Carpentry	Fitting
1. T-Lap Joint 2. Cross Lap Joint 3. Dovetail Joint 4. Mortise and Tenon Joint	1. Vee Fit 2. Square Fit 3. Half Round Fit 4. Dovetail Fit
Black Smithy	**Tin Smithy**
1. Round rod to Square 2. S-Hook 3. Round Rod to Flat Ring 4. Round Rod to Square headed bolt	1. Taper Tray 2. Square Box without lid 3. Open Scoop 4. Funnel
House Wiring	
1. Parallel / Series Connection of three bulbs 2. Stair Case wiring 3. Florescent Lamp Fitting 4. Measurement of Earth Resistance	

Note: At least two exercises to be done from each trade.

Reference:

1. Elements of workshop technology, Vol.1 by S. K. and H. K. Choudary.

PART B: IT WORKSHOP:

Course Objectives:

1. Understand the basic components and peripherals of a computer.
2. To become familiar in configuring a system.
3. Learn the usage of productivity tools.
4. Acquire knowledge about the netiquette and cyber hygiene.
5. Get hands on experience in trouble shooting a system

Course Outcomes:

At the end of the course the students can able to

1. Assemble and disassemble the systems
2. Use the Microsoft office tools
3. Install various software
4. Know about various search engines
5. Trouble shoot various Hardware and Software problems
6. Use MATLAB and LATEX software

LIST OF EXCERCISES

1. System Assembling, Disassembling and identification of Parts / Peripherals
2. Operating System Installation-Install Operating Systems like Windows, Linux along with necessary Device Drivers.
3. MS-Office / Open Office
   a) Word - Formatting, Page Borders, Reviewing, Equations, symbols.
   b) Spread Sheet - organize data, usage of formula, graphs, charts.
   c) Power point - features of power point, guidelines for preparing an effective presentation.
   d) Access- creation of database, validate data.
4. Network Configuration & Software Installation-Configuring TCP/IP, proxy and firewall settings. Installing application software, system software & tools.
5. Internet and World Wide Web-Search Engines, Types of search engines, netiquette, cyber hygiene.
6. Trouble Shooting-Hardware trouble shooting, Software trouble shooting.
7. MATLAB- basic commands, subroutines, graph plotting.
8. LATEX-basic formatting, handling equations and images.

Reference Books:

1. Computer Hardware, Installation, Interfacing, Troubleshooting and Maintenance, K.L. James, Eastern Economy Edition.
2. Microsoft Office 2007: Introductory Concepts and Techniques, Windows XP Edition ByGary B. Shelly, Misty E. Vermaat and Thomas J. Cashman (2007, Paperback).
3. LATEX- User's Guide and Reference manual, Leslie Lamport, Pearson, LPE, 2/e.
4. Getting Started with MATLAB: A Quick Introduction for Scientists and ngineers, Rudraprathap, Oxford University Press, 2002.
5. Scott Mueller's Upgrading and Repairing PCs, 18/e, Scott. Mueller, QUE, Pearson, 2008
6. The Complete Computer upgrade and repair book, 3/e, Cheryl A Schmidt, Dreamtech.
7. Comdex Information Technology course tool kit Vikas Gupta, WILEY Dreamtech.
8. Introduction to Information Technology, ITL Education Solutions limited, Pearson Education.

Code: B17 BS 1212

INNER ENGINEERING
(Common to CSE, ECE & IT)

Practice : 2 Periods

Course Objectives:

Student should learn:

1. Human values and tools to lead a happy, stress-free life.
2. Yoga asanas, Pranayama, Sudarshan Kriya & Meditation
3. At-least two creative arts out of photography, sketching, craft-making, singing, clay molding, upcycling etc.
4. Concentration pranayama, Ego bursting process.
5. To take up responsibility for society and teach classes of their choice to school children.
6. About good food habits for good health.

Course Outcomes:

Student should be capable of

1. To improve his concentration levels and improve his public speaking abilities.
2. To balance his academic and non-academic activities (Work Life Balance).
3. To widen his vision and increase his breadth of perspective in his journey of 4 years.
4. To improve his communications skills, leadership, teamwork and decision-making abilities.
5. To inculcate creativity & innovation, planning & organizing as part of their life.
6. Taking responsibility for themselves and people around them.
7. To make their journey more fun and enjoyable.

SYLLABUS

Unit-I

YES!+ Workshop:

Yoga Postures – Seven Layers To our Existence – Puzzles – Sources Of Energy – Live in the present Moment – Importance of Breath – Ujjai Breath – Pranayama – Sudarshana Kriya.

Unit-II

YES!+ Workshop:

Yoga Postures (Suryanamaskars) – Giving 100% in everything – Time management – Happiness point – Opposite Values – Pranayama – Sudarshan kriya

Unit-III

YES!+ Workshop:
Yoga Postures – Knowledge points (Acceptance, opinions discretion and handling mistakes) – Eye Gazing Process – Dance – Life Story process – Sudarshana Kriya (short) – Eternal life – Ego Bursting – Relationships – Parents – Studies – Compliments/Praising process.

Unit-IV

Creative Arts:
Photography – Sketching – Handy-crafts – Clay molding – Singing – Upcycling – Communing with nature – Creative writing.

Unit -V

Service:
Leadership in action – Contributing to society – Take up Responsibility –Empowerment – Public Speaking – Art of Teaching.

REFERENCE BOOKS:

1. Discourse on Patanjali Yoga Sutras by H.H Sri SriRavishankar
2. Human values and professional ethics byRRGaur,RSangal,GPBagaria
3. The Art of Photography by AL Judge
4. Hand made in India : Crafts of India by Ranjan Aditi

(**Note**: It is an uncredited course. It will not be included in the Grade Memo / Certificate. The Certificate will be issued based on the performance and attendance. This course attendance will be counted in the semester overall attendance.)

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)
(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi)
Accredited by NAAC with 'A' Grade
Recognised as Scientific and Industrial Research Organisation
CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204
ESTD: 1980

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R17)
II/IV B.TECH
(With effect from **2017-2018**Admitted Batch onwards**)**

ELECTRONICS & COMMUNICATION ENGINEERING

I-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 BS 2101	Mathematics - IV	3	3	1	--	4	30	70	100
B17 EC 2101	Electronic Devices and Circuits	3	3	1	--	4	30	70	100
B17 EC 2102	Switching Theory and Logic Design	3	3	1	--	4	30	70	100
B17 EC 2103	Signals and Systems	3	3	1	--	4	30	70	100
B17 EE 2104	Network Analysis	3	3	1	--	4	30	70	100
B17 EC 2104	Probability Theory and random Processes	3	3	1	--	4	30	70	100
B17 EC 2107	Electronic Devices and Circuits Lab	2	--	--	3	3	50	50	100
B17 EE 2106	Networks and Electrical Technology Lab	2	--	--	3	3	50	50	100
B17 BS 2106	Programming Skills-I	1	--	--	2	2	50	---	50
B17 BS 2107	English Proficiency-I	--	1	1	--	2	--	--	--
	Total	**23**	**19**	**7**	**8**	**34**	**330**	**520**	**850**

Code: B17BS2101

MATHEMATICS IV
(Common to CE,ECE,EEE& ME)

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

Students should learn

1. The concept of Analytic function, its implications and applications in flow problems.
2. Solution of one- dimensional wave equation, one-dimensional heat equation and two-dimensional Laplace equation by the use of 'separation of variables'.
3. Formation and solution of linear difference equations. Important concepts of Z-transform and its use to solve linear difference equations.
4. Basic concepts of certain discrete and continuous probability distributions.
5. Concepts of Sampling theory for analyzing large and small samples.

Course Outcomes:

Students will be capable of

1. Using the concept of Analytic function in applications including Electrostatics and Fluid dynamics.
2. Finding theoretical solution of certain Elliptic, Parabolic and Hyperbolic partial differential equations.
3. Using Z-transforms to solve linear difference equations with constant coefficients.
4. Fitting of probability frequency distribution to a given data.
5. Using the concepts of sampling theory to analyze data related to some large and small samples.

SYLLABUS

UNIT-I Functions of a Complex Variable

Review- Cartesian form and polar form of a complex variable, Real and imaginary parts of z^n, e^z, sin z, sinh z and log z **(no questions may be set).**

Limit and continuity of a function of the complex variable, derivative, analytic function, entire function, Cauchy- Riemann equations, finding an analytic function, Milne-Thomson method, Applications of analytic function to flow problems, and in Electrostatics. Conformal mapping: the transformations defined by $w = z+c$, $w = cz$, $w = 1/z$, the Bilinear transformation, $w = z^2$ and $w=e^z$.

UNIT-II Applications of Partial Differential Equations

Method of separation of variables, One –dimensional wave equation, the D'Alembert's solution, one-dimensional heat equation, two-dimensional heat flow in steady state (solution of two-dimensional Laplace equation in Cartesian coordinates only)

UNIT-III Difference Equations And Z-Transforms

Formation of a difference equation, Rules for finding complimentary function and particular integral for linear difference equations.

Definition of Z- transform, some standard Z- transforms, properties, transform of a function multiplied by n, initial value theorem and final value theorem(without proof), evaluation of inverse Z- transforms, convolution theorem (without proof), solution of linear difference equations by the use of Z- transforms.

UNIT-IV Probability Distributions

Binomial distribution, Poisson distribution, Normal distribution: Definition (pmf/pdf), notation, mean, variance, moment generating function, probability generating function and fitting of a distribution.

UNIT-V Sampling Theory

Sampling theory: Sampling distribution, standard error, testing of Hypothesis, level of significance, confidence limits, simple sampling of attributes, sampling of variables, estimation of mean and variance.

Large samples: testing of hypothesis for sample proportion, two proportions, single mean and two means.

Small samples: Degrees of freedom, Students' t- distribution, t-test for single mean, two means; Chi-squared distribution-testing the goodness of a fit.

Text Book:

1. Scope and Treatment as in "Higher Engineering Mathematics", by Dr.B.S.Grewal, 43rd Edition, Khanna Publishers.

Reference Books:

1. Advanced Engineering Mathematics, by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N.P.Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics, by H.K.Dass, S.Chand Company.
4. Higher Engineering Mathematics, by B.V.Ramana, Tata Mc Graw Hill Company.
5. Higher Engineering Mathematics, by Dr. M.K.Venkatraman, The National Publishing Company.

Code: B17 EC 2101

ELECTRONIC DEVICES AND CIRCUITS
(Common to ECE & EEE)

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course objectives:

1. To give exposure to the students about intrinsic and extrinsic semiconductors, semiconductor diodes, special purpose diodes like Zener diode, Photo diode, LED, Schottky barrier diode, PIN diode, varactor diode and tunnel diode etc.
2. To give exposure to the students about rectifier circuits using diodes.
3. To give exposure to the students on basics of BJT, JFET and MOSFET and biasing of BJT and FETs.
4. To give exposure to the students on the analysis of transistor at low and high frequencies.

Course outcomes:

After completion of the course the students will be able to

1. Understand the physical structure, principles of operation, electrical characteristics and circuit models of diodes, BJT's and FET's.
2. Use the concepts of semiconductor physics and electronic devices to design and fabricate simple electronic circuits.
3. Use this knowledge to analyze and design amplifier circuits and oscillator circuits to be used in various applications.
4. Extend the understanding of how electronic circuits and their functions fit into larger electronic systems.

SYLLABUS

UNIT-I: Transport Phenomena in Semi Conductors
Mobility and conductivity, intrinsic and extrinsic semiconductors, mass action law, charge densities in a semiconductors, Hall Effect, generation and recombination of charges, drift and diffusion currents, the continuity equation, injected minority carrier charge, potential variation in graded semiconductors.

UNIT- II: PN junction diode and Diode Rectifiers
Open circuited PN junction , PN junction as a rectifier, current components in a PN diode, V-I characteristics and its temperature dependence, transition capacitance, charge control description of a diode, diffusion capacitance, junction diode switching times, Zener diode, Tunnel Diode, Photo diode,Varactor diode, LED,Half wave, Full wave and Bridge Rectifiers with and without filters, Ripple factor and regulation characteristics

UNIT – III: Bipolar junction transistors

Introduction to BJT, operation of a transistor and transistor biasing for different operating conditions, transistor current components, transistor amplification factors: α,β,γ relation between α and β,γ early effect or base-width modulation, common base configuration and its input and output characteristics, common emitter configuration and its input and output characteristics, common collector configuration and its input and output characteristics, Comparison of CE, CB and CC Configurations, Break- down in transistors, Photo Transistor.

Transistor Biasing Circuits: The operating point, Bias stability, different types of biasing techniques, stabilization against variation in I_{co} , $V_{BE,}$ & β. Bias compensation, thermal runaway, thermal stability.

UNIT – IV: Field Effect transistors

JFET and its characteristics, pinch off voltage, FET small signal model, MOSFET and its characteristics, Biasing of FETs.

UNIT – V: Transistors at low and High frequencies

Transistor hybrid model, H-parameters, Analysis of transistor amplifier circuits using h-parameters, comparison of transistor amplifier configurations, analysis of single stage amplifier, effects of bypass and coupling capacitors, frequency response of CE amplifier, Emitter follower, High frequency model of transistor.

Text books:

1. Integrated Electronics: Analog and Digital Circuits and Systems: Jacob Millman, C Halkias, Chetan D Parikh. McGraw – Hill.
2. Electronic Devices and Circuits: N Salivahanan and Suresh Kumar, Third edition, TMH.

Reference Books:

1. Electronic Devices and Circuits Theory, Boylsted, 10^{th}Edition,Pearson
2. Electronic Principles : Albert Paul Malvino, McGraw-Hill.

Code: B17 EC 2102

SWITCHING THEORY AND LOGIC DESIGN

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. To provide insight of number systems and minimization of Boolean functions
2. To learn simplification of Boolean functions using different methods
3. To learn the design of various combinational circuits and their applications
4. To learn the design of various sequential circuits and their applications
5. To introduce the basics and design of Counters, Synchronous and Asynchronous sequential circuits

Course Outcomes:

By the end of the course the students will be able to

1. Understand various basic number systems, codes and basic logic gates.
2. Learn various types of Boolean expressions and theorems and simplificationsusing K-map and Tabulation methods.
3. Design and analyze combinational circuits using logic gates.
4. Understand basics of Flip-flops, design and analyze sequential circuits using those Flip-flops and gates.
5. Design of all types of counters and understand basics of Synchronous and Asynchronous sequential circuits, and analyze them.

SYLLABUS

UNIT-I-Number Systems, Codes and Boolean Algebra:

Number Systems, Base Conversion Methods, Complements of Numbers, 4 Bit Codes-BCD, Excess-3, 2421, 8421 codes. Even and Odd parity, Hamming code, Error detecting and Error correcting codes.

Fundamentals of Boolean Algebra and Logic Gates – AND, OR, NOT, NAND, NOR and XOR. Boolean theorems and proofs.

UNIT-II- Boolean Functions and Minimization:

Boolean SOP and POS functions-Canonical and Standard. Realization with Universal Gates– Simplification of Boolean functions using Karnaugh Map (up to 6 variables) and Quine McClusky methods.

UNIT-III-Combinational Logic Circuits and Design:

Logic Design of Combinational circuits – Binary Addition, Subtraction, Multiplexers, Demultiplexers, Decoders, Encoders, Code Conversion, Priority Encoders, Seven – segment Displays, Comparators and PLDs.

UNIT-IV-Sequential Logic Circuits and Design:
The Flip-flops: SR, RS and JK Flip-Flops, Race around problem, MSJK, T and D-Flip-flops. Flip Flops with preset and clear inputs. Excitation tables of all Flip- Flops and conversions from one type to another. Design of Shift Registers with SIPO, SISO, PIPO and PISO modes and universal shift register. Ring counter and Johnson counter.

UNIT-V- Asynchronous and Synchronous Sequential Circuits:
Design of Asynchronous counters for any modulous. Design of Synchronous counters using SR, JK, T and D-FFs. Basics of Asynchronous Sequential Circuits, Cycles, Races and Hazards. Analysis and Design of Synchronous Sequential Circuits with State Diagrams and State Reduction.

Text books:

1. An Approach to Digital Design, William I. Fletcher, PHI.Engineering
2. Switching and Finite Automata Theory, 2nd Edition, ZviKohavi, Tata McGraw-Hill, 1978.

Reference Books:

1. Fundamentals of Logic Design, Charles H. Roth, Thomson Publications, 5th Edition, 2009
2. Digital Design, Morris Mano, PHI, 3rd Edition, 2001.

Web Resourses:

http://www.ece.ubc.ca/~saifz/eece256.htm
http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT%20Guwahati/digital_circuit/frame/index.html

Code: B17 EC 2103

SIGNALS AND SYSTEMS

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. To introduce the fundamental concepts and techniques associated with the understanding of signals and systems.
2. To familiarize with techniques suitable for analyzing both continuous-time and discrete time LTI systems using transforms.
3. To familiarize with development of the mathematical skills to solve problems involving convolution, filtering, and sampling.

Course Outcomes:

After completion of the course the student will be able to

1. Understand the basic concepts of signals and systems.
2. Analyze the spectral characteristics of Continuous Time and Discrete Time periodic and aperiodic signals using Fourier analysis.
3. Analyze system properties based on impulse response and Fourier analysis.
4. Apply Laplace- transforms for analyzing Continuous -time signals and systems.
5. Apply Z- transforms for analyzing discrete-time signals and systems.
6. Understand the process of sampling and the effects of under sampling.

SYLLABUS

UNIT-I : Introduction to Continuous –Time and Discrete –Time signals and systems

Continuous –Time and Discrete –Time signals, Signal Energy and Power, Periodic Signals, Even and odd Signals, continuous- Time complex Exponential and Sinusoidal Signals, Discrete –Time complex Exponential and Sinusoidal Signals and their Periodicity , The Unit Impulse and Unit step Functions, The Continuous-Time Unit impulse and Unit step Sequence, Continuous –Time and Discrete –Time Systems, Interconnections of Systems, Basic System Properties, Continuous –Time and Discrete Time LTI Systems: – The Graphical interpretation of Convolution Integral and The Convolution Sum, Casual LTI Systems Described by Differential and Difference Equations, Singularity Functions.

UNIT-II : Fourier Series Representation of Periodic Signals

Introduction, Fourier Series Representation of continuous time Periodic Signals(Complex Exponential and Trigonometric Fourier Series only),convergence of the Fourier Series, Properties of continuous time Fourier Series, Fourier Series representation of discrete time periodic signals, Properties of discrete time Fourier Series (Elementary Level on DTFS).

UNIT-III : Continuous and Discrete time Fourier Transform

Introduction, Representation of Aperiodic signals, The continuous time Fourier Transform, The Fourier Transform for periodic signals, Properties of the continuous time Fourier Transform, Systems characterized by linear constant-coefficient differential equations. Discrete time Fourier Transform, Properties of the Discrete time Fourier Transform, Systems characterized by linear constant co-efficient differential equations (Elementary Level on DTFT).

UNIT-IV : Laplace Transform

Introduction, The Laplace Transform, the region of convergence for Laplace Transforms, The Inverse Laplace Transform, Properties of Laplace Transforms, The initial and Final value theorems, Analysis and characterization of LTI systems using the Laplace Transforms.

UNIT-V : Sampling Theorem and Z-transform

Introduction to Sampling Theorem,StatementofSampling Theorem for Low pass and Band pass signals(Theorem Proof for Low Pass signals only) ,reconstruction of a signal from its samples using interpolation, discussion on Oversampling, Critical sampling and Under sampling(aliasing). The Z-Transform(Bilateral and unilateral), The Inverse Z-Transform, Properties of Z-Transform, Initial and Final Value theorems, some common Z-transform pairs, Analysis and characterization of LTI systems using the Z-Transforms.

Text Books:

1. Signals Systems and Communication-B. P. Lathi, BS Publication.
2. Signals and Systems- Alan V. Oppenheim, Alan S. Willsky and Ian T. Young, PHI, 2ndEdn.

Reference Books:

1. Signals and Systems – P.RamakrishnaRao, TMH.
2. Signals and Systems- A.AnandaKumar,PHI.

Code: B17 EE 2104

NETWORK ANALYSIS

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. To learn the concept of network theory and definitions of circuit elements for modeling practical electric circuits.
2. To learn various theorems and techniques in electric circuit analysis and to know their significance and applications
3. To learn phasor concept and apply it to analysis of circuits in sinusoidal steady state.
4. To know the behavior of the steady states and transients states in RLC circuits.
5. To understand the two port network parameters.

Course Outcomes:

1. Gain the knowledge on basic network elements and learn various circuits analyzing techniques
2. Will learn the behavior of energy storing elements (Inductance & Capacitance) in circuits and analyses transient and steady state responses.
3. Will analyze the RLC circuits behavior in detailed.
4. Analyze the performance of periodic waveforms.
5. Gain the knowledge in characteristics of two port network parameters (Z, Y, ABCD, h & g).

SYLLABUS

UNIT-I :Analysis of DC Circuits:

Network elements classification, series and parallel combination of Resistance, Inductance and Capacitance. Star to delta transformation. Types of sources, Source transformation, Mesh analysis and Nodal analysis problem solving with resistances only including dependent sources.

UNIT-II :DC transients:

Inductor, Capacitor, source free RL, RC and RLC response, Evaluation of Initial conditions, Application of unit-step function to RL, RC and RLC circuits, concepts of Natural, Forced and Complete response.

UNIT-III :Steady State Analysis of A.C Circuits :

Average and Effective value of Voltage and Current, Response to sinusoidal excitation - pure resistance, pure inductance, pure capacitance, impedance concept, phase angle, series R-L, R-C, R-L-C circuits problem solving. Complex impedance and phasor notation for R-L, R-C, R-L-C Problem solving using mesh and nodal analysis, Instantaneous and Average Power, Complex Power.

UNIT-IV: Network Theorems:
Thevenin's, Norton's, Milliman's, Reciprocity, Superposition, Max Power Transfer, Tellegens theorems- problem solving using dependent sources also.

Resonance: Introduction, Definition of Q, Series resonance, Bandwidth of series resonance, Parallel resonance, Condition for maximum impedance, , Bandwidth of parallel resonance, general case resistance present in both branches.

UNIT-V: Two-port networks :
Relationship of two port networks, Z-parameters, Y-parameters, Transmission line parameters, h-parameters, Inverse h-parameters, Inverse Transmission line parameters, Relationship between parameter sets, Parallel connection of two port networks, Cascading of two port networks, series connection of two port networks, concept of duality,problem solving including dependent sources .

Textbooks:

1. Engineering Circuit Analysis, William H.Hayt Jr. and Jack E. Kemmerley, 5th Edition, McGraw Hill International Edition.
2. Network Analysis, M. E. Van Valkenburg, 3rd Edition, PHI.
3. Theory and Problems of Electric Circuits 4th Edition, MahmoodNahvi, JosephA. Edminister, Schaum's Outline Series Mcgraw-Hill

Reference Books:

1. Fundamentals of Electric circuits 5th edition Charles K. Alexander and Matthew Sadiku
2. Network Analysis &Synthesis ,Franklin F. Kuo; 2nd edition John Wiley & Sons Inc.
3. Circuit Theory Analysis and Synthesis. Edition 2014 AbhijitChakrabarthi,DhanpatRai&Co.
4. Network Analysis, 3rd Ed, A SudhakarShyammohan.SPalli Tata McGraw Hill Education Pvt Ltd.

Code:B17 EC 2104

PROBABILITY THEORY & RANDOM PROCESSES

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course objectives:

1. To introduce the fundamental concepts and theorems of probability theory.
2. To introduce fundamental concepts of random variables and their statistical descriptions
3. To discuss about various types of random processes, their properties, spectral representation and applications.
4. To understand the difference between time averages and statistical averages, Stationary and Ergodicity.
5. To apply elements of stochastic processes for problems in real life and understand the concept of Noise as applicable to linear Systems.

Course outcomes:

On completion of the course the student will be able to

1. Understand the axiomatic formulation of modern probability theory.
2. Characterize Probability Models and functions of Random variables based on single and multiple random variables.
3. Evaluate and apply moments and characteristic functions and understand the concept of Inequalities and probabilistic limits.
4. Understand the concept of Random process and determine covariance and spectral density of stationary random processes.
5. Demonstrate specific applications to Poisson and Gaussian process, representation of low pass and band pass noise models, Analyze the response of random inputs to linear time invariant systems.

SYLLABUS

UNIT-I Probability Theory
Definitions of Probability, Axioms of Probability, Probability Spaces, Properties of Probabilities, Joint and Conditional Probabilities, Independent Events

UNIT-II Random Variables
Probability Distribution Function, Probability Density Function, Joint Distribution of Two Variables, Conditional Probability Distribution and Density, Independent Random Variables, Normal Distribution, Cauchy's distribution, Exponential Distribution, Binomial Distribution, Poisson distribution, Functions of Random Variables.

UNIT-III Statistical Averages
Random Vectors, Statistical Averages, Characteristic Function of Random Variables, Inequalities of Chebyshev's and Schwartz, Convergence Concepts, Central Limit Theorem.

UNIT-IV Random Processes
Introduction, Definitions, Stationarity, Ergodicity, Covariance Function and their Properties, Spectral Representation, Weiner-Kinchine Theorem.

UNIT-V Linear Systems and Random Noise Processes
Classification of Linear systems, Response of Linear Systems to Random signals, Spectral characteristics of system Response, Gaussian processes, Poisson Processes**,** Low-pass and Band-pass Noise Representation.

Text Books:

1. Probability Theory and Random Processes, S. P. Eugene Xavier, S. Chand and Co. New Delhi, 1998 (2^{nd} Edition).
2. Probability Theory and Random Signal Principles, Peebles, Tata McGrew Hill Publishers.

Reference Books:

1. B.P. Lathi, “Signals, Systems & Communications”, B.S. Publications, 2003.
2. Signal Analysis, Papoulis, McGraw Hill N. Y., 1977.

Code: B17 EC 2107

ELECTRONICS DEVICES AND CIRCUITS LAB
(Common to ECE & EEE)

Lab : 3 Periods	**Int.Marks : 50**
Exam : 3Hrs.	**Ext. Marks : 50**
	Credits : 2

Course objectives:

1. To familiarize the students with various passive and active components like resistors, capacitors, inductors ,semiconductor diodes, Zener diodes, LEDs, BJTS, JFETs and UJTs.
2. To familiarize the students with operation of CROs , function generators and bread boards.
3. To observe and analyze the characteristics of devices like diodes, BJTs & FETs.
4. To analyze the behavior of BJT and JFET amplifiers.

Course outcomes:
At the end of the semester students should be able to

1. Design and fabricate simple circuits like diode rectifiers with filters for providing dc voltages in electronic circuits.
2. Design and fabricate amplifiers with required gain for use in various communication applications.
3. Design and fabricate simple electronic circuits for everyday applications like traffic control lights using relays, automatic counters using LDRs and Burglar alarms.

ELECTRONIC WORKSHOP PRACTICE

1. Identification ,Specifications and testing Of R,L,C components, colour codes, potentiometers, coils and bread boards
2. Identification ,Specifications and testing of devices like diodes, BJTs, JFETs, SCR and UJT.
3. Soldering of Simple Circuits using Active &Passive Components.
4. Study and operation of Transformers, Ammeters(Analog & Digital),Voltmeters(Analog &Digital) , Analog and Digital Multimeters and Function Generators, Regulated Power Supply, Decade Resistance, Inductance &Capacitance Boxes And CRO.

LIST OF HARDWARE EXPERIMENTS:
1. V-I Characteristics Of Semiconductor Diode (Ge& Si), LED and Zener Diode
2. Half Wave And Full Wave Rectifier With And Without Filter
3. Characteristics Of BJT In CE Configuration
4. JFET Characteristics
5. Transistor Biasing Circuits And Transistor As Switch
6. CE Amplifier
7. JFET Common Source Amplifier

LIST OF SIMULATION EXPERIMENTS

1. Simulation of V-I Characteristics Of Semiconductor Diode, LED and Zener Diode
2. Simulation of Regulation Characteristics Of ZENER Diode
3. Simulation of CC Amplifier
4. Simulation of JFET Characteristics
5. Simulation of BJT Characteristics In CB Configuration
6. Simulation of JFET Amplifier
7. Simulation of UJT Characteristics

NOTE: (Minimum of Twelve Experiments Should Be Conducted)

Reference Books:

1. Integrated Electronics: Analog and Digital Circuits and Systems: Jacob Millman, C Halkias, Chetan D Parikh. McGraw – Hill.

Code: B17 EE 2106

NETWORKS AND ELECTRICAL TECHNOLOGY LAB

Lab : 3 Periods		**Int.Marks**	**: 50**
Exam : 3Hrs.		**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

1. To learn to make simple electric circuits by using different sources, loads and components and verify basic laws.
2. To experimentally verify various theorems of circuit analysis.
3. To learn to find circuit models for two-terminal devices and two-port networks.
4. Conducting experiments on characteristics of generators & motors
5. Load tests on series, shunt, compound motors and compound generators-swinburne's, Hopkinson's test.
6. OC & SC tests on single phase transformers, Sumpner's test.

Course Outcomes:

1. Students will gain the skill to make and experiment with practical electric circuits.
2. Students will be able to measure voltage, current, power in practical electric circuits.
3. Students will know the significance of various theorems and their applications.
4. Students will be able to model devices for circuit analysis.
5. Students will be able to assess the behaviour of different electrical machines.
6. Students will be able to predetermine the efficiency and regulation of different machines.

LIST OF EXPERIMENTS

1. Maximum Power Transfer Theorem
2. Superposition Theorem
3. Thevinin’s Theorem
4. Series Resonanc
5. Ohm’s Law and Characteristics of Filament Lamp
6. Parameters of Iron Cored Inductor
7. Swinburne’s Test
8. Load Test on Dc Shunt Motor
9. Load Test on Dc Series Motor
10. Load Test on 3 Phase Slip ring Induction Motor
11. OC and SC Test on Single Phase Transformer
12. Voltage Regulation of An Alternator by Synchronous Impedance Method
13. Speed Control of Dc Shunt Motor

Reference Books:

1. Network Analysis, M. E. Van Valkenburg, 3rd Edition, PHI.

Code: B17 BS 2106

PROGRAMMING SKILLS-I
(PYTHON)
(Common to ECE & EEE)

Lab	**: 2 Periods**	**Int.Marks**	**: 50**
Exam	**: 3 Hrs.**	**Credits**	**: 1**

Course Objectives:

1. To know basic principles of Object Oriented Programming in the context of python programming language.
2. To study different data types of python, to design a programs.
3. To study and apply different types of concepts like Inheritance, Exception Handling, Turtle and design programs using these concepts.
4. To study different built in modules like sys, os , math & SQLite
5. To study server side scripting in python like CGI scripts

Course Outcomes:

1. Ability to apply object oriented concepts in programming.
2. Ability to define, understand and differentiate different types of data types and apply them.
3. Ability to recognize various concepts of python and develops the programs using them and also develop web based application.

SYLLABUS

UNIT-I:
Overview, Environment Set Up, Basic Syntax, Identifiers, Reserved Words, Lines and Indentation, Multi-Line Statements, Quotation, Comments, Multiple Statements on a Single Line Variable Types, Standard Data Types, Numbers (math, random, fraction) , Strings, Lists, Tuples , Dictionaries

UNIT-II:
Operators, Arithmetic Operators, Comparison (Relational) Operators, Assignment Operators, Logical Operators, Bitwise Operators, Membership Operators, Identity Operators, Decision Making :if, if-else, nested if , Loops: for, while, nested loops

UNIT-III:
Functions, Function Arguments: Required arguments, Keyword arguments, Default arguments, Variable-length arguments, The Anonymous Functions: lambda, Scope of Variables, Modules, sys, os , Date & Time

UNIT-IV:

Files & its operations, Exceptions, Standard Exceptions, Assertions, The try-finally Clause, Raising an Exception, User-Defined Exceptions, Classes and objects , OOPS, Data member , Function overloading, Instance variable, Inheritance, Instance, Instantiation, Operator overloading

UNIT-V:

HTML,CSS Basics, Data Base(SQLite), Database Connection, CRUD Application , CGI Architecture, Web Server Support and Configuration, GET and POST Methods, CGI Scripts.

UNIT-VI:

Project

References:

1. Dive Into Python 3
2. Think Python
3. Halterman python book
4. Dr. Andrew N. Harrington Computer Science Department, Loyola University Chicago© Released under the Creative commons Attribution-Noncommercial-Share Alike 3.0 United States License http://creativecommons.org/licenses/by-nc-sa/3.0/us/
5. https://www.python.org/
6. http://www.tutorialspoint.com/python3/
7. https://www.w3schools.com/

Code: B17 BS 2107

ENGLISH PROFICIENCY-I

(Common to All Branches)

Lecture	**: 1Period**		
Tutorial	**: 1 Periods**	**Int.Marks**	**: --**
Exam	**: ---**	**Credits**	**: --**

AIM:

Enriching the communicative competency of the students by adopting the activity-based as well as the class-oriented instruction with a view to facilitate and enable them to enhance their language proficiency skills.

Course Objectives:

The Students will be able to

1. Communicate their ideas and views effectively.
2. Practice language skills and improve their language competency.
3. Know and perform well in real life contexts.
4. Identify and examine their self attributes which require improvement and motivation.
5. Build confidence and overcome their inhibitions, stage freight, nervousness, etc.
6. Improve their innovative practices in speaking.

Course Outcomes:

The Students will

1. Improve speaking skills.
2. Enhance their listening capabilities.
3. Learn and practice the skills of composition writing.
4. Enhance their reading and understanding of different texts.
5. Improve their inter-personal communication skills.
6. Be confident in presentation skills.

SYLLABUS

UNIT-1:LISTENING

Selected Motivational Speeches

Selected Moral Stories

UNIT-2:SPEAKING

Book Review

Skit Presentation

PowerPoint Presentations

Describing event/place/thing

Extempore

Group Discussion

Picture Perception and Describing Test

UNIT-3:READING

Speeded Reading
Reading Comprehension

UNIT-4:WRITING

Paragraph Writing
Literary Appreciation – Understanding the Language of Literature

UNIT-5:PROJECT

Ad Making

References:

1. Classic Short Stories-A Reader's Digest Selection
2. English for Colleges by Brendan J. Carroll, Macmillan Publications
3. The World's Great Speeches- edited by B.S. Sekhar, Jeet Publications
4. Fundamentals of Technical Communication by Meenakshiraman, Sangeta Sharma of OUP
5. English and Communication Skills for Students of Science and Engineering, by S.P. Dhanavel, Orient Blackswan Ltd. 2009
6. Enriching Speaking and Writing Skills, Orient Blackswan Publishers
7. The Oxford Guide to Writing and Speaking by John Seely OUP

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R17)
II/IV B.TECH
(With effect from **2017-2018**Admitted Batch onwards**)**

ELECTRONICS & COMMUNICATION ENGINEERING

II-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 EC 2201	Electronic Circuit Analysis	3	3	1	--	4	30	70	100
B17 EE 2203	Control Systems	3	3	1	--	4	30	70	100
B17 EC 2202	Electromagnetic Field Theory and Transmission Lines	3	3	1	--	4	30	70	100
B17 EC 2203	Analog Communications	3	3	1	--	4	30	70	100
B17 EC 2204	Computer Architecture and Organization	3	3	1	--	4	30	70	100
B17 BS 2201	Management Science	3	3	1	--	4	30	70	100
B17 EC 2207	Electronic Circuit Analysis Lab With Simulation	2	--	--	3	3	50	50	100
B17 EC 2208	Analog Communication Lab	2	--	--	3	3	50	50	100
B17 BS 2205	Programming Skills–II	1	--	--	2	2	50	--	50
B17 BS 2204	Professional Ethics & Human Values	--	2	--	--	2	--	--	--
B17 BS 2206	English Proficiency-II	--	1	1	--	2	--	--	--
	Total	**23**	**21**	**7**	**8**	**36**	**330**	**520**	**850**

Code: B17EC2201

ELECTRONIC CIRCUIT ANALYSIS

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

The aim of this course is to

1. Understand the concept of multistage amplifiers and analyze them.
2. Learn the classification of feedback amplifiers and analyze them.
3. Compare voltage, power and tuned voltage amplifiers and analyze them.
4. Understand the principle of oscillator and analyze different types of sinusoidal oscillators.
5. Understand the concept and analyze applications of op-amp.

Course Outcomes:

After the completion of the course, students will be able to:

1. Know the equivalent circuit of multistage amplifier and its analysis. [K3]
2. Identify the different feedback topologies and analyze them. [K1]
3. Explain the principle of oscillator and design different types of sinusoidal oscillators. [K3]
4. Define the difference between voltage and power amplifiers and design different classes and know that Tuned amplifiers amplify a narrow band of frequencies and will also be able to analyze them.[K1,K2, K3]
5. Identify that Op-amp not only amplifies but also performs different operations and analyze some of its applications.[K1,K2]

SYLLABUS

UNIT – I: Multistage Amplifiers

Transistor at high frequencies, CE short circuit current gain and concept of Gain Bandwidth product. BJT and FET RC coupled amplifiers at low and high frequencies. Frequency response and calculation of Band Width of Multistage Amplifiers.

UNIT – II: Feed Back Amplifiers

Concept of Feed Back Amplifiers - Effect of Negative Feedback on the amplifier characteristics. Four feedback topologies, Method of analysis of Voltage Series, Current Series, Voltage Shunt and Current Shunt feedback Amplifiers.

UNIT – III: Sinusoidal Oscillators

Condition for oscillations and types of Oscillators – RC Oscillators: RC Phase Shift and Wien bridge Oscillators. LC Oscillators: Hartley, Colpitts, Clapp, Tuned Collector and Crystal Oscillators.

UNIT – IV: Power and Tuned Voltage Amplifiers

Classification of Power Amplifiers. Series fed, Transformer coupled class-A and class-B power amplifiers. Push Pull Class-A, Class-B and Class-AB Power Amplifiers. Cross-over Distortion in Pure Class-B Power Amplifier and Class-AB Power Amplifier- Trickle Bias, Derating Factor and Heat Sinks – Complementary Push Pull Amplifier. Analysis of Single tuned, Double tuned and Stagger Tuned Amplifiers with gain and Bandwidth Calculations.

UNIT – V: Operational Amplifiers

Concept of Differential Amplifier. Differential Amplifier supplied with a constant current source. Calculation of common mode rejection ratio. Block diagram and Ideal characteristics of an Op-Amp. Applications of Op-Amp: Inverting and Non-Inverting amplifiers, Integrator, Differentiator, Summing, Subtracting and Logarithmic Amplifiers. Definition and Measurement of OP-Amp Parameters.

Text Books:

1. Integrated Electronics- Millman and Halkias.
2. Op-amps and Linear Integrated Circuits – Gayakwad.

Reference Books:

1. Electronic Devices and Circuits – Mottershead.
2. Electronic Devices and Circuits by Salivahanan. Tata McGraw-Hill pub.

Code: B17 EE 2203

CONTROL SYSTEMS

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course objectives:

1. To learn the modeling of linear systems using transfer functions and obtain transfer functions for physical electrical and mechanical systems.
2. To learn to represent systems using block diagrams and signal flow graphs and derive their transfer functions.
3. To learn the significance of time response and find it for system analysis in transient and steady state.
4. To learn the concept of stability and know different techniques of stability analysis.
5. To learn the concept of frequency response and its application for control system analysis.

Course Outcomes:

1. Students will be able to model electrical and mechanical physical systems by applying laws of physics.
2. Students will be able to represent mathematical models of systems using block diagrams & Signal Flow Graphs and derive their transfer functions.
3. Students will be able to analyze systems in time domain for transient and steady-state behavior.
4. Students will learn the concept of stability and use RH criterion and Root locus methods for stability analysis.
5. Students will learn to obtain frequency response plots of systems and use them for system analysis and stability assessment.

SYLLABUS

UNIT-I

Introduction to control systems- Open loop and closed loop systems- Transfer Functions of Linear Systems– Impulse Response of Linear Systems – Mathematical Modeling of Physical Systems – Equations of Electrical Networks – Modeling of Mechanical Systems – Equations of Mechanical Systems, Analogous Systems.

UNIT-II

Block Diagrams of Control Systems – Signal Flow Graphs (Simple Problems) – Reduction Techniques for Complex Block Diagrams and Signal Flow Graphs (Simple Examples)- Feedback Characteristics of Control Systems

UNIT-III

Time Domain Analysis of Control Systems – Time Response of First and Second Order Systems with Standard Input Signals – Steady State Error Constants – Effect of Derivative and Integral Control on Transient and Steady State Performance of Feedback Control Systems.

UNIT-IV

Concept of Stability– Routh-Hurwitz Criterion, Relative Stability Analysis, the Concept and Construction of Root Loci, Analysis of Control Systems with Root Locus (Simple Problems to understand theory).

UNIT-V

Frequency Domain Analysis of control systems - Bode Plots- Log Magnitude versus Phase Plots- Polar Plots -Correlation between Time and Frequency Responses - Nyquist Stability Criterion -Assessment of Relative Stability -All Pass and Minimum Phase Systems - Constant M and N Circles.

Text Books:

1. I. J. Nagrath and M. Gopal, ' Control Systems Engineering', New Age International Publishers(6^{th} Edition).
2. Benjamin C. Kuo, ' Automatic Control Systems' , PHI (5th Edition).

Reference Books:

1. Katsuhiko Ogata, 'Modern Control Engineering', , PHI (4^{th} Edition).
2. Richard C. Dorf and Robert H. Bishop, 'Modern Control Systems', Addison-Wesley Publishers(8^{th} Edition)

Code: B17 EC 2202

ELECTROMAGNETIC FIELD THEORY & TRANSMISSION LINES

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. To introduce the concepts of static electric field , steady magnetic field and time varying electromagnetic fields in real time applications.
2. To introduce Maxwell's equations and their applications in practical situations.
3. To introduce the fundamental theory of electromagnetic wave propagation in bounded and unbounded media.
4. To study the propagation of energy in practical transmission lines and wave guides.

Course Outcomes:

After the completion of the course, Students should have the

1. Ability to apply the knowledge of mathematics, Science and engineering to the analysis and design of systems involving electric and magnetic fields as well as electromagnetic Waves.
2. Ability to identify, formulate and solve engineering problems in the area of electric and Magnetic fields and waves.
3. Ability to use Maxwell's equations to solve electromagnetic field problems.
4. Ability to apply the knowledge of electromagnetic fields in practical transmission lines and waveguides.

SYLLABUS

UNIT – I :Electrostatics:

Introduction, Coulomb's law and electric field intensity, electric field due to different types of charge distributions, Field due to infinite line charge and finite line charge, Field due to infinite sheet charge, electric flux density, gauss's law and applications, Energy and potential, electric field in terms of potential gradient, electric dipole, stored energy in static electric field and energy density, convection and conduction currents, continuity equation, conductors in electric field, relaxation time, dielectrics in electric field, Laplace's and Poisson's equations, uniqueness theorem, different capacitance configurations, Boundary conditions on $\vec{E}$ & $\vec{D}$ at the interface between two media, Related Problems.

UNIT – II :Magneto statics:

Introduction, Biot-savart's law, Ampere's circuital law, applications of Ampere's circuital law, Point form of Ampere,s circuital law, magnetic flux density, Gauss's law for magnetic fields, scalar and vector magnetic potentials, forces due to magnetic fields, magnetization in materials, inductance, boundary conditions on $\vec{H}$ & $\vec{B}$ at the interface between two media, energy stored in steady magnetic field, Related problems.

UNIT – III Time varying fields and Maxwell's equations:
Introduction, Faraday's law of electromagnetic induction, Transformer emf and motional emf, Maxwell's equations in integral and differential forms, word statements, Maxwell's equations using phasor notation, Boundary conditions on $\overrightarrow{E}, \overrightarrow{D}$,$\overrightarrow{H}$ &$\overrightarrow{B}$ at the interface between two media,Retarded Potentials, Related problems.

UNIT – IV Electromagnetic Waves:
Introduction, Wave equations for free space and for a conductive medium, uniform plane waves, properties of uniform plane waves, Relation between E and H in uniform plane wave, wave propagation in lossless and lossy media, Propagation in good conductors and good dielectrics, depth of penetration, polarization, Reflection of plane waves by a perfect conductor for normal and Oblique incidences, Reflection of plane waves by a perfect dielectric for normal and Oblique incidences, Brewster angle and critical angle, Poynting's theorem, Related Problems.

UNIT – V:Transmission lines and Rectangular Wave guides:
Transmission lines **-** Introduction, types of transmission lines, equivalent circuit of transmission line, Primary and secondary constants of the line, Transmission line equations, characteristic impedance and expression for characteristic impedance, Reflection coefficient, standing wave ratio, lossless line, distortion less line, input impedance of transmission line, shorted and open circuited lines, impedance transformation with $\frac{\lambda}{8}, \frac{\lambda}{4} and \ \frac{\lambda}{2}$ lines, Construction of smith chart, applications of smith chart, Single stub matching, Related problems.
Rectangular Waveguides **-** Introduction, TM modes in rectangular waveguides, TE modes in rectangular waveguides, Impossibility of TEM mode in waveguides, Characteristics of TE and TM modes, cutoff frequency, cutoff wavelength, phase and group velocities, characteristic wave impedance, dominant mode, related problems.

Text Books:

1. Principles of Electromagnetics - N.O.Sadiku, Oxford University Press,4th edition.
2. EM Waves and Radiating Systems – E.C.Jordan and K.G.Balmain, Printice Hall India

Reference Books:

1. Engineering Electromagnetics – W.A.Hayt and JABuck, Tata McGraw Hill.
2. Electromagnetics with applications – Kraus and Fleisch, McGraw Hill.

Code: B17 EC 2203

ANALOG COMMUNICATIONS

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. To understand the fundamental concepts of communication systems.
2. To familiarize with the concepts of linear or amplitude modulation (AM, DSB-SC, SSB and VSB) and demodulation techniques.
3. To familiarize with the concepts of angular or non-linear modulation (FM and PM) and demodulation techniques.
4. To understand and compare different analog modulation schemes.
5. To familiarize with techniques for generating and demodulating narrow-band and wide-band frequency and phase modulated signals.
6. To provide a good understanding of the behavior of analog communications in the presence of noise.
7. To develop a clear insight into the relations between the input and output signals in various stages of a transmitter and a receiver of AM & FM systems.
8. To classify and discuss different types of transmitters and receivers as applicable to analog communication systems.

Course Outcomes:

Upon successful completion of this course the student will be able to

1. Understand the need for modulation and the concepts of Amplitude Modulation and Demodulation techniques and evaluate various parameters in time and frequency Domain.
2. Understand the concepts of Angle Modulation and Demodulation techniques and Evaluate various parameters of Angle modulated waveform in Time and Frequency Domain.
3. Analyze and compare the performance of various analog modulation techniques in the presence of noise.
4. Analyze different characteristics of transmitters.
5. Analyze different characteristics of receivers.

SYLLABUS

UNIT-I: Amplitude Modulation Systems:

Need for Frequency Translation, A Method for Frequency Translation, Amplitude Modulation, AM Modulators, Envelope detector, Square law demodulator, Maximum Allowable Modulation for Rectifier Detection, Spectrum and Power Efficiency, DSB-SC Modulation and its Spectrum, Balanced Modulator, Synchronous Detectors, SSB Modulation, SSB Modulators (Filter Method, Phase Shift Method), SSB Demodulator, Vestigial Side Band Modulation, Quadrature Amplitude Modulation, Frequency Division Multiplexing.

UNIT-II: Angle Modulation:

Angle Modulation, Phase and Frequency Modulation, Relationship between Phase and Frequency Modulation, Phase and Frequency Deviation, Spectrum of an FM Signal, Bandwidth of Sinusoidally Modulated FM Signal, Effect of the Modulation Index on Bandwidth, Spectrum of Constant Bandwidth FM, Phasor Diagram for FM Signals.FM Generation**:** Parameter variation method, Armstrong's Indirect method, Frequency Multiplication and application to FM, FM Demodulator, FM Demodulation using PLL, Pre – emphasis and De – emphasis.

UNIT-III: Noise in AM and FM Systems:

Sources of Noise, Resistor Noise, Shot Noise, Noise in AM Systems, Noise in Frequency Modulation Systems, Comparison between AM and FM with respect to Noise, Pre-Emphasis and De-emphasis and SNR Improvement, Threshold in Frequency Modulation.

UNIT-IV: Radio Transmitters:

Classification of Radio Transmitters, AM and FM Transmitters, Radio Telegraph and Telephone Transmitters, SSB Transmitters.

UNIT-V: Radio Receivers:

Radio receiver Types-Tuned Radio Frequency Receiver, Super Heterodyne Receiver, **AM Receivers:** RF Section and Characteristics, Frequency Changing and Tracking, Intermediate Frequency and IF Amplifiers, Detection and Automatic Gain Control (AGC), **FM Receivers:** Amplitude Limiting, FM Demodulators, Comparison with AM Receivers. **Communication Receivers:** Extensions of Super-heterodyne Principles, Additional Circuits, **SSB and ISB Receivers**.

Text Books:

1. Principles of Communication Systems, H. Taub and D. L. Schilling, McGraw Hill, 1971.
2. Communication Systems, Simon Haykins (2nd Edition).
3. Electronic Communication Systems, G. Kennedy, McGraw Hill, 1977 (2nd Edition).

Reference Books:

1. Electronic Communication Systems, G. Kennedy, McGraw Hill, 1977 (2nd Edition).
2. Modern Digital and Analog Communication Systems, B. P. Lathi (2nd Edition).

Code: B17EC2204

COMPUTER ARCHITECTURE AND ORGANIZATION

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. To understand Basic building blocks of a Digital Computer.
2. To understand the designing of a CPU.
3. Analyze the concept of interfacing of peripheral devices.
4. To understand the concept of Memories and design of Main memory.

Course Outcomes:

After successfully completing the course students will be able to:

1. Understand how computers represent and manipulates data.
2. Develop the general architecture design of a digital computer.
3. Learn the art of Microprogramming.
4. Develop independent learning skills to interface main memory & I/O.

SYLLABUS

UNIT -I:Register Transfer and Micro operations:
Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro operations, Logic Micro operations, Shift Micro operations, Arithmetic Logic Shift Unit.

UNIT -II: Basic Computer Organization:
Instruction Codes, Computer Registers, Computer Instructions, Timing and Control, Instruction Cycle, Memory Reference Instructions, Input - Output and Interrupt, Complete ComputerDescription.

UNIT -III:Micro programmed Control:
Control Memory, Address Sequencing, Microinstruction Formats, Micro program Example, Design of Control Unit.
CPU Organization: Introduction, General Register Organization, Stack Organization Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Program Control, Reduced Instruction Set Computer (RISC).

UNIT-IV: Input – Output Organization:
Peripheral Devices, Input - Output Interface, Asynchronous Data Transfer, Modes of Transfer, Priority Interrupt, Direct Memory Access (DMA), Input- Output processor, CPU-IOP communication.

UNIT- V:Memory Organization:
Memory Hierarchy, Main Memory, Auxiliary Memory, Associative Memory, Cache Memory, Virtual Memory.

Text Books:

1. Computer System Architecture, M. Morris Mano, PHI Publications, (3rd Edition May 1996).
2. Computer Architecture and Organization. John P.Hayes. McGraw Hill International, (3rd Edition).

Reference Books:

1. Computer Organization, V. Carl Hamacher, Zvonko G. Vranesic and Safwat G. Zaky, McGraw Hill International, (4th Edition).
2. Digital Computer Fundamentals, Thomas C. Bartee.

Code: B17 BS 2201

MANAGEMENT SCIENCE
(Common to ECE & EEE)

Lecture	**: 3 Periods**	**Int.Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objective:

To enlighten the technical students with functional management related issues like Principles of Management, Marketing Management, Human Resource Management, Production Management , Financial Management and Strategic Management techniques.

Course Outcomes:

1. Create awareness about the concepts like Evolution of Management thought, functions & principles of management.
2. Provide all round information to the students about matters related to concepts & functions related to Marketing.
3. Acquire in-depth knowledge about the concepts and functions of HRM.
4. Understand about aspects of Production Management and Financial Management.
5. Gain knowledge about Strategy formulation & implementation, SWOT analysis in order to compete with the competition & to gain competency advantage.

SYLLABUS

UNIT-I: Introduction to Management
Concept, Nature and importance of Management, Functions of management, Evolution of Management thought, Fayol's principles of Management, Theories of Motivation, Decision making process.

UNIT-II: Marketing Management
Concept, Functions of marketing, Marketing Mix, Marketing strategies based on Product life cycle, Channels of distribution.

UNIT-III: Human Resource Management (HRM)
Concepts of HRM, Personal Management and Industrial Relations, Basic functions of HR Manager-Man power planning, Recruitment , Selection, Placement, Training, Development, Compensation and Performance Appraisal.

UNIT-IV: Production Management
Production planning &control(PPC), Objectives, Functions, Stages of PPC, Plant location(Site Selection).

Financial Management

Types of capital- Fixed and Working Capital, Methods of Raising finance. Long-term, Medium-term and Short-term financial sources.

UNIT-V:Strategic Management

Vision, Mission, Goals, objectives, policy, strategy, Elements of corporate planning process, Environmental scanning, SWOT analysis Steps in strategy formulation and implementation of Generic strategy alternatives

Text Books:

1. Dr. Arya Sri – Management Science**,** TMH 2011

Reference Books:

1. Marketing Management- PHILIP KOTLER.
2. HRM & IR- P.SUBBA RAO
3. Business Policy & Strategic Management- FRANCIS CHERUNILAM
4. Financial Management - I.M.Pandey.

Code: B17 EC 2207

ELECTRONIC CIRCUIT ANALYSIS LAB WITH SIMULATION

Lab	**: 3 Periods**	**Int.Marks**	**: 50**
Exam	**: 3Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

This laboratory course enables students to get practical experience in design, assembly and evaluation of analog electronic circuits. They will use Multisim to test their electronic designs.

Course Outcomes:

Students will be able to:

1. Acquire a basic knowledge on simple applications of operational amplifier.
2. Observe the amplitude and frequency responses of negative feedback amplifier and two stage RC coupled amplifier.
3. Design and test sinusoidal oscillators.
4. Design and test a power amplifier.
5. Design, construct and take measurement of the analog electronic circuits to compare experimental results in the laboratory with theoretical analysis.
6. Use Multisim to test their electronic design.

LIST OF EXPERIMENTS

1. Design of LC Oscillators (Hartley Oscillator,Colpitts Oscillator)
2. Design of RC Oscillators (Wien Bridge Oscillator,RC phase Shift Oscillator)
3. Design of Basic Applications of Operational Amplifier.
4. Frequency response of Two Stage RC Coupled Amplifier.
5. Frequency response ofCurrent Series Feedback Amplifier(with and without feedback)
6. Measurement of resonant frequency, bandwidth and quality factor of single Tuned Voltage Amplifier.
7. Calculation of Collector Circuit efficiency of Class B Push Pull Power Amplifier.
8. Applications of Operational Amplifiers.

LIST OF EXPERIMENTS
(Simulation)

9. Design of LC Oscillators (Hartley Oscillator,Colpitts Oscillator)
10. Design of RC Oscillators (Wien Bridge Oscillator,RC phase Shift Oscillator)
11. Design of Basic Applications of Operational Amplifier.
12. Frequency response of Two Stage RC Coupled Amplifier.
13. Frequency response ofCurrent Series Feedback Amplifier(with and without feedback)
14. Measurement of resonant frequency, bandwidth and quality factor of single Tuned Voltage Amplifier.
15. Calculation of Collector Circuit efficiency of Class B Push Pull Power Amplifier.
16. Applications of Operational Amplifiers.

Reference Book :

1. Integrated Electronics- Millman and Halkias.

Code: B17 EC 2208

ANALOG COMMUNICATION LABORATORY

Lab : 3 Periods		**Int.Marks**	**: 50**
Exam : 3Hrs.		**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives

1. The purpose of this course is to provide the student with a practical perspective of various Analog communication modules.
2. This course also helps the student to implement various analog modulation and demodulation schemes using discrete components.
3. To be familiar with the design of various parameters of LPF , BPF and HPF .
4. To design IF and RF amplifiers and plot their frequency response.
5. To be familiar with different types of experiments like pre-emphasis, de-emphasis and DSB-SC waveform generators.

Course Out Comes

1. Design and implement modulation and demodulation circuits for amplitude modulation technique.
2. Design and implement modulation and demodulation circuits for frequency modulation technique.
3. Design second order passive and active filters for various frequency bands.
4. Construct the circuit and study the characteristics of different transmitter and receiver circuits such as Harmonic generator, RF Amplifier, IF Amplifier, pre-emphasis and de-emphasis.

SYLLABUS

Generation of AM Signal and measurement of Modulation Index.
Diode Detector for AM Signals.
Generation of FM Signal.
FM Detector.
Receiver Measurements/RF Amplifier
Balanced Modulator.
Passive/Active Filters (LPF, HPF, BPF).
Attenuator/Equalizer/ Twin-T-Network.
Frequency Multiplier/Limiter.
SSB Generation and Detection.
Pre-emphasis and De-emphasis.
IF Amplifier.

Reference :

1. Principles of Communication Systems, H. Taub and D. L. Schilling, McGraw Hill, 1971.

Code: B17 BS 2205

PROGRAMMING SKILLS-II
(JAVA)
(Common to ECE & EEE)

Lab	**: 2 Periods**	**Int.Marks**	**: 50**
Exam	**: 3 Hrs.**	**Credits**	**:1**

Course Objectives:

1. To know basic principles of Object Oriented Programming in the context of java programming language.
2. To study different types of arrays to design programs.
3. To study and apply different types of java concepts like multithreading ,packages ,exception handling ,interfaces and design programs using these concepts.
4. To study and apply various features of AWT components.
5. To know the basic concepts of networking in the context of java programming.

Course Outcomes:

1. Ability to define different procedural and object oriented concepts and will be able to differentiate between them.
2. Ability to define, understand and differentiate different types of arrays and apply them.
3. Ability to recognize various concepts of java and develops the programs using them.
4. Ability to identify and differentiate the various features of AWT components to construct container based programs.
5. Ability to describe and explain the concept of networking.

SYLLABUS

UNIT-I:
Overview, Environment Set Up, Basic Syntax, Identifiers, Reserved words, Data Types, Literals, Basic Operators

UNIT-II:
Control Statements in Java: if...else statement, for, while, do-while, for-each, Nested for loops, switch, break, continue, return, Objects & Classes, Access Specifiers, Input & Output, Arrays, Strings

UNIT-III:
Methods, Relationship between objects, Object-Oriented Programming: Encapsulation, Abstraction, Inheritance, Polymorphism, Interfaces, Type Casting, Packages

UNIT-IV:

Exception Handling: try, catch, final, finally, throw, throws, Built-in, User-defined Exceptions, Files: Read, Write and Append operations using text streams & byte streams

UNIT-V:

Collection Framework, Generics

UNIT-VI:

Threads: life cycle, single tasking, multi tasking, Deadlocks, Thread Priorities, Daemon Threads, Serialization

References:

1. JAVA Complete Reference by Herbert Schildt
2. Core JAVA An Integrated Approach by Dr. R. Nageswara Rao.
3. http://spoken-tutorial.org/

Code: B17BS2204

PROFESSIONAL ETHICS & HUMAN VALUES

(Common to CSE, ECE & IT)

Lecture	**:2 Periods**	**Int. Marks**	**: --**
Exam	**: ---**	**Credits**	**: --**

Course Objectives:

1. To inculcate Ethics and Human Values into the young minds.
2. To develop moral responsibility and mould them as best professionals.
3. To create ethical vision and achieve harmony in life.

Course outcomes:

By the end of the course student should be able to understand the importance of ethics and values in life and society.

SYLLABUS

UNIT – I

Ethics and Human Values: Ethics and Values, Ethical Vision, Ethical Decisions, **Human Values** – Classification of Values, Universality of Values.

UNIT – II

Engineering Ethics: Nature of Engineering Ethics, Profession and Professionalism, Professional Ethics , Code of Ethics, Sample Codes – IEEE, ASCE, ASME and CSI.

UNIT – III

Engineering as Social Experimentation:

Engineering as social experimentation, Engineering Professionals – life skills, Engineers as Managers, Consultants and Leaders Role of engineers in promoting ethical climate, balanced outlook on law.

UNIT – IV

Safety Social Responsibility and Rights:

Safety and Risk, moral responsibility of engineers for safety, case studies – Bhopal gas tragedy, Chernobyl disaster, Fukushima Nuclear disaster, Professional rights, Gender discrimination, Sexual harassment at work place.

UNIT – V

Global Issues:

Globalization and MNCs, Environmental Ethics, Computer Ethics, Cyber Crimes, Ethical living, concept of Harmony in life.

Text Books:

1. Govindharajan, M., Natarajan, S. and Senthil Kumar, V.S., Engineering Ethics, Prentice Hall of India, (PHI) Delhi, 2004.
2. Subramainam, R., Professional Ethics, Oxford University Press, New Delhi, 2013.

Reference Books:

1. Charles D, Fleddermann, “Engineering Ethics”, Pearson / PHI, New Jersey 2004 (Indian Reprint)

Code: B17 BS 2206

ENGLISH PROFICIENCY-II

(Common to All Branches)

Lecture	**: 1Period**		
Tutorial	**: 1 Periods**	**Int.Marks**	**: --**
Exam	**: ---**	**Credits**	**: --**

AIM:
To equip the students with the components of the language required and help them gain adequate knowledge so as to become employable and competent.

Course Objectives:
Students will be able to
1. Enhance their Interpretative skills
2. Understand how to prepare a text.
3. Comprehend various types of writing discourses and respond.
4. Produce effectively different write-ups related to various business contexts.
5. Strengthen their emotional make-up.
6. Perceive various writing discourses.

Course Outcomes:

The students will

1. Develop the skills of taking and making notes
2. Interpret the pictures appropriately and effectively.
3. Read, comprehend and infer a given piece of writing effectively.
4. Learn and practice the skills of Research writing.
5. Communicate well through various forms of writing.
6. Be confident in giving presentations and dealing with people.

SYLLABUS

UNIT-1:SPEAKING
Analyzing proverbs
Enactment of One-act play

UNIT-2:READING
Reading Comprehension
Summarizing Newspaper Article

UNIT-3:WRITING
Note Taking &Note Making
Precis Writing
Essay Writing
Letter Writing
Picture Description
Literary Appreciation– Learning the Language of Literature

UNIT-4:VOCABULARY
Indian-origin English Words
Phrasal Verbs for Day-to-Day Communication
Commonly used Idiomatic Expressions

UNIT-5:PROJECT
Research Writing

Reference Books:

1. English for Colleges by Brendan J. Carroll, Macmillan Publications
2. Effective Technical Communication by M.AshrafRizwi. Tata Mcgraw Hill
3. Enriching Speaking and Writing Skills, Orient Blackswan Publishers
4. The Oxford Guide to Writing and Speaking by John Seely OUP
5. Six Weeks to Words of Power by Wilfred Funk. W.R.Goyal Publishers
6. English for Engineers and Scientists by Sangeeta Sharma and Binod Mishra, PHI Learning

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi) Accredited by NAAC with 'A' GradeRecognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

Estd. 1980

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R17)

III/IV B.TECH

(With effect from 2017-2018 Admitted Batch onwards)

Under Choice Based Credit System

ELECTRONICS AND COMMUNICATION ENGINEERING

I-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 EC 3101	Pulse and Digital Circuits	3	3	1	--	4	30	70	100
B17 EC 3102	Linear ICs and Applications	3	3	1	--	4	30	70	100
B17 EC 3103	Electronic Measurements And Instrumentation	3	3	1	--	4	30	70	100
B17 EC 3104	Digital Communication	3	3	1	--	4	30	70	100
B17 EC 3105	Antennas and Propagation	3	3	1	--	4	30	70	100
B17 EC 3106	Computer Network Engineering	3	3	1	--	4	30	70	100
B17 EC 3107	Linear Integrated Circuits and Pulse Circuits Lab with Simulation	2	--	--	3	3	50	50	100
B17 EC 3108	Digital ICs Laboratory with simulation	2	--	--	3	3	50	50	100
B17 BS 3101	Problem Solving & Linguistic Competence	1	--	3	--	3	30	70	100
B17 BS 3102	Basic Coding	1	--	--	3	3	50	50	100
Total		**24**	**18**	**9**	**9**	**36**	**360**	**640**	**1000**

Code: B17 EC 3101

PULSE AND DIGITAL CIRCUITS

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives: The main objectives of this course are:

1. To provide insight of the applications of Integrator, differentiator circuits.
2. To introduce the design of various clippers circuits and to provide insight of the applications of clamper circuits.
3. To introduce the analysis of various Bistable, Monostable, Astable Multivibrators and Schmitt trigger for various applications.
4. To introduce various Time Base Generators.
5. To provide insight of the synchronization techniques for sweep circuits and to provide insight of different logic families; realize logic gates using diodes and transistors.

Course Outcomes: By the end of the course the learners (students) will be able to:

1. Understand the applications of Integrator, differentiator circuits.
2. Design of different clipping circuits and understand the applications clamper circuits.
3. Analyze different Bi-stable, Monostable, AstableMultivibrators and Schmitt trigger for various applications.
4. Understand Different Time Base Generators.
5. Analyze synchronization techniques for sweep circuits and to understand different logic families; realize logic gates using diodes and transistors.

SYLLABUS

UNIT-I: Linear Wave Shaping:
High pass, low pass RC circuits-response to sinusoidal, step, pulse, square and ramp inputs, The High pass RC circuit as a differentiator and the Low pass RC circuit as an integrator, Attenuators.

UNIT-II: Non-linear wave shaping:
Diode clippers, Clippers at two independent levels, Transfer characteristics of clippers, Transistor clipper, Emitter coupled clipper, Clamping operation, diode clamping circuits with source resistance and diode resistance -transient and steady state response for a square wave input, clamping circuit theorem.

UNIT-III: Bi-stable multi vibrators:
Transistor as a Switch, Transistor switching timings, A basic binary circuit-explanation.
Fixed-bias transistor binary, self-biased transistor binary, binary with commutating capacitors-analysis, Non-saturated binary–symmetrical triggering, and Schmit ttrigger circuit-emitter coupled binary circuit.

Mono-stable multi vibrator: Basic circuit-collector coupled monostable multivibrator-explanation.
Astable multi vibrator: The collector coupled Astable multivibrator-explanation.

UNIT-IV: Time –Base Generators:

Voltage sweep -- Simple Exponential sweep Generator. Errors that define Deviation from linearity, UJT Relaxation Oscillator – Methods of linearising a Voltage Sweep – Bootstrap and Miller Circuits – Current Sweep – Linearising a current Sweep by adjusting the driving Waveform.

UNIT-V: Synchronization and frequency division:

Pulse synchronization of relaxation devices, frequency division in the sweep circuit, Synchronization of Astable multivibrator, Monostable multivibrator, synchronization frequency division with a sweep circuit.

Digital logic Families: Introduction, RTL,DTL, TTL, ECL, NMOS logic, PMOS logic, CMOS logic-analysis

Text Books:

1. Pulse, Digital and switching wave forms by Milliman and Taub, McGraw Hill.
2. Pulse and Digital Circuits by A. Anand Kumar, PHI.

Reference Books:

1. Pulse and Digital Circuits by MS PrakashRao, Tata McGraw Hill.
2. Pulse and Digital Circuits by Venkatrao K., Ramasudha K., Manmadharao. G, Pearson Education, 2010.

Code: B17 EC 3102

LINEAR ICS AND APPLICATIONS

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. To understand the internal diagram and characteristics of operational amplifier.
2. To learn about the linear and non-linear applications of operational amplifier.
3. To know the concepts of Active filters and waveform generator
4. To understand the industrial applications using 555 timer, PLL.
5. To understand the concepts of Analog to Digital Converters and Digital to Analog Converters

Course Outcomes: Upon completion of the course, students will be able to

1. Understand the external behavior and characteristics of operational amplifier.
2. Design and analyze linear and non-linear circuits using operational amplifier.
3. Design and analyze oscillators and active filters using operational amplifier.
4. Design and analyze various applications using IC 565 and IC 555.
5. Understand the operation of Analog to Digital and Digital to Analog Converters.

SYLLABUS

UNIT-I: Applications of Operational Amplifiers:
Basics of Op-Amp, Block Diagram, open loop and closed loop op-amp configurations,Frequency compensation Techniques, Logarithmic Amplifier, Instrumentation Amplifiers, Voltage to Current and Current to Voltage Converters. Op-amp As a Comparators, Schmitt trigger, Wave form Generators, Sample and Hold Circuits, Rectifiers, Peak Detection

UNIT-II: Active Filters:
Butterworth type LPF, HPF, BPF, BEF, All-pass Filters, Higher Order Filters and their Comparison, Switched Capacitance Filters.

UNIT-III: Oscillators:
Op-Amp Phase Shift, Wien-bridge and Quadrature Oscillator, VoltageControlled Oscillators, Analog Multiplexers.

UNIT-IV: Special ICs:
555 Timers, 556 Function Generator ICs and their Applications, Three Terminal IC Regulators, IC 565 PLL and its Applications, Voltage to Frequency and Frequency to Voltage Converters.

UNIT-V: Digital to Analog and Analog to Digital Converters:
DAC techniques, Weighted resistor DAC, R-2R ladder DAC, inverted R-2R DAC, Different types of ADCs-parallel Comparator type ADC, Counter type ADC, Successive approximation ADC and ADC specifications.

Text Books:

1. Microelectronics- Jacob Millman.
2. Op-Amps and Linear ICs- RamakanthGayakwad, PHI, 1987.
3. Linear Integrated Circuits- D.RoyChowdhury, New Age International(p) Ltd, 2nd Edition,2003.

Reference Books:

1. Integrated Circuits- Botkar, Khanna Publications.
2. Applications of Linear ICs- Clayton.

Code: B17 EC 3103

ELECTRONIC MEASUREMENTS AND INSTRUMENTATION

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives: The student will

1. Select the instrument to be used based on the requirements.
2. Understand and analyze different signal generators and analyzers
3. Understand the design of oscilloscopes for different applications.
4. Understand the principle of operation and working of various types of bridges for
5. measurement of parameters

Course Outcomes: The student will be able to

1. Evaluate basics of measurement systems, principle of basic meter
2. Evaluate how a signal can be generated using different types of meters.
3. Investigate a signal / waveform with different oscillators.
4. Use bridges of many types and measure appropriate parameters.
5. Design different transducers for measurement of different parameters.

SYLLABUS

UNIT-I:

Performance characteristics of instruments, Static characteristics, Accuracy, Resolution, Precision, Expected value, Error, Sensitivity.Errors in Measurement, Dynamic Characteristics-speed of response, Fidelity, Lag and Dynamic error. DC Voltmeters-Multirange, Range extension voltmeters, AC voltmeters, True RMS responding voltmeter, Electronic Multimeter.

UNIT-II:

Transducers- active & passive transducers : Resistance, Capacitance, inductance; Strain gauges, LVDT, Piezo Electric transducers, Resistance Thermometers, Thermocouples, Thermistors, Sensistors. Introduction to smart sensors.

UNIT-III:

Oscilloscopes CRT features, vertical amplifiers, horizontal deflection system, sweep, trigger pulse, delay line. Dual beam CRO, .Dual trace oscilloscope, sampling oscilloscope, digital storage oscilloscope, Lissajous method of frequency measurement, standard specifications of CRO.

UNIT – IV:

AC Bridges Measurement of inductance- Maxwell's bridge, Anderson bridge. Measurement of capacitance –Schearing Bridge. Wheatstone bridge.Wien Bridge, Errors and precautions in using bridges.

UNIT – V:

Signal Generator- fixed and variable, AF oscillators, Standard and AF sine and square wave signal generators, Function Generators, Square pulse, Random noise, sweep, Arbitrary waveform. Wave Analyzers, Harmonic Distortion Analyzers, Spectrum Analyzers.

Text Books:

1. Electronic instrumentation, second edition - H.S.Kalsi, Tata McGraw Hill, 2004.
2. Modern Electronic Instrumentation and Measurement Techniques – A.D. HelfrickandD.W. Cooper, PHI, 5th Edition, 2002.

Reference Books:

1. Electronic Instrumentation & Measurements - David A. Bell, PHI, 2nd Edition, 2003.
2. Electronic Test Instruments, Analog and Digital Measurements - Robert A.Witte, PearsonEducation, 2nd Ed., 2004.

Code: B17 EC 3104

DIGITAL COMMUNICATION

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. To introduce the elementary concepts of digital communication systems.
2. To get introduced with emphasis on different modulation techniques.
3. Understand the effect of noise on signal transmission.
4. To learn about optimum detection and probability of error.
5. To compare the performance of two digital modulation techniques and introduce the elementary concept of spread spectrum modulation system.

Course Outcomes: By the end of the course the learners (students) will be able to

1. Understand the basic concepts of sampling and digital communication systems.
2. Understand the concept of binary and M-ary modulation techniques.
3. Understand the problems of noise and can design any digital communication system for the real time environment.
4. Designing of optimal receiver and understanding the concept of probability of error.
5. Analyze the error performance of two digital modulation techniques and understand the concept of spread spectrum communication system

SYLLABUS

UNIT-I: Pulse Modulation and Digital Representation of Analog Signal:
Sampling, Pulse Amplitude Modulation and Concept of Time Division Multiplexing, Pulse Width Modulation, Pulse Position Modulation, Digital representation of analog signal: Quantization of signals, Quantization error, Pulse Code Modulation, Companding, T1 Digital system, Differential Pulse Code Modulation, Delta Modulation, Adaptive Delta Modulation, Continuously Variable Slope Delta Modulation.

UNIT-II: Digital Modulation and Transmission:
Binary Phase-Shift Keying, Differential Phase-Shift Keying, Differentially-Encoded PSK (DEPSK), Quadrature Phase-Shift Keying (QPSK), M-ary PSK, Binary Frequency Shift-Keying, Comparison of BFSK and BPSK, M-ary FSK, Minimum Shift Keying (MSK), Duo-binary Encoding.

UNIT-III: Mathematical Representation of Noise:
Some Sources of Noise, Frequency-domain representation of Noise, Spectral Components of Noise, Response of a Narrowband Filter to Noise, Effect of a Filter on the Power Spectral Density of Noise, Superposition of Noises, Linear Filtering, Noise Bandwidth, Quadrature Components of Noise, Power Spectral Density of Quadrature Components of Noise.

UNIT-IV: Optimal Reception of Digital Signal:
A Base-band Signal Receiver, Probability of Error, Optimum Receiver for both Baseband and Passband - Calculation of optimum filter Transfer function, Optimum filter realization using Matched filter, Probability of Error of the Matched Filter, Optimum filter realization using Correlator, Optimal of Coherent Reception: PSK, FSK, QPSK, Comparison of Modulation Systems.

UNIT-V: Noise in Pulse Code Modulation and Delta Modulation Systems:
PCM Transmission, Calculation of Signal-to-Noise Ratio in PCM, Delta Modulation(DM) Transmission, Calculation of Signal-to-Noise Ratio in DM, Comparison of PCM and DM.
Introduction to Spread Spectrum Modulation: Direct Sequence (DS) Spread Spectrum, Use of Spread Spectrum with Code Division Multiple Access (CDMA), Ranging using DS Spread Spectrum, Frequency Hopping (FH) Spread Spectrum, Generation and Characteristics of PN Sequences.

Text Books:

1. Principles of Communication Systems by Herbert Taub, Donald L Schilling and GoutamSaha,3rd edition, Tata McGraw-Hill Publications, 2008 New Delhi.
2. Digital Communications by Simon Haykins John Wiley, 2005

Reference Books:

1. Principles of Digital Communications- J.Das, SK.Mullick, P.K.Chatterjee.
2. Modern Analog and Digital Communications by B.P.Lathi, Oxford reprint, 3rd Edition, 2004.

Code: B17 EC 3105

ANTENNAS & PROPAGATION

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. Understand the radiation mechanism of antennas and to learn about basic parameters like impedance, gain, directivity, bandwidth, effective length, beam width and radiation pattern etc.
2. Derive fields and power radiated by elemental antenna, half wave dipole, quarter wave monopole and values of their radiation resistance.
3. Understand the necessity of antenna arrays and to learn about theory of uniform linear arrays, broad side and end fire arrays, non-uniform linear arrays like binomial arrays and pattern multiplication.
4. Have knowledge about practical LF, HF, VHF, UHF and Microwave antennas and be able to design practical antennas.
5. Have knowledge about various antenna measurements and be able to conduct different types of antenna measurements.
6. Have knowledge about various types of radio wave propagation like Ground wave, Ionospheric, space wave and Duct propagation and be able to design different types of communication links.

Course Outcomes: After completion of the course the student will be able to

1. Understand Radiation mechanism and functions of antennas, identify antenna parameters derive expressions for antenna parameters .
2. Analyze and design wire and aperture antennas for different applications.
3. Analyze and design Antenna arrays.
4. Capable of performing various antenna measurements and come up with conclusions about antenna parameters and performance.
5. Identify characteristics of radio wave propagation and be able to design different types of communication links for different frequency bands

SYLLABUS

UNIT-I: Fundamentals of Antennas & Radiation from Antennas:
Definition of antennas, functions of Antennas, properties of antennas, antenna parameters, polarization, basic antenna elements, radiation mechanism, radiating fields of alternating current element, radiated power and radiation resistance of current element, different types of current distribution on linear antennas, radiated fields, radiated power and radiation resistance of half-wave dipole and quarter – wave monopole, directional characteristics of dipole antennas.

UNIT-II: Linear Arrays:
Uniform linear arrays, field strength of a uniform linear arrays, locations of principal maximum, null and secondary maxima, first side lobe level, analysis of broad side and end fire , Pattern multiplication, binomial arrays, effect of earth on vertical patterns, methods of excitation of antennas, impedance matching techniques, transmission loss between transmitting and receiving antennas – Friis formula, antenna noise temperature and signal-to-noise ratio, Introduction to array synthesis Methods.

UNIT-III:Practical Antennas – LF,MF,HF,VHF& UHF antennas

Classification of antennas according to type of radiation and type of current distribution of antennas – Isotropic, Omni directional & directional antennas, standing wave and travelling wave antennas, Classification according to frequency of operation – LF, MF, HF, VHF & UHF, brief introduction to LF & MF antennas, earth mat, counterpoise earth, top capacitance hat.

HF, VHF & UHF Antennas - V Antennas, Inverted V Antennas, Rhombic antennas, folded dipole, Yagi-Uda antenna, Log periodic antenna, Loop and Helical Antennas.

UNIT – IV: Microwave antennas:

Introduction, types of reflector antennas, corner reflector, parabolic reflector, feed systems for parabolic reflector, horn antennas, slot antennas and impedance of slot antennas, Babinet's principle and microstrip antennas.

Antenna measurements**:** Introduction, measurement ranges, antenna impedance measurements, antenna gain and directivity measurement, measurement of radiation pattern, beamwidth and SLL.

UNIT-V: Wave Propagation

Types of radio wave propagation, ground wave propagation and Somerfield's analysis of ground wave propagation, wave tilt of ground wave, structure of ionosphere, refractive index of ionosphere, mechanism of wave bending by ionosphere, critical frequency, MUF, Skip distance, fading and remedial measures, effect of earth's magnetic field on ionosphere propagation, faraday rotation, tropospheric (space wave) propagation, range of space wave propagation, effective earth radius, field strength of space wave, atmospheric effects on space wave propagation, duct propagation and scatter propagation.

Textbooks:

1. EM waves and Radiating systems – by E. C. JORDAN and K. G. Balmain – PHI , New Delhi.
2. Antenna theory- by C. A. Balanis, john wiley.

Reference Books:

1. Antennas – By J.D. Kraus, McGrawhill.
2. Antenna and wave propagation – by G.S.N Raju, Pearson eduction.

Code: B17 EC 3106

COMPUTER NETWORK ENGINEERING

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. To familiarize with the fundamental concepts of computer networking and network engineering reference models.
2. To introduce basic concepts of analog and digital transmission techniques, switching techniques.
3. To understand error control and flow control mechanisms.
4. To familiarize with different multiple access protocols such as ALOHA, CSMA.
5. To familiarize with different networking devices and congestion control algorithms.
6. To familiarize with TCP and UDP header formats.

Course Outcomes:

Upon completion of the course, students will be able to

1. Explain basic computer network principles and layers of the OSI model and TCP/IP.
2. Explain the concepts of transmission media, switching and multiplexing techniques.
3. Explain and analyze the error control and flow control methods.
4. Explain different multiple access control protocols and IEEE standards for LANs and MANs.
5. Identify the different types of connecting devices andexplain the basic concepts of congestion control algorithms and internetworking.
6. Explain TCP and UDP header formats.

SYLLABUS

UNIT-I

Uses of Computer Networks, Line Configuration, Topology, Transmission mode, Categories of Networks-LAN, MAN, WAN; Network Software- Protocol Hierarchies, Design issues of layers, Connection Oriented and Connectionless services; Reference Models- The OSI Reference Model, The TCP/IP Reference Model, The B-ISDN ATM Reference Model.

UNIT-II

Theoretical basis for Data communication, Transmission media- Guided and Unguided Transmission media; The Telephone System-Structure of Telephone system, Trunks and Multiplexing, Frequency Division Multiplexing, Time Division Multiplexing, Switching-Circuit Switching, The Switch Hierarchy, Crossbar switches, Space Division Switches, Time Division Switches; Narrow band ISDN, Broadband ISDN and ATM- Virtual Circuits versus Circuit Switching.

UNIT-III

DATA LINK LAYER

Design issues, Error Detection and Correction, Elementary Data link protocols, Sliding window protocols, HDLC, **Medium access sub layer**-The Channel allocation problem, Multiple Access Protocols-ALOHA, Carrier Sense Multiple Access protocols; IEEE standard for 802 LANs, Satellite Networks

UNIT-IV

NETWORK LAYER

Design considerations, Difference between Gateways, Ethernet switch, Router, Hub, Repeater, Congestion Control algorithms- General principles of Congestion Control, Congestion prevention policies. The Leaky bucket algorithm and Token bucket algorithm, The Network Layer in the Internet- The IP Protocol, IP Addresses.

UNIT-V

TRANSPORT LAYER

The Transport layer Service, Elements of Transport protocols, The Internet Transport Protocols- UDP, TCP.

APPLICATION LAYER

The Domain Name System, Electronic mail, The World Wide Web.

Text Books:

1. Data Communications and Networking by BehrouzA.Forouzan, 2nd edition, Tata McGraw Hill.
2. Computer Networks — Andrew S Tanenbaum, 3rd Edition, Pearson Education/PHI.

Reference Books:

1. An Engineering Approach to Computer Networks-S.Keshav, 2nd Edition,Pearson Education
2. Understanding communications and Networks, 3rd Edition, W.A. Shay, Thomson

Code: B17 EC 3107

LINEAR INTEGRATED CIRCUITS & PULSE CIRCUTS LAB WITH SIMULATION

Lab	**: 3 Periods**	**Int.Marks**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

This laboratory course enables students to get practical experience in design, assembly and evaluation of Linear integrated circuits & Pulse Circuits. They will use Multisim to test their electronic designs.

Course Outcomes: Students will be able to:

1. Design and conduct experiments on RC low pass and high pass circuits.
2. Observe operation of UJT Sweep Generator.
3. Design and test different types of Multi vibrators
4. Acquire a basic knowledge on simple applications of operational amplifier.
5. Design, construct Schmitt trigger using operational amplifier.
6. Use Multisim to test their electronic designs.

LIST OF EXPERIMENTS

1. Linear Wave Shaping
   a) Passive RC Differentiator
   b) Passive RC Integrator
2. Non Linear Wave shaping
   a) Clipping Circuits
   b) Clamping Circuits
3. Self biasbistableMultivibrator
4. Schmitt Trigger Using µA 741
5. UJT Sweep Generator
6. AstableMultivibrator using 555 timer
7. Multiplexer
8. Shift Registers

LIST OF EXPERIMENTS
(Simulation)

1. Linear Wave Shaping
   a) Passive RC Differentiator
   b) Passive RC Integrator
2. Non Linear Wave shaping
   a) Clipping Circuits
   b) Clamping Circuits

3. Self biasbistableMultivibrator
4. Schmitt Trigger Using μA 741
5. UJT Sweep Generator
6. AstableMultivibrator using 555 timer.
7. Multiplexer
8. Shift Registers

Reference: Lab Manuals

Code: B17 EC 3108

DIGITAL IC'S LABORATORY WITH SIMULATION

Lab	**: 3 Periods**	**Int.Marks**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

1. Learn and understand the basics of digital electronics, Boolean algebra, and able to design the simple logic circuits and test/verify the functionality of the logic circuits.
2. Design combinational and sequential logic circuits using digital ICs.
3. This laboratory course enables student to get practical experience in design, assembly and evaluation of digital integrated circuits and HDL lab. Students use digital trainer kit and Xilinx ISE simulator to test their electronic designs.

Course Outcomes: Upon completion of the course, students will be able to

1. Synthesize, simulate and implement a digital design in a configurable digital circuit with computer supported aid tools and digital trainer kit.
2. Acquire Knowledge of analysis and synthesis of combinational and sequential circuits with simulators and digital trainer kits.
3. Build high level programming (HDL programming) skills for digital circuits.
4. Adapt digital circuits to electronics and telecommunication field.

LIST OF EXPERIMENTS

A. **HARDWARE**

1. Verify the operation of following digital components using Digital Trainer Kit
   a. Full adder using gates
   b. Full subtract or using gates
2. Design and verify the logic functions of multiplexer and de-multiplexers using digital trainer kit
3. Design code convertors using digital trainer kit
   a. BCD TO SEVEN segment display
   b. Priority encoder
4. Verify the operation of following flip-flops using Digital Trainer Kit
   a. JK flip flop
   b. D flip flop
   c. T flip flop
5. Design a following synchronous counters using Digital Trainer Kit
   a. Mod 16 counter
   b. Mod 8 counter
   c. Decade counter
6. Verify the functioning of shift register using Digital Trainer Kit

B. SOFTWARE

7. Verify the operation of following digital components using ISE Simulator
   a. Full adder
   b. Full subtractor
8. Verify the operation of multiplexer and priority encoder using ISE Simulator
9. Design ALU and verify the operation using ISE Simulator
10. Design RAM for read/write operations using ISE Simulator

Equipment Required:

1. Personal Computer with necessary peripherals and Xilinx Vivado ISE software
2. Digital trainer kits.

Reference: Lab Manuals

Code: B17BS3101

PROBLEM SOLVING & LINGUISTIC COMPETENCE
(Common to all Branches)

Tutorial	**: 3 Periods (VA-2+QA-1)**	**Int.Marks**	**: 30**
Exam	**: 3 Hrs.**	**Ext.Marks**	**: 70**
		Credits	**: 1**

Part-A: Verbal and Soft Skills-I

Course Objectives:

1. To introduce concepts required in framing grammatically correct sentences and identifying errors while using Standard English.
2. To familiarize the learner with high frequency words as they would be used in their professional career.
3. To inculcate logical thinking in order to frame and use data as per the requirement.
4. To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.
5. To familiarize students with soft skills and how it influences their professional growth.

Course Outcomes:

The student will be able to
1. Detect grammatical errors in the text/sentences and rectify them while answering their competitive/ company specific tests and frame grammatically correct sentences while writing.
2. Answer questions on synonyms, antonyms and other vocabulary based exercises while attempting CAT, GRE, GATE and other related tests.
3. Use their logical thinking ability and solve questions related to analogy, syllogisms and other reasoning based exercises.
4. Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.
5. Apply soft skills in the work place and build better personal and professional relationships making informed decisions.

SYLLABUS

Grammar: (VA**)**
Parts of speech(with emphasis on appropriate prepositions, co-relative conjunctions, pronouns-number and person, relative pronouns), articles(nuances while using definite and indefinite articles), tenses(with emphasis on appropriate usage according to the situation), subject – verb agreement (to differentiate between number and person) , clauses(use of the appropriate clause , conditional and relative clauses), phrases(use of the phrases, phrasal verbs) to-infinitives, gerunds, question tags, voice, direct & indirect speech, degrees of comparison, modifiers, determiners, identifying errors in a given sentence, correcting errors in sentences.

Vocabulary: (VA)
Synonyms and synonym variants(with emphasis on high frequency words), antonyms and antonym variants(with emphasis on high frequency words), contextual meanings with regard to inflections of a word, frequently confused words, words often mis-used, multiple meanings of the same word (differentiating between meanings with the help of the given context), foreign phrases, homonyms, idioms, pictorial representation of words, word roots, collocations.

Reasoning: (VA)
Critical reasoning (understanding the terminology used in CR- premise, assumption, inference, conclusion), Analogies (building relationships between a pair of words and then identifying similar relationships), Sequencing of sentences (to form a coherent paragraph, to construct a meaningful and grammatically correct sentence using the jumbled text), odd man (to use logical reasoning and eliminate the unrelated word from a group), YES-NO statements (sticking to a particular line of reasoning Syllogisms.

Usage: (VA)
Sentence completion (with emphasis on signpost words and structure of a sentence), supplying a suitable beginning/ending/middle sentence to make the paragraph coherent, idiomatic language (with emphasis on business communication), punctuation depending on the meaning of the sentence.

Soft Skills:
Introduction to Soft Skills – Significance of Inter & Intra-Personal Communication – SWOT Analysis –Creativity & Problem Solving – Leadership & Team Work - Presentation Skills Attitude – Significance – Building a positive attitude – Goal Setting – Guidelines for Goal Setting – Social Consciousness and Social Entrepreneurship – Emotional Intelligence - Stress Management, CV Making and CV Review.

Text Books:

1. Oxford Learners's Grammar – Finder by John Eastwood, Oxford Publication.
2. R S Agarwal's books on objective English and verbal reasoning
3. English Vocabulary in Use- Advanced , Cambridge University Press.
4. Collocations In Use, Cmbridge University Press.
5. Soft Skills & Employability Skills by SaminaPillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. English Grammar in Use by Raymond Murphy, CUP
2. Websites: Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from ‗IMS, Career Launcher and Time' institutes for competitive exams.
4. The Art of Public Speaking by Dale Carnegie
5. The Leader in You by Dale Carnegie
6. Emotional Intelligence by Daniel Golman
7. Stay Hungry Stay Foolish by RashmiBansal
8. I have a Dream by RashmiBansal

Part-B: Quantitative Aptitude -I

Course objectives:

The objective of introducing quantitative aptitude-1 is:
1. To familiarize students with basic problems on numbers and ratio‘s problems.
2. To enrich the skills of solving problems on time, work, speed, distance and also measurement of units.
3. To enable the students to work efficiently on percentage values related to shares, profit and loss problems.
4. To inculcate logical thinking by exposing the students to reasoning related questions.
5. To expose them to the practice of syllogisms and help them make right conclusions.

Course Outcomes:

1. The students will be able to perform well in calculating on number problems and various units of ratio concepts.
2. Accurate solving problems on time and distance and units related solutions.
3. The students will become adept in solving problems related to profit and loss, in specific, quantitative ability.
4. The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry.
5. The students will learn to apply Logical thinking to the problems of syllogisms and be able to effectively attempt competitive examinations like CAT, GRE, GATE for further studies.

SYLLABUS

Numbers, LCM and HCF, Chain Rule, Ratio and Proportion Importance of different types of numbers and uses of them: Divisibility tests, Finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three different methods, Problems related to Ratio and Proportion.

Time and work, Time and Distance Problems on man power and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.

Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends
Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, marked price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership, Sleeping partner concept and problems, Problems on shares and dividends, and stocks.
Introduction, number series, number analogy, classification, Letter series, ranking, directions Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series

and letter series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.

Data sufficiency, Syllogisms Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency, Syllogisms Problems.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi

References:

1. Barron‘s by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt.Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from ‗IMS, Career Launcher and Time‘ institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

Websites:

1.www.m4maths.com
2. www.Indiabix.com
3. www.800score.com
4. Official GRE site
5. Official GMAT site

Code: B17 BS 3102

BASIC CODING
(Common to ECE & EEE)

Lab	**: 3 Periods**	**Int.Marks**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 1**

Course Objectives:

1. To develop programming skills among the students.
2. To familiarize the student with Control Structures, Loop Structures.
3. To familiarize the student with Basic searching and sorting Methods.
4. To familiarize the student with Functions, Recursions and Storage Classes.
5. To familiarize the student with Structures and Unions.
6. To familiarize the student with Operating System concepts.
7. To familiarize the student with Networking concepts.

Course Outcomes:

At the end of the course students will be able to

1. Know about Control Structures, Loop Structures and branching in programming.
2. Know about various searching and sorting methods.
3. Know about Functions, Recursions and Storage Classes.
4. Know about Structures and Unions.
5. Know different Operating System concepts.
6. Differentiate OSI Model Vs. TCP/IP suite.

SYLLABUS

UNIT-I Review of Programming constructs
Programming Environment, Expressions formation, Expression evaluation, Input and Output patterns, Control Structures, Sequential branching, Unconditional branching, Loop Structures, Coding for Pattern Display.

UNIT-II Introduction to Linear Data, strings and pointers
Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding, Working on character data, Compiler defined methods, Substitution coding for defined methods, Row Major representation, Column Major representation, Basic searching and sorting Methods.

UNIT-III Functions, Recursions and Storage Classes
Functions – Introduction to modular programming – Function Communication - Pass by value, Pass by reference – Function pointers – Recursions – Type casting – Storage classes

Practice: programs on passing an array and catching by a pointer, function returning data, comparison between recursive and Iterative solutions.
Data referencing mechanisms: Pointing to diff. data types, Referencing to Linear data, Runtime-memory allocation, Named locations vs pointed locations, Referencing a 2D-Matrix

UNIT-IV User-defined datatypes, Pre-processor Directives and standard storage
Need for user-defined data type – structure definition – Structure declaration – Array within a Structure – Array of Structures – Nested Structures - Unions – Declaration of Union data type, StructVs Union - Enum – Pre-processor directives , Standard storage methods, Operations on file, File handling methods, Orientation to Object oriented programming

Practice: Structure padding, user-defined data storage and retrieval programs

UNIT-V Operating system principles and Database concepts
Introduction to Operating system principles, Process scheduling algorithms, Deadlock detection and avoidance, Memory management, Networking: Introduction to Networking, OSI Model Vs. TCP/IP suite, Datalink layer, Internet layer, DVR Vs. LSR, Transport Layer, Application Layer

References:

1. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
3. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), EnzoMarinari (Author), Giovanni Organtini, World Scientific .
4. ObjectOrientedProgrammingin C++: N. Barkakati, PHI.
5. ObjectOrientedProgrammingthrough C++ byRobatLaphore.
6. https://www.geeksforgeeks.org/.
7. https://www.tutorialspoint.com/

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R17)

III/IV B.TECH

(With effect from 2017-2018 Admitted Batch onwards)

Under Choice Based Credit System

ELECTRONICS AND COMMUNICATION ENGINEERING

II-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 EC 3201	Microprocessors and its Applications	3	3	1	--	4	30	70	100
B17 EC 3202	Microwave Engineering	3	3	1	--	4	30	70	100
B17 EC 3203	VLSI Design	3	3	1	--	4	30	70	100
B17 EC 3204	Digital Signal Processing	3	3	1	--	4	30	70	100
B17 EC 3205	Radar Engineering	3	3	1	--	4	30	70	100
#OE	OPEN ELECTIVE	3	3	1	--	4	30	70	100
B17 EC 3208	Microprocessors and Microcontrollers Lab	2	--	--	3	3	50	50	100
B17 EC 3209	VLSI Lab	2	--	--	3	3	50	50	100
B17 BS 3201	Employability Skills	1	--	3	--	3	30	70	100
B17 BS 3203	Advanced Coding	1	--	--	3	3	50	50	100
B17 BS 3206	IPR & PATENTS	--	--	2	--	2	--	--	--
Total		**24**	**18**	**11**	**9**	**38**	**360**	**640**	**1000**

OPEN ELECTIVE	B17EC3206	Microcontrollers
	B17CS3214	Oops through Java
	B17CS3215	Data Mining
	B17ME3210	Industrial Robotics
	B17EE3209	Power Electronics
	B17EC3207	Bio Medical Engineering
	B17CS3216	Artificial Neural Networks

Code: B17 EC 3201

MICROPROCESSORS AND ITS APPLICATIONS

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. To understand the architecture of 8085 Microprocessor
2. To be familiar with 8085 assemble language programming
3. To understand the concept of interfacing peripheral devices and memory to 8085 Microprocessor
4. To understand the architecture of 8086/8088 Microprocessor
5. To be familiar with 8086 assemble language programming

Course Outcomes: By the end of the course the learners (students) will

1. Understand and analyze architecture of the 8085 microprocessor
2. Be familiar with the 8085 Assembly Language Programming
3. Be familiar with Hardware and software requirements in interfacing and designing 8085 microprocessor based products for practical applications
4. Understand and analyze architecture of the 8086 microprocessor
5. Be familiar with the 8086 Assembly Language Programming

SYLLABUS

UNIT-I: 8085 Architecture:

Bus structure of 8085,internal architecture and functional description of INTEL 8085 Microprocessor pin out & signals, flag register, Fetch cycle ,memory Read /Write and I/O Read /Write Cycles with Timing Diagrams , Stack memory organization, Interrupt structure of 8085,Vectored, non-vectored, maskable and non maskable interrupts, pending interrupts, execution of SIM and RIM instructions.

UNIT-II: 8085 Programming:

Introduction to 8085 Assembly Language Programming, Programming model of 8085 and function of each register, Addressing modes of 8085 with examples, I/O addressing, Stack memory operation using PUSH and POP instructions, Classification of 8085 instructions with examples, Instruction set, Sample Programs, Subroutines, CALL and RET instructions, and Interrupt Service Routines.

UNIT-III: 8085 Interfacing:

Interfacing of semiconductor Memory and I/O devices to 8085, Classification of Read /Write and Read only memories, Interfacing of SRAMs, DRAMs and EPROMs using 74LS138.Functional description of PPI(8255),PIT(8253/8254) and USART(8251A).Interfacing of parallel I/O (8255), Timer/Counter (8253/8254), Serial I/O (8251A) with 8085 Microprocessor.

UNIT-IV: 8086/8088 Architecture:
Internal Architecture and Functional description of INTEL 8086/8088 microprocessor, and their comparisons. Memory segmentation and physical memory address generation, pipeline architecture and instruction queue .Register organisation, Status flags and machine control flags of 8086, pin out and signals in detail, Memory read /write and I/O read/Write Bus cycles with timing diagrams, 8086 memory Banks,8086 minimum and maximum modes of operation.

UNIT-V: 8086 Programming:
Introduction to 8086 Assembly language programming, programmable register array of 8086 and function of each register, Data addressing modes of 8086 with examples, fixed and variable I/O addressing. Stack memory operation, classification of 8086 instructions, sample 8086 assembly language programs using data transfer, Arithmetic and logic instructions, Introduction to ARM.

Text Books:

1. Architecture Programming and Applications. Ramesh S.Goankar.New Age International Pvt.Ltd.,(3rd Edition)'
2. Microprocessors and interfacing ,DouglasV.Hall, Tata McGraw-Hill Revised 2nd Edition.

Reference Book:

1. Microprocessors: The 8086/8088, 80186/80286, 80386/80486 and the Pentium Family.NileshB.Bahadure, Phi Learning Pvt.Ltd.,2010.

Code: B17 EC 3202

MICROWAVE ENGINEERING

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. The purpose of this course is to provide the operational characteristics and conceptual understanding of active and passive components at microwave frequencies.
2. This course also emphasizes formulation and application of scattering matrix for the analysis of different microwave passive components.
3. Further, this course also provides the understanding of measurement techniques of different parameters.

Course Outcomes: By the end of the course the learners (students) will be able to

1. Explain the working principle of different passive waveguide components used atmicrowave frequencies.
2. Apply the properties of scattering matrix for solving the scattering matrix of different passive microwave components for both ideal and practical considerations and analyse their operation.
3. Understand the conceptual and operational characteristics of different microwave Tube circuits.
4. Explain the operational characteristics of different microwave solid state devices.
5. Understand and implement different experimental procedures involving measurement of microwave parameters

SYLLABUS

UNIT-I: Microwave Components and its applications:
Introduction, Microwave Spectrum and Bands, Applications of Microwaves, Coupling Mechanisms – Probe, Loop, Aperture types. Waveguide Discontinuities – Waveguide irises, Tuning Screws and Posts, Matched Loads. Waveguide Attenuators – Resistive Card, Rotary Vane types; Waveguide Phase Shifters – Dielectric, Rotary Vane types, E-plane and H-plane Tees, Magic Tee, Hybrid Ring; Directional Couplers – 2Hole, Bethe Hole types, Ferrite Components– Faraday Rotation, Gyrator, Isolator, Circulator, Related Problems.

UNIT-II: Scattering Matrix:
Scattering Matrix – Significance, Formulation and Properties, Scattering Matrix of Isolator, circulator, directional coupler, E Plane Tee, H plane Tee and Magic Tee.

UNIT-III:Qualitative treatment on Microwave Tubes:
Limitations and Losses of conventional tubes at microwave frequencies.Re-entrant Cavities,Microwave tubes – O type and M type classifications. O-type tubes :2 Cavity Klystrons – Structure, Velocity Modulation Process and Applegate Diagram, Bunching Process and Small Signal Theory, Applications, Reflex Klystrons – Structure, Applegate Diagram and Principle of working, Electronic Admittance; Electronic and Mechanical Tuning, Applications, Related Problems.

HELIX TWTS: Significance, Types and Characteristics of Slow Wave Structures; Structure of TWT (Qualitative treatment).
M-type Tubes Introduction, Cross-field effects, Magnetrons – Different Types, 8-Cavity Cylindrical Travelling Wave Magnetron – Hull Cut-off Condition, Modes of Resonance and PI-Mode Operation, Separation of PI-Mode, o/p characteristics.

UNIT-IV: Microwave Solid state Devices:
Negative resistance phenomenon, Gunn Diode, domain formation, Tunnel Diode- principle of operation, IMPATT- principle of operation, TRAPATT, PIN Diodes and its applications (Qualitative analysis only).Detector diode or point contact diode and its characteristics.

UNIT-V:Microwave Measurements:
Microwave Test bench, Measurement of Power, VSWR, Frequency, Guide Wavelength, Unknown load impedance, S parameters of reciprocal and non reciprocal devices

Text Books:

1. Foundations for Microwave Engineering, R. R. Collin, McGraw Hill.
2. Microwave Devices and Circuits, Third Edition, Samuel Y. Liao, Pearson Education.

Reference Books:

1. Microwave Engineering, Annapurna Das, Sisir K. Das, Tata McGraw-Hill Education
2. Microwave Engineering, 4th Edition, David M. Pozar, November 2011.
3. Microwave and Radar Engineering, GottapuSasibhushanaRao, Pearson Education, New Delhi, 2014.
4. Microwave and Radar Engineering-M.Kulkarni, Umesh Publications, 3rd Edition.

Code: B17 EC 3203

VLSI DESIGN
(Common to ECE & EEE (Open Elective))

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives: Student will be introduced to

1. Use mathematical methods and circuit analysis models in analysis of CMOS digital electronics circuits, including logic components and their interconnections. Learn the various fabrication steps of NMOS and CMOS.
2. Apply CMOS technology-specific layout rules in the placement and routing of transistors and interconnect and to verify the functionality, timing, power and parasitic effects.
3. Learn some basic electrical properties of MOSFET and scaling models and limitations of scaling of MOS circuits.
4. The concepts and techniques of modern integrated circuit design and testing (CMOS VLSI). Learn basic concepts of FPGA.
5. Introduction to Low power CMOS Logic circuits and also someoptimisation techniques.

Course Outcomes: By the end of the course the learners (students) will be able to

1. Apply the Concept of design rules during the layout of a circuit.
2. Model and simulate digital VLSI systems using hardware design language.
3. Synthesize digital VLSI systems from register-transfer or higher level descriptions
4. Understand current trends in semiconductor technology, and how it impacts scaling and performance.
5. Understand the basic concepts of FPGA and low power VLSI design

SYLLABUS

UNIT-I: Introduction :

Introduction to IC Technology,Fabrication process: NMOS, PMOS and CMOS.I_{ds} versus V_{ds}Relationships,Aspects of MOS transistor Threshold Voltage,MOS transistor Transconductance,OutputConductance and Figure of Merit.NMOSInverter,Pull-up to Pull-down Ratio for NMOS inverter driven by another NMOS Inverter,and through one or more pass transistors,Alternative forms of pull-up,The CMOS Invertor,Latch-up in CMOS circuits,Comparison between CMOS and Bi-CMOS technology.

UNIT-II: MOS and Bi-CMOS Circuit Design Processes:

MOS Layers, Stick Diagrams, Design Rules and Layout, General observations on the Design rules, 2µm Double Metal, Double Poly, CMOS/BiCMOS rules, 1.2µm Double Metal, Double Poly CMOS rules, Layout Diagrams of NAND and NOR gates and CMOS inverter, Symbolic Diagrams-Translation to Mask Form.

UNIT-III: Basic Circuit Concepts:

Sheet Resistance, Sheet Resistance concept applied to MOS transistors and Inverters, Area Capacitance of Layers, Standard unit of capacitance, The Delay Unit, Inverter Delays, Driving large capacitive loads,Propagation Delays, Wiring Capacitances, Choice of layers

Scaling of MOS Circuits: Scaling models, Scaling factors for device parameters, Limits due to sub threshold currents, current density limits on logic levels and supply voltage due to noise and current density.Switchlogic,Gate logic.

UNIT-IV: Test and Testability:
Design for Testability, Practical design for Test (OFT) Guidelines, Scan Design Techniques and Built-In-Self Test.
FPGA Based Systems: Introduction, Basic concepts, FPGA architecture.

UNIT-V: Introduction to Low Power VLSI Design:
Introduction to Deep submicron digital IC design,Low power CMOS Logic circuits:Over view of power consumption,Low –Power design through voltage scaling,Estimation and optimisation of switching activity,Reduction of switching capacitance, interconnectDesign,Power Grid and Clock Design.

Text Books:

1. Essentials of VLSI Circuits and Systems By Kamran Eshraghian, Douglas and A. Pucknell and SholehEshraghian, Prentice-Hall of India Private Limited,2005 Edition.
2. CMOS Digital Integrated Circuits Analysis and Design, Sung-Mo Kang, Yusuf Leblebici, Tata McGraw Hill Education,2003.

Reference Books:

1. “FPGA Based System Design”- Wayne Wolf, Pearson Education, 2004, Technology and Engineering.

Code: B17 EC 3204

DIGITAL SIGNAL PROCESSING

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

This course introduces students to the fundamental principles of Digital Signal Processing and develops essential analysis and design tools required for signal processing systems & implementations. Also this subject is an introduction to the graduate-level courses in a broad range of disciplines spanning communications, speech processing & image processing.

The topics include SS basics, sampling theorem, Z-transform, analysis of Discrete-time Linear Time-Invariant Systems, Realization structures, Frequency domain representation of signals and systems, DTFT, DFS, Discrete Fourier Transform (DFT), linear/circular convolutions, Fast Fourier Transform (FFT) algorithms, FIR & IIR digital filter design, Multi-rate DSP and a few DSP applications.

Course Outcomes: At the end of this course, the students will be able to;

1. Describe the DSP fundamental theory and components, Develop an understanding of DSP advantages, limitations and fundamental tradeoffs. Carry-out LTI system analysis using convolution & Z-transform
2. Carryout data analysis &spectrum analysis using FFT
3. Design of IIR digital filters to meet specifications
4. Design of FIR digital filters to meet specifications
5. Knows multi-rate signal processing aspects & DSP applications

SYLLABUS

UNIT-I: Discrete-Time Signals and Systems: (Oppenheim &Proakis**)**
Introduction to Digital Signal Processing, Basic elements of a DSP system, Advantages of Digital SP over Analogy SP, Discrete-time signals and systems, DT-LTI systems described by Linear constant–coefficient difference equations, Properties & Analysis of DT-LTI systems, Discrete linear convolution, Frequency domain representation of DT Signals and Systems, DTFT, Review of the Z-transform, Properties, Inverse Z-transform, Analysis of DT-LTI systems in Z-Domain, System function, One-sided Z-transform, Solution of difference equations, Structures and Realization of Digital Filters, Direct-I, II, series and parallel forms.

UNIT-II: Discrete Fourier Transform (DFT) and Fast Fourier Transform Algorithms (FFT): (Oppenheim &Proakis)
Frequency analysis of discrete time signals, DFS, Properties of DFS, Sampling of DTFT, DFT, Properties of DFT, Circular and linear convolution of sequences using DFT, Efficient computation of DFT, Radix–2 Decimation–in–Time(DIT) & Decimation–in–Frequency(DIF) FFT Algorithms, Inverse FFT.

UNIT-III: Design of IIR Digital Filters: (Oppenheim &Proakis**)**
General considerations in Filter design, Analog filter approximations– Butterworth and Chebyshev, Frequency response specifications; Design of IIR digital filters from analog filters, Bilinear Transformation Method, Impulse Invariance Technique, and Low-pass filter Design examples.

UNIT-IV:**Design of FIR Digital Filters:** (Oppenheim &Proakis**)**
Characteristics of FIR Digital Filters, Design of Linear Phase FIR digital Filters using Windows, Effect of Window selection & filter length on filter frequency response, Design examples, Comparison of IIR and FIR Filters.

UNIT-V: DSP Applications and Fundamentals of Multirate Digital Signal Processing: (SK Mitra**)**
Overview of DSP applications, Spectral analysis of sinusoidal signals using FFT, Subband coding of speech signals, Signal compression, Finite precision arithmetic effects.

Introduction to Multirate DSP, Basic sampling rate alteration devices: upsampler, downsampler, Time and Frequency domain characterization of up/down samplers, Interpolator and decimator. Interactive programing based examples.

Text Books:

1. Alan V. Oppenheim, Ronald W. Schafer, "Digital Signal Processing" – PHI Ed., 2006
2. John G. Proakis, D.G. Manolakis, "Digital Signal Processing: Principles, Algorithms and Applications", 3rd Ed., PHI, 1996.

Reference Books:

1. Sanjit K. Mitra, "Digital Signal Processing: A Computer Based Approach", Tata McGraw Hill.
2. Lawrence R.Rabiner, Bernard Gold, "Theory and application of digital signal processing", Prentice Hall.

Code: B17 EC 3205

RADAR ENGINEERING

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives

1. To provide insight of basic working principle of Radar
2. To apply different methods to measurement the Range, angle information etc. of the target from the radar,
3. To introduce different types of Radar systems and other types of tracking Radars,
4. To provide insight of advantages, limitations and applications of various Radar.
5. To provide insight of basics of various navigational aids and their working principles, applications, limitations and different methods to overcome their limitations

Course Outcomes:

By the end of the course the learners (students) will be able to

1. Able to understand the basic working principles of various Radars .
2. Apply various mathematical equations to measure the Range and angle information of the targets from the radar.
3. Analyze and design of radar signals, MTI, Pulse Doppler radar and various tracking Radars.
4. Analyze various Radar systems, advantages, limitations and their applications.
5. Analyze various Navigational Aids like LORAN, DECCA and VOR.

SYLLABUS

UNIT-I:AN INTRODUCTION TO RADAR:

Origin of Radar, Basic Principle of Radar, Range to a target, Pulse Repetition Frequency and Range Ambiguities, Radar Block Diagram and Operation,Radar Equation, Integration of Radar Pulses ,Probability of Detection and Probability of False Alarm, CW Radar and applications, Radar Antenna Parameters, System Losses and Propagation Effects, Applications of Radar.

UNIT-II: MTI AND PULSE DOPPLER RADAR:

Pulse Doppler Radar, Butterfly effect, Coherent and Non Coherent Moving Target Indication Radar, Delay line Cancellers, Limitation to MTI performance, Moving target Detector, MTI from moving platform

UNIT-III:TRACKING RADAR:

Types of Tracking Radars, Sequential Lobing, Conical Scan, Monopulse tracking Radar, Low angle tracking, Synthetic Aperture Radar (SAR), Active and Passive Aperture Phased array Radars,. MST Radar, ECM, ECCM

UNIT-IV:RADAR TRANSMITTERS&RECEIVERS:
Noise Figure and Noise Temperature, Types of Duplexers, Types of Mixers, Radar Displays, Receiver Protectors, Match Filter & Antennas

UNIT-V: **FUNDAMENTALS OF NAVIGATIONAL AIDS:**
Principles of Direction Finders, Sense Finders, VOR, Aircraft Homing and ILS, Radio Altimeter, LORAN and NDB.

Text Books:

1. Introduction to Radar Systems – Merrill I. Skolnik, THIRD EDITION, Tata McGraw-Hill, 2001.
2. Radar Systems and Radio Aids to Navigation-Prof A.K.Sen and Dr.A.B.Bhattacharya

Reference Books:

1. Introduction to Radar Systems – Merrill I. Skolnik, SECOND EDITION, McGraw-Hill, 1981.
2. Radar Engineering and Fundamentals of Navigational Aids, G S N Raju, IK International Publishers, 2008.
3. Fundamentals of RADAR, SONAR and Navigation Engineering – K.K.Sharma

Code: B17EC3206

MICROCONTROLLERS
(Open Elective)

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. To understand the basic architectures of various processors.
2. Study the architecture and addressing modes of 8051
3. Impart knowledge about assembly language programs of 8051
4. Analyze the concept of inter facing of peripheral devices and Memory.
5. Introductory programs on embedded C

Course Outcomes: After successfully completing the course students will be able to:

1. Understand instruction execution sequence with clock.
2. Gain comprehensive knowledge about architecture and addressing modes of 8051
3. Learn the art of programming in assembly language for various embedded system applications.
4. Develop independent learning skills to interface memory and PPI with 8051
5. Create the IO interfacing techniques with 8051

SYLLABUS

UNIT-I: Introduction to 8051
Microprocessors and Microcontrollers, RISC & CISC CPU Architectures, Harvard & Von-Neumann CPU architecture. 8051Microcontroller: Introduction, Architecture of 8051, Pin diagram of 8051, Memory organization, External Memory interfacing, stacks.

UNIT-II: Addressing modes and Instruction set:
Introduction, Instruction syntax, Data types, Subroutines, Addressing modes,Assembler directives, Instruction set, Instruction timings, example programs in assembly language.

UNIT-III: 8051 Interrupts and Timers/counters:
Basics of interrupts, 8051 interrupt structure, Timers and Counters, 8051 timers/counters, special function registers, programming 8051 timers in assembly language.

UNIT-IV: 8051 Interfacing and Applications:
Basics of I/O concepts, I/O Port Operation, Interfacing 8051 to LCD, Keyboard, parallel and serial ADC, DAC, Stepper motor interfacing and DC motor interfacing and programming.

UNIT-V: 8051 Serial Communication:
Data communication, Basics of Serial Data Communication, 8051 Serial Communication, connections to RS-232, 8255A Programmable Peripheral Interface: Architecture of 8255A, I/O devices interfacing with 8051 using 8255A, Introduction to embedded C.

Text Books:

1. “The 8051 Microcontroller and Embedded Systems – using assembly and C ”-, Muhammad Ali Mazidi and Janice Gillespie Mazidi and Rollin D. McKinlay; PHI, 2013 / Pearson, 2013
2. “8051 Microcontrollers”-MCS51 Family and its variants, Satish Shah, Oxford university press, 2010

Reference Books:

1. “The 8051 Microcontroller Architecture, Programming & Applications”, 2e Kenneth J. Ayala Penram International, 1996 / Thomson Learning 2005.
2. “The 8051 Microcontroller”, V.Udayashankar and MalikarjunaSwamy, TMH, 2009
3. Microcontrollers: Architecture, Programming, Interfacing and System Design”,Raj Kamal, “Pearson Education, 2005

Code: B17CS3214

OOPS THROUGH JAVA
(Common to ECE & EEE)
(Open Elective)

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. Understanding the OOP's concepts, classes and objects, threads, files, applets, swings and act.
2. This course introduces computer programming using the JAVA programming language with object- oriented programming principles.
3. Emphasis is placed on event-driven programming methods, including creating and manipulating objects, classes, and using Java for network level programming and middleware development

Course Outcomes:

1. Understand Java programming concepts and utilize Java Graphical User Interface in Programwriting.
2. Write, compile, execute and troubleshoot Java programming for networkingconcepts.
3. Build Java Application for distributedenvironment.
4. Design and Develop multi-tier applications.
5. Identify and Analyze Enterpriseapplications

SYLLABUS

UNIT-I:
Introduction to OOP, procedural programming language and object oriented language, principles of OOP, applications of OOP, history of java, java features, JVM, program structure.
Variables, primitive data types, identifiers, literals, operators, expressions, precedence rules and associativity, primitive type conversion and casting, flow of control.

UNIT-II:
Classes and objects, class declaration, creating objects, methods, constructors and constructor overloading, garbage collector, importance of static keyword and examples, this keyword, arrays, command line arguments, nested classes.

UNIT-III:
Inheritance, types of inheritance, super keyword, final keyword, overriding and abstract class. Interfaces, creating the packages, using packages, importance of CLASSPATH and java.lang package. Exception handling, importance of try, catch, throw, throws and finally block, user-defined exceptions, Assertions.

UNIT-IV:
Multithreading: introduction, thread life cycle, creation of threads, thread priorities, thread synchronization, communication between threads. Reading data from files and writing data to files, random access file,

UNIT-V:

Applet class, Applet structure, Applet life cycle, sample Applet programs. Event handling: event delegation model, sources of event, Event Listeners, adapter classes, inner classes.

AWT: introduction, components and containers, Button, Label, Checkbox, Radio Buttons, List Boxes, Choice Boxes, Container class, Layouts, Menu and Scrollbar.

Text Books:

1. The complete Reference Java, 8^{th} edition, Herbert Schildt,TMH.
2. Programming in JAVA, SachinMalhotra, SaurabhChoudary,Oxford.
3. Introduction to java programming, 7^{th} edition by Y Daniel Liang,Pearson.

Reference Books:

1. Swing: Introduction, JFrame, JApplet, JPanel, Componets in Swings, Layout Managersin
2. Swings, JList and JScrollPane, Split Pane, JTabbedPane, JTree, JTable, DialogBox.

Code: B17CS3215

DATA MINING
(Open Elective)

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. Students will be enabled to understand and implement classical models and algorithms in data warehousing and datamining.
2. They will learn how to analyze the data, identify the problems, and choose the relevant models and algorithms toapply.
3. They will further be able to assess the strengths and weaknesses of various methods and algorithms and to analyze theirbehavior.

Course Outcomes:

1. Understand stages in building a Data Warehouse
2. Understand the need and importance of preprocessingtechniques
3. Understand the need and importance of Similarity and dissimilaritytechniques
4. Analyze and evaluate performance of algorithms for AssociationRules.
5. Analyze Classification and Clusteringalgorithms

SYLLABUS

UNIT –I

Introduction: Why Data Mining? What Is Data Mining?1.3 What Kinds of Data Can Be Mined?1.4 What Kinds of Patterns Can Be Mined? Which Technologies Are Used? Which Kinds of Applications Are Targeted? Major Issues in Data Mining. Data Objects and Attribute Types, Basic Statistical Descriptions of Data, Data Visualization, Measuring Data Similarity and Dissimilarity

UNIT –II

Data Pre-processing: Data Preprocessing: An Overview, Data Cleaning, Data Integration, Data Reduction, Data Transformation and Data Discretization

UNIT –III

Classification: Basic Concepts, General Approach to solving a classification problem, Decision Tree Induction: Working of Decision Tree, building a decision tree, methods for expressing an attribute test conditions, measures for selecting the best split, Algorithm for decision treeinduction.

UNIT –IV

Classification: Alterative Techniques, Bayes' Theorem, Naïve Bayesian Classification, Bayesian Belief Networks

Association Analysis: Basic Concepts and Algorithms: Problem Defecation, Frequent Item Set generation, Rule generation, compact representation of frequent item sets, FP-Growth Algorithm. **(Tan &Vipin)**

UNIT –V

Cluster Analysis: Basic Concepts and Algorithms: Overview: What Is Cluster Analysis? Different Types of Clustering, Different Types of Clusters; K-means: The Basic K-means Algorithm, K-means Additional Issues, Bisecting K-means, Strengths and Weaknesses; Agglomerative Hierarchical Clustering: Basic Agglomerative Hierarchical Clustering Algorithm DBSCAN: Traditional Density Center-Based Approach, DBSCAN Algorithm, Strengths and Weaknesses. **(Tan &Vipin)**

Text Books:

1. Introduction to Data Mining: Pang-Ning Tan & Michael Steinbach, VipinKumar,Pearson.
2. Data Mining concepts and Techniques, 3/e, Jiawei Han, Michel Kamber,Elsevier.

Reference Books:

1. Data Mining Techniques and Applications: An Introduction, Hongbo Du, CengageLearning.
2. Data Mining :VikramPudi and P. RadhaKrishna,Oxford.
3. Data Mining and Analysis - Fundamental Concepts and Algorithms; Mohammed J. Zaki, Wagner Meira, Jr, Oxford
4. Data Warehousing Data Mining & OLAP, Alex Berson, Stephen Smith,TMH.

Code: B17ME3210

INDUSTRIAL ROBOTICS
(Common to ECE & EEE)
(Open Elective)

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course objectives:

1. To give students practice in applying their knowledge of mathematics, science, and Engineering and to expand this knowledge into the vast area of robotics.
2. The students will be exposed to the concepts of robot kinematics, Dynamics, Trajectory planning.
3. Mathematical approach to explain how the robotic arm motion can be described.
4. The students will understand the functioning of sensors and actuators.

Course Outcomes:

Upon successful completion of this course you should be able to:

1. Identify various robot configuration and components,
2. Select appropriate actuators and sensors for a robot based on specific application
3. Carry out kinematic and dynamic analysis for simple serial kinematic chains.
4. Perform trajectory planning for a manipulator by avoiding obstacles

SYLLABUS

UNIT-I

Introduction: Automation and Robotics, CAD/CAM and Robotics – An over view of Robotics – present and future applications – classification by coordinate system and control system.

UNIT – II

Components Of The Industrial Robotics: Function line diagram representation of robot arms, common types of arms. Components, Architecture, number of degrees of freedom – Requirements and challenges of end effectors, determination of the end effectors, comparison of Electric, Hydraulic and Pneumatic types of locomotion devices.

UNIT – III

Motion Analysis: Homogeneous transformations as applicable to rotation and translation – problems.

Manipulator Kinematics: Specifications of matrices, D-H notation joint coordinates and world coordinates Forward and inverse kinematics – problems.

Differential transformation and manipulators, Jacobians– problems Dynamics: Lagrange – Euler and Newton – Euler formulations – Problems.

UNIT IV

General considerations in path description and generation.Trajectory planning and avoidance of obstacles, path planning, Skew motion, joint integrated motion –straight line motion – Robot programming, languages and software packages-description of paths with a robot programming language..

UNIT V

Robot Actuators and Feed Back Components:

Actuators: Pneumatic, Hydraulic actuators, electric & stepper motors.

Feedback components: position sensors – potentiometers, resolvers, encoders – Velocity sensors.

Robot Applications in Manufacturing:

Material Transfer - Material handling, loading and unloading- Processing - spot andcontinuous arc welding & spray painting - Assembly and Inspection.

Text Books:

1. Industrial Robotics / GrooverM P /PearsonEdu.
2. Robotics and Control / Mittal R K &NagrathI J /TMH.

Reference Books:

1. Robotics / Fu K S/ McGrawHill.
2. Robotic Engineering / Richard D. Klafter, PrenticeHall
3. Robot Analysis and Control / H. Asada and J.J.E. Slotine/ BSP BooksPvt.Ltd.
4. Introduction to Robotics / John J Craig / PearsonEdu.

Code: B17EE3209

POWER ELECTRONICS
(Open Elective)

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives: Students will

1. Understand the concepts of Power Semiconductor devices and their applications.
2. Understand the need of Energy conversion and effective implementation methods.

Course outcomes:

Students are able to

1. Explain the principle of operation of thyristor, modern power semiconductor devices and necessity of series and parallel connection of thyristors.
2. Explain the operation of Firing and Commutation techniques.
3. Evaluate the phase controlled rectifiers with different loads.
4. Analyse different Choppers, Cyclo-converter and AC voltage Controller configurations.
5. Investigate harmonic reduction techniques for inverters based on PWM techniques.

SYLLABUS

UNIT I: MODERN POWER SEMI CONDUCTOR DEVICES

Thyristors – Silicon Controlled Rectifiers (SCRs) – BJT – Power MOSFET – Power IGBT and their characteristics. Basic theory of operation of SCR – Static characteristics and Dynamic characteristics of SCR - Turn on and Turn off times – Turn on and turn off methods. Two transistor analogy of SCR -Series and parallel connections of SCRs Snubber circuit details – Numerical problems.

UNIT II: THYRISTORFIRING AND COMMUTATIONCIRCUITS

SCR trigger circuits-R, RC and UJT triggering circuits. The various commutation methods of SCRs-Load commutation- Resonant Pulse Commutation- Complementary Commutation-Impulse Commutation- External Pulse Commutation Techniques. Protection of SCRs

UNIT III: PHASE CONTROLLED RECTIFIERS

Principles of phase controlled rectification -Study of Single phase and three-phase half controlled and full controlled bridge rectifiers with R, RL, RLE loads. Effect of source inductance. Dual converters- circulating current mode and circulating current free mode-control strategies. Numerical problems.

UNIT IV:CHOPPERS, CYCLONVERTER AND AC VOLTAGE CONTROLLER

Classification of Choppers A, B, C, D and E, Switching mode regulators-Study of Buck, Boost and Buck-Boost regulators, Cukregulators . Principle of operation of Single phase bridge type Cycloconverter and their applications. Single phase AC Voltage Controllers with R and RL loads.

UNIT-V INVERTERS

Principle of operation of Single phase Inverters -Three phase bridge Inverters (180^0 and 120^0 modes)-voltage control of inverters-Single pulse width modulation- multiple pulse width modulation, sinusoidal pulse width modulation. Harmonic reduction techniues- Comparison of Voltage Source Inverters and Current source Inverters.

Text Books:

1. Power electronics - P.S. Bimbhra- Khanna Publishers, 4th Edition
2. Power electronics – M.D. Singh & K.B. Kanchandhani, Tata McGraw – Hill Publishing Company, 2nd edition.

Reference Books:

1. Power Electronics: Circuits Devices and Applications – M.H. Rashid, Prentice Hall of India, 3rd edition.
2. Power Electronics – VedamSubramanyam, New Age International (p) Limited, Publishers.
3. Power Electronics – P.C. Sen, Tata McGraw-Hill Publishing.
4. Thyristorised power Controllers – G.K. Dubey, S.R Doradra, A. Joshi and R.M.K. Sinha, New Age international Pvt Ltd. Publishers latest edition

Code: B17EC3207

BIO MEDICAL ENGINEERING
(Open Elective)

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives: The objectives of this course are to:

1. Describe the origin, properties and suitable models of important biological signals such as ECG and EEG.
2. Introduce students to basic signal processing techniques in analyzing biological signals.
3. Develop the mathematical and computational skills relevant to the field of biomedical signal processing.
4. Develop a thorough understanding on basics of ECG signal compression algorithms.
5. Increasethestudent'sawarenessofthecomplexityofvariousbiologicalphenomena and cultivate an understanding of the promises, challenges of the biomedical engineering.

Course outcomes: At the end of the course, students will be able to:

1. Possess the basic mathematical skills necessary to analyze ECG and EEG signals.
2. Possess the basic scientific skills necessary to analyze ECG and EEG signals.
3. Possess the basic computational skills necessary to analyze ECG and EEG signals.
4. Apply classical and modern filtering and compression techniques for ECG and EEG signals.
5. Develop a thorough understanding on basics of ECG and EEG feature extraction.

SYLLABUS

UNIT-I:
Introduction to Biomedical Signals: The nature of Biomedical Signals, Examples of Biomedical Signals, Objectives and difficulties in biomedical analysis, Basic electrocardiography, ECG lead systems, ECG signal characteristics, Simple signal conversion systems, Conversion requirements for biomedical signals, Signal conversion circuits.

UNIT-II:
Signal Averaging: Basics of signal averaging, signal averaging as a digital filter, a typical averager, software for signal averaging, limitations of signal averaging, Adaptive Principal noise canceller model, 60-Hz adaptive cancelling using a sine wave model, other applications of adaptive filtering.

UNIT-III:
Data Compression Techniques: Turning point algorithm, AZTEC algorithm, Fan algorithm, Huffman coding, data reduction algorithms The Fourier transform, Correlation, Convolution, Power spectrum estimation, Frequency domain analysis of the ECG.

UNIT-IV:

Cardiological signal processing: Basic Electrocardiography, ECG data acquisition, ECG lead system, ECG signal characteristics (parameters and their estimation), Analog filters, ECG amplifier, and QRS detector, Power spectrum of the ECG, Bandpass filtering techniques, Differentiation techniques, Template matching techniques, A QRS detection algorithm, Real-time ECG processing algorithm, ECG interpretation, ST segment analyzer, Portable arrhythmia monitor.

UNIT-V:

Neurological signal processing: The brain and its potentials, The electrophysiological origin of brain waves, The EEG signal and its characteristics (EEG rhythms, waves, and transients), Correlation, Detection of EEG rhythms, Template matching for EEG, spike and wave detection

Text Books:

1. Biomedical Digital Signal Processing- Willis J. Tompkins, PHI 2001.
2. Biomedical Signal Processing Principles and Techniques- D C Reddy, McGrawHill publications 2005

Reference Books:

1. Biomedical Signal Analysis-Rangaraj M. Rangayyan, John Wiley & Sons 2002

Code: B17CS3216

ARTIFICIAL NEURAL NETWORKS
(Open Elective)

Lecture	**: 3 Periods**	**Int. Marks**	**: 30**
Tutorial	**: 1 Period.**	**Ext. Marks**	**: 70**
Exam	**: 3 Hrs.**	**Credits**	**: 3**

Course Objectives:

1. To Introduce the concept of Artificial Neural Networks , Characteristics, Models of Neuron, Learning Rules, Learning Methods, Stability and Convergence
2. To study the basics of Pattern Recognition and Feed forward Neural Networks
3. To study the basics of Feedback neural networks and Boltzmann machine
4. To introduce the Analysis of Feedback layer for different output functions, Pattern Clustering and Mapping networks
5. To study the Stability, Plasticity, Neo cognitron and Different applications of Neural Networks

Course Outcomes

1. This Course introduces Artificial Neural Networks and Learning Rules and Learning methods
2. Feed forward and Feedback Neural Networks are introduced
3. Applications of Neural Networks in different areas are introduced.

SYLLABUS

UNIT-I : Basics of Artificial Neural Networks
Introduction: Biological Neural Networks, Characteristics of Neural Networks, Models of Neuron, Topology, Basic Learning Rules
Activation and Synaptic Dynamics: Activation Dynamic Models, Synaptic Dynamic Models, Learning Methods, Stability & Convergence, Recall in Neural Networks

UNIT-II: Functional Units of ANN for Pattern Recognition Tasks: Pattern Recognition problem Basic Fundamental Units, Pattern Recognition Tasks by the Functional Units
Feed forward Neural Networks: Analysis of Pattern Association Networks, Analysis of Pattern Classification Networks, Analysis of Pattern Mapping Networks

UNIT-III:
Feedback Neural Networks: Analysis of linear auto adaptive feed forward networks, Analysis of pattern storage Networks, Stochastic Networks & Stimulated Annealing, Boltzmann machine

UNIT-IV:
Competitive Learning Neural Networks**:** Components of a Competitive Learning Network, Analysis of Feedback layer for Different Output Functions, Analysis of Pattern Clustering Networks and Analysis of Feature Mapping Network
Architectures for Complex Pattern Recognition Tasks**:** Associative memory, Pattern mapping Stability – Plasticity dilemma: ART, temporal patterns, Pattern visibility: Neocognitron

UNIT-V:
Applications of Neural Networks: Pattern classification, Associative memories, Optimization, Applications in Image Processing, Applications in decisionmaking

Text Book:

1. B.Yagnanarayana"Artificial Neural Networks", PHI

Reference Books:

1. LaureneFausett ,"Fundamentals of Neural Networks", PearsonEducation
2. Simon Haykin , "Neural Networks", SecondEdition

Code: B17 EC 3208

MICROPROCESSORS AND MICROCONTROLLERS LAB

Lab	**: 3 Periods**	**Int.Marks**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

1. To understand the basics of Microprocessors 8085
2. To understand the basics of Microprocessors 8086 and Microcontroller 8051.
3. To understand the internal organization of INTEL 8085,
4. To understand the internal organization of INTEL 8085,8086 Microprocessors and Microcontroller 8051.
5. Developing Assembly Language Programs using the instruction sets of microprocessors and microcontroller and to study the interfacing of the processor with various peripheral devices.

Course Outcomes: The objective of this course is

1. To become familiar with the instruction set of Intel microprocessors and microcontroller.
2. To familiarize with Assembly language programming.
3. The accompanying lab is designed to provide practical hands-on experience with microprocessor software applications and interfacing techniques.

SYLLABUS

Experiments Based On ALP (8085):

1. a. Assume that byte of data is stored at memory location 'X'. Write an ALP which tests bit 5 of thisdata. Write 'FF' in the location 'X+1' if the bit 5 is '1' and '00'if bit 5 is '0'.
   b. Check the zero condition of this number and write '00' at location'Y' if it is '0'and 'FF' at 'Y' if non zero.
   c. For data value in the location 'X' compute the number of logic 1's and store the result in the location 'Y+1'.
2. a. Write an ALP to swap the contents of location 'X' and 'X+1' using BC & HL Register pairs.
   b. By using above logic, write an ALP to transfer a block of data into another block.
3. a. Write an ALP to add and subtract two eight bit Number stored in the location 'X' and 'X+1' by assuming that content of 'X' is greater than content of 'X+1'
   b. Modify this program to add two 16 bit numbers without using DAD instruction.
4. Two 8 bit numbers 34H and 43H are stored in locations 'X' and 'X+1' compute the product of these two numbers using
   a. Repetitive addition method b. Shift and add method
5. The number of the bytes of a block of data is in location 'X' and data starts from location 'X+1' onwards defining a stack pointers. Write an ALP to arrange this sequence of data in reverse order. Keep the reverse sequence from 'Y' onwards.
6. The number of bytes of a block of data is location 'X' and data starts from location 'X+1'onwards. Arrange this block of data in ascending order by using bubble sorting technique
7. Using 8279 write an ALP to generate the message of 4 characters. Activate the LED's individually and make the display ON &OFF for every 0.5 seconds

Experiments Based On ALP (8086):

1. Write an 8086 ALP to addition of two-32 bit numbers stored in the memory location 6000H and 6004H.Store the result at location 6008H.

2. Write an 8086 ALP to Subtraction of two-32 bit numbers stored in the memory location 6000H and 6004H.Store the result at location 6008H.

3. Write an 8086 ALP to Multiply two 16 bit numbers stored in the memory location 9000H and 9002H. Store the result at location 9005H.

4. Write an 8086 ALP to divide 32bit dividend with 16 bit devisor stored in the memory location 5000H and 5004H respectively. Store the quotient at 5006H and the remainder in location 5008H.

5. Write an 8086 program to add four digit BCD numbers present in memory locations 15000 H and 15002 H.Store the result at memory location 15004 H.

6. Write an 8086 program to sort the given block of data using bubble sorting technique. Assume number bytes of block of data stored in the memory location 3000H and Actual block of data starts from 3001H onwards.

Experiments based on Interfacing and Microcontroller (8051):

Programs on Data transfer instructions using 8051 Microcontroller
Programs on Arithmetic and Logical instructions using 8051 Microcontroller

References:

1. Lab Manual

Code: B17EC3209

VLSI LAB

Lab	**: 3 Periods**	**Int. Marks**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 2**

Course Objectives:

1. Learn and understand the basics of NMOS and CMOS logic and able to design the schematic diagrams of basic combinational and sequential circuits using CMOS logic with necessary EDA tools (Mentor Graphics/Cadence Tools)
2. Draw the layout diagrams of combinational and sequential to perform the following experiments using CMOS 130nm Technology with necessary EDA tools (Mentor Graphics/Cadence Tools)
3. This laboratory course enables student to get practical experience in design and evaluation of performance metrics.

Course Outcomes: Upon completion of the course, students will be able to

5. Learn the work flow of mentor graphic tools/Cadence tools forlogic gates, Combinational and Sequential circuits.
6. Simulate combinational and sequential circuits with EDA tools
7. Acquire Knowledge of analysis of combinational and sequential circuits using CMOS 130nm Technology.
8. Acquire practical experience in drawing layouts using Cadence/Mentor Graphics CAD tools.

List of Experiments:

1. Design and implementation of an inverter
2. Design and implementation of universal gates (NAND, NOR)
3. Design and implementation of AND, OR gates
4. Design and implementation of EXOR gate using minimum no. of transistors
5. Design and implementation of 2 to 1 Multiplexer
6. Design and implementation of full adder
7. Design and implementation of full subtractor
8. Design and implementation of D-latch
9. Design and implementation 3-bit asynchronous counter
10. Design and Implementation of static 1-bit RAM cell

Equipment Required:

1. Mentor Graphics/Cadence tools software-latest version
2. Personal computer with necessary peripherals.

References:

1. Lab Manual

Code: B17BS3201

EMPLOYABILITY SKILLS
(Common to all Branches)

Theory	**: 3 Periods (VA-2+QA-1)**	**Int.Marks**	**: 30**
Exam	**: 3 Hrs.**	**Ext.Marks**	**: 70**
		Credits	**: 1**

Part-A: Verbal Aptitude and Soft Skills-II

Course objectives:

1. To expose the students to bettering sentence expressions and also forming equivalents.
2. To instill reading and analyzing techniques for better comprehension of written discourses.
3. To create awareness among the students on the various aspects of writing, organizing data, preparing reports, and applying their writing skills in their professional career.
4. To inculcate conversational skills, nuances required when interacting in different situations.
5. To build/refine the professional qualities/skills necessary for a productive career and to instill confidence through attitude building.

Course Outcomes:

The students will be able to

1. Construct coherent, cohesive and unambiguous verbal expressions in both oral and written discourses.
2. Analyze the given data/text and find out the correct responses to the questions asked based on the reading exercises; identify relationships or patterns within groups of words or sentences
3. Write paragraphs on a particular topic, essays (issues and arguments), e mails, summaries of group discussions, reports, make notes, statement of purpose(for admission into foreign universities), letters of recommendation(for professional and educational purposes).
4. Converse with ease during interactive sessions/seminars in their classrooms, compete in literary activities like elocution, debates etc., raise doubts in class, participate in JAM sessions/versant tests with confidence and convey oral information in a professional manner.
5. Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, tailor make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.

SYLLABUS

UNIT -I (VA)

Sentence Improvement (finding a substitute given under the sentence as alternatives), Sentence equivalence (completing a sentence by choosing two words either of which will fit in the blank), cloze test (reading the written discourse carefully and choosing the correct options from the alternatives and filling in the blanks), summarizing and paraphrasing.

UNIT- II (VA)

Types of passages (to understand the nature of the passage), types of questions (with emphasis on inferential and analytical questions), style and tone (to comprehend the author‘s intention of writing a passage), strategies for quick reading(importance given to skimming, scanning), summarizing ,reading between the lines, reading beyond the lines, techniques for answering questions related to vocabulary (with emphasis on the context), supplying suitable titles to the passage, identifying the theme and central idea of the given passages.

UNIT- III (VA)
Punctuation, discourse markers, general Essay writing, writing Issues and Arguments(with emphasis on creativity and analysis of a topic), paragraph writing, preparing reports, framing a ‗Statement of purpose‘, ‗Letters of Recommendation‘, business letter writing, email writing, writing letters of complaints/responses. picture perception and description, book review.

UNIT-IV (VA)
Just a minute sessions, reading news clippings in the class, extempore speech, telephone etiquette, making requests/suggestions/complaints, elocutions, debates, describing incidents and developing positive non verbal communication, story narration, product description.

UNIT-V (SS)
Employability Skills – Significance — Transition from education to workplace - Preparing a road map for employment – Getting ready for the selection process, Awareness about Industry / Companies – Importance of researching your prospective workplace - Knowing about Selection process - Resume Preparation: Common resume blunders – tips, Resume Review, Group Discussion: Essential guidelines – Personal Interview: Reasons for Rejection and Selection.

Reading/ Listening material:

1. Guide to IELTS, Cambridge University Press
2. Barron‘s GRE guide.
3. Newspapers like ‗The Hindu, ‗Times of India, ‗Economic Times.
4. Magazines like Frontline, Outlook and Business India.
5. News channels NDTV, National News, CNN

Text Books:

1. Objective English and Verbal Reasoning by R S Agarwal.
2. Communication Skills by Sanjay Kumar and PushpaLatha, Second Edition, OUP.
3. Business Correspondence and Report Writing – A Practical Approach to Business and Technical Communication by R C Sharma and Krishna Mohan.
4. Soft Skills & Employability Skills by SaminaPillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
5. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. Oxford Guide to Effective Writing and Speaking by John Seely.
2. Collins Cobuild English Grammar by Collins
3. The Art of Public Speaking by Dale Carnegie
4. The Leader in You by Dale Carnegie
5. Emotional Intelligence by Daniel Golman
6. Stay Hungry Stay Foolish by RashmiBansal

Part-B: Quantitative Aptitude-II

Course objectives:

The objective of introducing quantitative aptitude-II is:

1. To refine concepts related to quantitative aptitude. – SOLVING PROBLEMS OF DI and accurate values using averages, percentages.

2. To inculcate logical thinking by exposing the students to puzzles and reasoning related questions.

3. To familiarize the students with finding out accurate date and time related problems.

4. To enable the students solve the puzzles using logical thinking.

5. To expose the students to various problems based on geometry and mensuration.

Course Outcomes:

1. The students will be able to perform well in calculating different types of data interpretation problems.

2. The students will perform efficaciously on analytical and logical problems using various methods.

3. Students will find the angle measurements of clock problems with the knowledge of calendars and clock.

4. The students will skillfully solve the puzzle problems like arrangement of different positions.

5. The students will become good at solving the problems of lines, triangulars, volume of cone, cylinder and so on.

SYLLABUS

UNIT I: Averages, mixtures and allegations, Data interpretation Understanding of AM,GM,HM-Problems on averages, Problems on mixtures standard method. Importance of data interpretation: Problems of data interpretation using line graphs, Problems of data interpretation using bar graphs, Problems of data interpretation using pie charts, Problems of data interpretation using others.

UNIT II: Puzzle test, blood Relations, permutations, Combinations and probability Importance of puzzle test, Various Blood relations-Notation to relations and sex making of family Tree diagram, Problems related to blood relations, Concept of permutation and combination, Problems on permutation, Problems on combinations, Problems involving both permutations and combinations, Concept of probability-Problems on coins, Problems on dice, Problems on cards, Problems on years.

UNIT III: Periods,Clocks, Calendars, Cubes and cuboids Deriving the formula to find the angle between hands for the given time, finding the time if the angle is known, Faulty clocks, History of calendar-Define year, leap year, Finding the day for the given date, Formula and method to find the day for the given date in easy way, Cuts to cubes, Colors to cubes, Cuts to cuboids, Colors to cuboids.

UNIT IV: Puzzles Selective puzzles from previous year placement papers, sitting arrangement, problems- circular arrangement, linear arrangement, different puzzles.

UNIT V: Geometry and Mensuration Introduction and use of geometry-Lines, Line segments, Types of angles, Intersecting lines, Parallel lines, Complementary angles, supplementary angles, Types of triangles-Problems on triangles, Types of quadrilaterals-Problems on quadrilaterals, Congruent triangles and properties, Similar triangles and its applications, Understanding about circles-Theorems on circles, Problems on circles, Tangents and circles, Importance of mensuration-Introduction of cylinder, cone, sphere, hemi sphere.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi
4. More puzzles by shakunataladevi
5. Puzzles by George summers.

Reference Books:

1. Barron‘s by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt.Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from ‗IMS, Career Launcher and Time‘ institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

Code: B17 BS 3203

ADVANCED CODING
(Common to ECE & EEE)

Lab	**: 3 Periods**	**Int.Marks**	**: 50**
Exam	**: 3 Hrs.**	**Ext. Marks**	**: 50**
		Credits	**: 1**

Course Objectives

1. To understand the basics of modular programming
2. To learn about ADT, Linked Lists and Templates.
3. To investigate different methods to find time complexities.
4. To learn about Java collections and Libraries

Course Outcomes

At the end of the course, a student should be able to:

1. Acquire coding knowledge on essential of modular programming
2. Acquire Programming knowledge on linked lists
3. Acquire coding knowledge on ADT
4. Acquire knowledge on time complexities of different methods
5. Acquire Programming skill on Java libraries and Collections

SYLLABUS

UNIT I Review Coding essentials and modular programming

Introduction to Linear Data, Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding.

Introduction to modular programming: Formation of methods, Methods: Signature and definition, Inter-method communication, Data casting & storage classes, Recursions

UNIT II Linear Linked Data

Introduction to structure pointer, Creating Links Basic problems on Linked lists, Classical problems on linked lists. Circular Linked lists, Operations on CLL, Multiple links, Operations on Doubly linked lists

UNIT III Abstract Data-structures

Stack data-structure, Operations on stack, Infix/Prefix/Post fix expression evaluations, Implementation of stack using array, Implementation of stack using linked lists.

Queue data-structure: Operations on Queues, Formation of a circular queue, Implementation of queue using stack, Implementation of stack using array, Implementation of stack using linked lists

UNIT IV Running time analysis of code and organization of linear list data

Code evaluation w.r.t running time, Loop Complexities, Recursion complexities, Searching techniques: sequential Vs. binary searching.

Organizing the list data, Significance of sorting algorithms, Basic Sorting Techniques: Bubble sort, selection sort, Classical sorting techniques: Insertion sort, Quick sort, Merge sort.

UNIT V Standard Library templates and Java collections
Introduction to C++ language features, Working on STLs, Introduction to Java as Object Oriented language, Essential Java Packages, Coding logics.

Note: This course should focus on Problems

References:

1. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
3. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), EnzoMarinari (Author), Giovanni Organtini, World Scientific .
4. ObjectOrientedProgrammingin C++: N. Barkakati, PHI.
5. ObjectOrientedProgrammingthrough C++ byRobatLaphore.
6. https://www.geeksforgeeks.org/.
7. https://www.tutorialspoint.com/

Code: B17BS3206

IPR & PATENTS
(Common to CSE, ECE & IT)

Tutorial	**: 2 Periods**	**Credits**	**: 0**

Course Objectives:

1. To introduce the idea of tangible and intangible property and its protection.
2. To familiarize with the frameworks for protection of intellectual property.
3. To layout the procedures to claim rights over intellectual property.

Course Outcomes:

After successful completion of the course, the student shall be able to

1. Identify various types of intangible property that an engineering professional could generate in the course of his career.
2. Distinguish between various types of protection granted to Intellectual Property such as Patents, Copy Rights, Trademarks etc.,
3. List the steps involved in getting protection over various types of intellectual property and maintaining them.
4. Take precautions in writing scientific and technical reports without plagiarism.
5. Help micro, small and medium entrepreneurs in protecting their IP and respecting others IP as part of their business processes.

SYLLABUS

UNIT I

Intellectual Property Law: Basics - Types of Intellectual Property - Innovations and Inventions - Trade related Intellectual Property Rights – Agencies Responsible for Intellectual Property Registration – Infringement - Compliance and Liability Issues

UNIT II

Principles of Copyright – Subject Matters of Copyright – Rights Afforded by Copyright Law –Copyright Ownership–Copyright Formalities and Registration – Limitations – Infringement of Copyright - Plagiarism and difference between Copyright infringement and Plagiarism

UNIT III

Introduction to Trade Mark – Trade Mark Registration Process – Post registration procedures – Trade Mark maintenance– Infringement – Dilution of Ownership of Trade Mark – Likelihood of confusion – Trade Mark claims – Trade Marks Litigation – International Trade Mark Law

UNIT IV

Introduction to Patent Law – Rights and Limitations – Rights under Patent Law – Patent Requirements – Ownership and Transfer – Patent Application Process and Granting of Patent – Patent Infringement and Litigation – International Patent Law – Double Patenting

UNIT V

Introduction to Trade Secrets – Maintaining Trade Secret – Physical Security – Employee Access Limitation – Employee Confidentiality Agreement – Trade Secret Law – Unfair Competition – Trade Secret Litigation – Breach of Contract – Applying State Law.

Text Books:

1. KompalBansal&ParikshitBansal "Fundamentals of Intellectual Property for Engineers", BS Publications
2. PrabhuddhaGanguli: "Intellectual Property Rights" Tata McGraw –Hill, New Delhi
3. R. Radha Krishnan, S. Balasubramanian: "Intellectual Property Rights: Text and Cases",Excel Books, New Delhi.

Reference Books:

1. Deborah E.Bouchoux: "Intellectual Property". Cengage learning , NewDelhi
2. Richard Stim: "Intellectual Property", Cengage Learning, New Delhi.